

YONAS

A QUICK REVISSION ON

Common short cases

and

Procedures in pediatrics

VOLUME I

First edition

2010 ec

**UNIVERSITY OF GONDAR
COLLAGE OF MEDICINE AND HEALTH SCIENCE**

A surreal landscape painting. In the foreground, a large, circular structure resembling a library or a giant bookshelf is built into a lush green forest. The structure is composed of many horizontal layers, each filled with books. A river flows through the center of the structure, starting from the top and winding its way down. The river is a light blue color, contrasting with the green of the forest. In the background, the forest continues to a distant horizon under a soft, hazy sky. In the bottom right corner of the painting, the word 'YERNA' is visible.

በጠበቡ ጠቢባን ፤
ተጠበን ጠበን ፤
ለዚህ ቢቃን !!!

መታሰቢያነቱ

በ 2009 / 2010 ዓ.ም በነበረው አለመረጋጋት ምክንያት
ህይወታቸውን ላጡ ኢትዮጵያውያን በሙሉ ይሁንልን !!!

የጌታ ፈቃድ ይሁን ብለን ዝም አልን !!!

የሐ.ሥ. 21 ፤ 14



ዮናስ ገሻዬ

YONAS GASHAYE (MED IV)

The author

Preface

*It is observed that students, especially medicine 4th year, public health officer and other health science students, often find difficult to prepare themselves for clinical physical examinations and how to do common procedures in pediatrics after theory papers. They need to know basic physical examination methods with relevant steps, technique, differential diagnosis and investigations of the specific cases. They also need to know indications, contraindications, instruments, basic procedures and complications of post procedures. Keeping this in mind, this book “*YONAS* a quick revision on COMMON SHORT CASES and PROCEDURES IN PEDIATRICS” has been brought out to go through quickly prior to examinations and procedures. This also useful to 5th year medical students (SMS) and Interns as well.*

I hope this book will earn its value in its own way in a student circle.

I thank everybody who are back bone of this title. Any criticisms are well accepted.

For your comment, suggestion or question, inbox at dagnachewdg.21@gmail.com

Yonas Gashaye (DG)

Acknowledgement

I am happy to bring out this new book of clinical and practical importance, YONAS a quick revision on common short cases and procedures in pediatrics, first edition. This is due to constant help and support of many.

*I sincerely thank to:- Yenew Berhan (SMS), Zelalem Chanie (SMS), Yonathan Mulugeta (SMS), Firiehiwot Alemu (SMS), Tedros Haile (4th yr), Enyew Sereke (4th yr), Oliad Bekako (4th yr), Yenas Mekonen (4th yr), Mehari Manaye (4th yr), Tadesse Sileshi (4th yr), Amanuel Mengist (4th yr), Deresse (4th yr), Elleni Alebel (4th yr), Eshetie Dargie (4th yr), Yidnekachew (4th yr), all **Mekdi** (batch of 2006 ec) and **Ebola** (batch of 2007 ec) batches, for their direct or indirect contribution to this edition.*

Special thanks to:-

- 1. GOD and St. Marry*
- 2. My mom, dad, sisters and brothers*
- 3. Amelework Yimer and all her family*
- 4. Sileshi Kebede and all his family*
- 5. Priest Getnet Bizuneh and all his family*
- 6. Tigist Birhanu*
- 7. Yenew Berhan*
- 8. Aster Yimer*
- 9. All my uncles (Dessalegn, Worku, Ayinalem, Zelalem & Tilahun)*

Yonas Gashaye (DG)

Acronyms used in this book

- *CVS* = *cardio vascular system*
- *MSS* = *musculoskeletal system*
- *Yr* = *year*
- *Pt* = *patient*
- *Ec* = *Ethiopian calendar*
- *P/E* = *physical examination*
- *IDA* = *iron deficiency anemia*
- *SMS* = *senior medical student*
- *AIDS* = *acquired immunodeficiency syndrome*
- *TORCHS* = *toxoplasmosis, rubella, cytomegalovirus, herpes simplex virus, syphilis & others*
- *Hgb* = *hemoglobin*
- *Hct* = *hematocrit*
- *gm* = *gram*
- *dL* = *deciliter*
- *mL* = *milliliter*
- *L* = *liter*
- *Exp* = *expiration*
- *Insp* = *inspiration*
- *URT* = *upper respiratory tract*
- *Echo* = *echocardiography*
- *ECG* = *electrocardiography*
- *CT* = *computed tomography*
- *MRI* = *magnetic resonance imaging*
- *CXR* = *chest x-ray*
- *VOG* = *University of Gondar*
- *GCMHS* = *Gondar Collage of Medicine and Health Science*
- *BBS* = *bronchial breath sound*
- *CHF* = *congested heart failure*
- *GERD* = *gastro esophageal reflux disease*
- *TB* = *tuberculosis*
- *Vs* = *versus*
- *SLE* = *systemic lupus erythematosus*
- *RA* = *rheumatic arterities*
- *Med* = *medicine*

CONTENTS

HEENT	10
RESPIRATORY SYSTEM EXAMINATION	18
CVS EXAMINATION	49
ABDOMINAL EXAMINATION	76
MSS EXAMINATION.....	101
MASS EXAMINATION	105
WOUND EXAMINATIN	109
LOWER MOTOR EXAMINATION	113
MALNUTRITION.....	120
ANEMIA	128
IRON DEFICIENCY ANEMIA.....	138
RICKETS	146
DEHYDRATION	152
EDEMA.....	161
DOWN SYNDROME	164
BURN.....	169
SHOCK	176
POISONING	189
COMMON PEDIATRIC PROCEDURES.....	203
PERIPHERAL IV INSERTION	204
LUMBAR PUNCTURE	211
BONE MARROW ASPIRATION AND BIOPSY.....	217
INTRAOSSEOUS INFUSION	223
FEMORAL VENOUS CATHETERIZATION	228
UMBILICAL VEIN CATHETERIZATION	236
EXCHANGE TRANSFUSION OF NEW BORN	241
SUBDURAL TAP.....	248
THORACENTESIS.....	251
CHEST TUBE INSERTION	257
PERICARDIOCENTESIS	263

NASOGASTRIC TUBE INSERTION	269
ABDOMINAL PARACENTESIS	273
TIP (TRIADS)	277
REFERENCES	279

<i>Seizure</i>	} 2nd edition
<i>Coma</i>	
<i>Skin lesion</i>	
<i>Growth & development</i>	
<i>Fluid & electrolyte</i>	
<i>History taking, neonatal physical examination and long cases volume two</i>	

SECTION – ONE

COMMON SHORT

CASES

HEENT

Question

Do HEENT examination

- Head, Eye, Ear, Nose, Mouth And Throat

HEAD

Look for

- Hair – amount, color, texture, distribution, pluckability
- Shape – caput quadratum, frontal bossing, ...
- Size – normal, big or small
- Masses
- Depression
- Tenderness of scalp
- Fontanel
 - Size & closure
 - Surface
 - ✓ Flat or slightly bulged – normal
 - ✓ Depressed – dehydration
 - ✓ Bulged
- Facial puffiness

NB.

- ✓ *Posterior fontanel is (1x1 cm) closed at the age of 2 months*
- ✓ *Anterior fontanel is (2±1 x2±1 cm), maximally enlarged & closed at the age of 6 months & 18 months respectively*

Ddx

➤ Macrocephaly

- Rickets
- Hydrocephalus (congenital or acquired)
- Hypothyroidism
- Achondroplasia
- Storage disease
- Intracranial hemorrhage
- Gigantism
- Familial- autosomal dominant

➤ Microcephaly

- Severe malnutrition
- AIDS
- TORCH infection
- Hyperthyroidism
- Craniosynstosis
- Dawn, Patau or Edward syndrome

➤ Frontal bossing

- Rickets
- Thalasemia major
- Congenital syphilis
- Achondroplasia

- Hurler's syndrome
- Cleidocranial syndrome
- Ectodermal dysplasia
- Pyknodysostosis
- Bulged fontanel
 - Crying infant
 - Meningitis
 - Intracranial bleeding
 - Hydrocephalus
 - Tumor
 - Pseudotumor cerebri
 - Hyperparathyroidism
- Wide fontanel
 - Rickets
 - Hypothyroidism
 - Hydrocephalus
 - Congenital rubella syndrome
 - Osteogenesis imperfecta
 - Prematurity
 - Down , Patau syndrome, Edward syndrome
- Craniotables
 - Physiological
 - Rickets
 - Congenital syphilis
 - Hydrocephalus
 - Osteogenesis imperfecta

EYE

Look for

- ✓ Excessive tearing / discharge→*conjunctivitis, keratitis, ...*
- ✓ Periorbital edema→*Nephrotic syndrome, kwashiorkor, ...*
- ✓ Orbital mass →*retinoblastoma...*
- ✓ Sunkening of eyeball→*dehydration, marasmus, ...*
- ✓ Palpebral fissure→*normal, slant (down syndrome)*
- ✓ Sclera →*icteric, non-icteric, bitot's spot*
- ✓ Conjunctiva→*pink, pale, injection, hemorrhage*
- ✓ Lid lag, lid retraction→*Graves' disease*
- ✓ Xanthelasma →*dyslipidemia*

Lid lag

- Steady the patient's head with one hand
- Ask the patient to look at your finger
- Ask the patient to look up and down following your finger
- The lid may lag while the eyeball move downward and the upper sclera become visible
 - ✓ Thyrotoxicosis
 - ✓ Graves' disease

Lid retraction

- Visibility of upper sclera at rest
- Due to spasm of upper eyelid

Xanthelasma

- Subcutaneous lipid deposit at periorbital area
- Signify presence of lipid disorder (dyslipidemia)

Ddx

- Orbital mass
 - Rhabdomyosarcoma
 - Neuroblastoma (metastasis)
 - Orbital cellulitis
 - Retinoblastoma

EAR

Look for

- ✓ Counter of pinna
- ✓ Set of the ear – normal, low or high
- ✓ Discharge
- ✓ Skin color change
- ✓ Swelling
- ✓ Mastoid & tragus tenderness

NB; Low set ear

- ✓ *Draw an imaginary line b/n inner & outer canthi w/c bisect the ears*
- ✓ *Normally divides into upper 1/3rd & lower 2/3rd portions*
- ✓ *When less than 20% comes above this line, it is low set ear*

Ddx

➤ Low set ear

- Dawn syndrome
- Turner syndrome
- Trisomy 17-18, 13-15
- Renal agenesis (potter facies)
- Treacher-Collins syndrome
- Cri-du-chat syndrome

NOSE

Look for

- ✓ Nasal bridge, septum, polyp
- ✓ Discharge
- ✓ Active bleeding
- ✓ Sinus tenderness

NB

- *Both the ethmoidal and maxillary sinuses are present at birth, but only the ethmoidal sinus is pneumatized*
- *The maxillary sinuses are not pneumatized until 4 year of age*
- *The sphenoidal sinuses are present by 5 year of age*
- *Frontal sinuses begin development at age 7–8 year and are not completely developed until adolescence*

MOUTH & THROAT

Look for

- ✓ Open mouth, protruded tongue
- ✓ Lip ulcer, fissures and cracks
- ✓ Central cyanosis
- ✓ Mouth odor - haluthosis, fetor hepaticus, acetone breath, ...
- ✓ Buccal mucosa – wet, dry, pink, pale, ...
- ✓ Gum bleeding, ulcers
- ✓ Dentition
- ✓ Tooth caries, extractions and dentures
- ✓ Tongue coating, fissures, atrophy
- ✓ Uvula – central or not, color (pink, erythematous, whitish, ...)

Ddx

- Macroglossia(big tongue)
 - Cretinism
 - Glycogen storage disease (Pompe disease)
 - Hurler's disease
 - Down syndrome
 - ✓ Tongue is normal but oral cavity is small and the child usually protrudes the tongue
 - Mass lesion growth from the tongue
 - ✓ Rhabdomyosarcoma
 - ✓ Neurofibromatosis

➤ Delayed dentition

- Rickets
- Hypothyroidism
- Hypopituitarism
- Down syndrome
- Cleidocranial dysplasia
- Constitutional delay

RESPIRATORY SYSTEM EXAMINATION

*Ques
tions*

- *Do respiratory system examination*
 - ✓ *i.e. both anterior and posterior chest*
- *Do anterior or posterior chest exam*
 - ✓ *i.e. only the one you ordered*
- *Differential diagnosis for your findings*
- *Investigation for your differential diagnosis*
- *Management principle*

LUNG ANATOMY

- Anteriorly, the apex of each lung rises about 2-4 cm above the inner third of the clavicle
- The lower border of the lung crosses the
 - ✓ 6th rib at the mid-clavicular line,
 - ✓ 8th rib at the mid-axillary line and
 - ✓ 10th rib at the posterior axillary line
- The right lung has 3 lobes; upper, middle and lower lobes
- The left lung has 2 lobes; upper and lower lobes
- The trachea bifurcates into main bronchi at the level of sternal angle anteriorly and the 4th thoracic spinous process posteriorly

Approaches to examine

- ☐ Inspection
- ☐ Palpation
- ☐ Percussion
- ☐ Auscultation

INSPECTION

Look for

- ☐ *Breathing pattern*
- ☐ *Quality of voice – wheeze, Stridor,*
- ☐ *Chest shape*
- ☐ *Signs of respiratory distress*
- ☐ *Chest expansion – symmetrical or not*
- ☐ *Cyanosis*
- ☐ *Clubbing*

1) Breathing pattern

- Rapid shallow breathing
 - ✓ Due to hypoxia in respiratory diseases
- Slow breathing
 - ✓ Occurs in drug-induced respiratory depression(E.g. Barbiturate poisoning)
- Kussmaul's breathing
 - ✓ Fast, deep and labored breathing usually occurs in metabolic acidosis
- Paradoxical respiration
 - ✓ The abdomen sucks inwards with inspiration due to diaphragmatic paralysis (it normally pouches outwards due to diaphragmatic descent)

2) Chest shape

- Barrel chest
 - ✓ Increased antero-posterior diameter of the chest in comparison to lateral diameter of the chest
- Funnel chest (Pectusexcavatum)
 - ✓ Depression in the lower end of the sternum
- Pigeon chest (Pectuscarinatum)
 - ✓ Anteriorly displaced sternum with depressed costal cartilage
- Harrison's sulcus – *see at rickets*
- Kyphosis→ exaggerated forward curvature of the spine
- Scoliosis→ lateral curvature of the spine
- Kypho-scoliosis→ forward and lateral bending of the spine

3) Signs of respiratory distress

- Tachypnea
- Flaring of alanasae
- Use of accessory muscles of respiration
- Intercostal retraction
- Subcostal retraction
- Chest indrawing
- Cyanosis

4) Cyanosis

- Is bluish discoloration of the skin and mucus membrane resulting from an increased quantity of deoxygenated Hgb

- Becomes evident when the absolute concentration of deoxygenated Hgb is $\geq 5\text{gm/dl}$ of capillary blood
- In patients with anemia, cyanosis does not occur until even greater levels of arterial desaturation is reached
- Example
 - ❖ Which one of the following is more susceptible for cyanosis?
 - A. Patient A: Hgb = 20 gm/dl
 - B. Patient B: Hgb = 10 gm/dl
 - C. Patient C: Hgb = 4gm/dl
 - ❖ The answer is “A”; Why?
 - ✓ Approximately 5% of Hgb is deoxygenated in normal blood
 - ✓ Then calculate how many grams of Hgb will be deoxygenated among the given value of Hgb with the given percent of deoxygenated Hgb. *(NB. For cyanosis to be occur, deoxygenated Hgb level should be $\geq 5\text{ gm/dl}$)*
 - ❖ Patient “c” never become cyanosed; because even 100% of the given Hgb(4gm/dl) is deoxygenated: $4\text{mg/dl} < 5\text{mg/dl}$

☐ Central cyanosis

- ✓ Bluish discoloration of lips and tongue due to hypoxia

☐ Peripheral cyanosis (acrocyanosis)

- ✓ Bluish discoloration of the distal parts of extremities due to vasoconstriction

5) Clubbing

- Is selective bulbous enlargement of the distal segments of fingers and toes due to proliferation of connective tissue
- Grading
 1. Grade I – spongy feeling on pressing the nail bed
 2. Grade II – obliteration of the hyponychial angle
 3. Grade III – drum stick appearance of the fingers
 4. Grade IV – drum stick appearance with hypertrophic osteoarthropathy
- Differential diagnosis
 - ❑ Cardiac diseases
 - Infective endocarditis
 - Cyanotic congenital heart disease
 - ❑ Pulmonary
 - Lung abscess
 - Empyema
 - Cystic fibrosis
 - Primary or secondary lung cancer
 - Lymphoid interstitial pneumonia
 - Bronchiectasis
 - Tuberculosis
 - ❑ Gastro intestinal diseases
 - Crohn's disease
 - Ulcerative colitis

- Celiac disease
- Liver cirrhosis
- ☐ Genetic
- ☐ Idiopathic

PALPATION

Approach = (3T + C/S)

- ☐ *Tracheal position*
- ☐ *Tenderness of chest wall*
- ☐ *Subcutaneous emphysema*
- ☐ *Tactile fremitus*
- ☐ *Chest expansion*

Technique

1. Tracheal position

- Feel for the trachea by putting the 2nd and 4th fingers on each edge of sternal notch and use the 3rd finger to assess the trachea is central or deviated to one side
- A slight deviation of the trachea to the right side may be found in healthy individuals
- Causes of tracheal displacement
 - ☐ Deviation away from side of the lung lesion
 - Unilateral massive pleural effusion
 - Unilateral pneumothorax
 - Mediastinal mass
 - ☐ Deviation towards the side of the lung lesion

- Upper lobe collapse
- Upper lobe fibrosis
- Pneumonectomy

2. Chest wall tenderness - Look at his face while palpating

3. Subcutaneous emphysema- Crackling sensation

4. Tactile fremitus (TF)

- For cooperative patients (age ≥ 7 year) or while crying
- Ask the patient to say “ninety-nine” or “arba-arat” repeatedly while the palm of hand over the chest wall
- Compare bilaterally in symmetrical fashion
 - ☐ Reduced TF on affected lung occurs in
 - Pneumothorax
 - Pleural effusion
 - Fibrotic lung disease
 - ☐ Increased TF on affected lung occurs in
 - Lung consolidation due to pneumonia

5. Chest expansion

- For cooperative patients (age ≥ 7 yr)a measuring tape is put around the mid-thorax perpendicular to vertebrae and patient is asked to breathe-in maximally and the difference between full inspiration & expiration is recorded, normally it is 2 cm

Anterior chest (to check symmetry)

- Place your thumbs along each costal margin, your hands holding the lateral rib cage
- Slide your thumbs medially to raise skin folds

Posterior chest (to check symmetry)

- Place your thumbs at the level of and parallel to the 10th rib, your hands grasping the lateral ribcage
- Slide your thumbs medially in order to raise loose skin folds between your thumbs and the spine
- Ask the patient to inhale deeply
- Watch divergence of your thumbs during inhalation
- Observe for symmetry and degree of chest expansion

☐ Reduced chest wall movement on affected side

- Consolidation
- Lung collapse
- Atelectasis
- Pleural effusion
- Pneumothorax
- Localized pulmonary fibrosis, ...

☐ Symmetrically reduced chest expansion

- Bilateral pleural effusion
- Bilateral lung collapse
- Emphysema
- Bilateral fibrotic lung disease, ...

PERCUSSION

- ❑ Percussion of the chest sets the chest wall & underlying tissues into motion, producing audible sounds & palpable vibration
- ❑ It helps to determine whether the underlying tissues are air-filled, fluid-filled or solid-filled

Approach

- ✓ *Percussion notes*
- ✓ *Diaphragmatic excursion*

1) Percussion notes

- Resonance – normal
- Relative dullness – fluid filled alveoli or consolidation
- Stony dullness – fluid in the pleural space
- Hyper resonance–air in the pleural space

2) Diaphragmatic excursion

- For cooperative patients (age ≥ 7 year)
- The distance between the level of dullness on full expiration and full inspiration
- Normal range of diaphragmatic excursion is 5-6 cm

Technique

- Percuss posterior chest from apices to lung base
- Determine the level of dullness at full expiration and then at full inspiration
- Determine the difference in cm
- ☐ Reduced diaphragmatic excursion on the affected side of the lung
 - Pleural effusion
 - Pulmonary edema
 - Consolidation
 - Lung collapse
 - Atelectatic
 - Fibrosis of the lung
- ☐ Reduced diaphragmatic excursion bilaterally
 - Hyperinflation
 - Bilateral pleural effusion
 - Bilateral lung collapse ...

AUSCULTATION

Approach

- ☐ *Breath sounds*
- ☐ *Adventitious sounds*
- ☐ *Pleural friction rub*
- ☐ *Transmitted sounds*

1) Breath sounds

- Normal breath sounds are classified according to their intensity, pitch and duration of their inspiratory and expiratory phases

Breath sounds	Duration	Intensity	Pitch	Location where normally heard	Time gap b/n insp&exp phases?
Vesicular	Insp>exp phase	soft	relatively low	over most of both lungs	no
Broncho-vesicular	insp = exp phase	intermediate	intermediate	often b/n 1 st & 2 nd ribs anteriorly & b/n scapula	no
Bronchial	Exp>insp phase	loud	relatively high	over the manubrium	yes
Tracheal	Insp = exp phase	very loud	relatively high	over the trachea in the neck	yes

2) Adventitious sounds

Crackles / crepitation

- Produced by sudden changes in gas pressure related to the sudden opening of previously closed small airways
- Intermittent & non-musical brief, high-pitched sounds
- Can be coarse or fine & unilateral or bilateral

Wheeze

- High pitched sounds with hissing or shrill quality
- Usually maximal during expiration and is accompanied by prolonged expiration
- Can be diffused, scattered or localized and unilateral or bilateral

Rhonchi

- Low-pitched sounds with snoring quality
- Disappear while coughing

Stridor

- Is a noisy breathing due turbulent air flow
- Cardinal sign of URT obstruction in children
- Can be
 - ❑ Inspiratory
 - At or above the vocal cords
 - Due to the collapse of the soft tissue with negative pressure during inspiration

- ☐ Expiratory
 - Due decreased airway caliber with expiration
 - Emanates from intra-thoracic trachea and bronchi
- ☐ Biphasic
 - Indicates unchanging airway caliber due to a fixed lesion
 - Characteristics of mid-tracheal lesions

3) Friction rub

- Occurs in pleural inflammation with creaking or rubbing quality

4) Transmitted sounds

- Common in consolidation
- Done in cooperative patients
- *Look for*
 - ☐ *Bronchophony*
 - ☐ *Aegophony*
 - ☐ *Whispering pectoriloquy*

Bronchophony

- Ask the patient to say repeatedly “ninety-nine”; louder & clearer sounds are heard on the chest wall
- Normally muffled and indistinct

Aegophony

- Ask the pt to say “ee-ee-ee” ; E-to-A change with nasal or bleating quality is heard
- Normally hear a muffled long “E” sound

Whispering pectoriloquy

- Ask the pt to whisper “one-two-three” ; louder & clearer whispered sounds are heard
- Normally heard faintly and indistinctly

SOME COMMON LUNG PROBLEMS

CONSOLIDATION

Signs

- Reduced chest expansion
- Increased tactile fremitus
- Relative dullness
- Bronchial breath sounds
- Crackles / crepitation
- Transmitted sounds are positive

Cause

- Pneumonia (commonest cause)
- Pulmonary edema or hemorrhage
- Aspiration
- Bronchogenic carcinoma

Investigation

- CBC – leukocytosis
- CXR
 - ✓ Air bronchogram
 - ✓ Hyper dense (homogeneous opacity)
 - ✓ Silhouette sign – if lobar pneumonia
- CT scan

PLEURAL EFFUSION

- ❑ Accumulation of fluid in the pleural space

Signs

- Trachea - deviated away from a massive effusion
- Reduced chest expansion
- Stony dullness
- Absent air entry
- May be BBS above the effusion

Pleural effusion

- Is collection of fluid in the pleural space
 - Simple fluid = hydrothorax
 - Blood = hemothorax
 - Lymph = chylothorax
 - Pus = empyema
- Can be exudative or transudative
 - To differentiate the two, we use “Lights criteria”

Exudative – if full fill one of the following Lights criteria

- ✓ Pleural protein-to-serum protein ratio > 0.5
- ✓ Pleural LDH-to-serum LDH ratio > 0.6 or
- ✓ Pleural LDH is above $\frac{2}{3}$ of upper normal serum level

Transudative – if not full fill any of the above parameters

Ddx: Exudative Vs Transudative

Exudative	Transudative
▪ <i>Parapneumonic effusion</i> ▪ <i>Metastatic cancer</i> ▪ <i>Lymphoma</i> ▪ <i>Tuberculosis</i> ▪ <i>Pulmonary infarction</i> ▪ <i>Traumatic effusion</i> ▪ <i>Connective tissue diseases (RA, SLE)</i> ▪ <i>Acute pancreatitis</i> ▪ <i>Drugs (cytotoxins, hydralazine..)</i>	▪ <i>Congestive heart failure</i> ▪ <i>Hypoalbuminemia from PLE</i> ▪ <i>Nephrotic syndrome</i> ▪ <i>Constrictive pericarditis</i> ▪ <i>Hypothyroidism</i> ▪ <i>Meig's syndrome, ...</i>

Bilateral pleural effusion

- CHF
- Nephrotic syndrome
- Pulmonary infarction
- Lupus
- Rheumatoid Arthritis
- Malignancy
- Tuberculosis

- ➡ Why pleural effusion is common in the right pleural space?
- ➡ Differential diagnosis for unilateral Vs bilateral crepitation ...

Investigations

❖ CBC

❖ CXR

- May not be visible if the amount is <250 ml
- Expected to be seen
 - ✓ Blunted/ obliterated costophrenic angle
 - ✓ Radio-opaque density extending from the base
 - ✓ Meniscus sign at upper surface of the fluid
 - ✓ Mediastinal shift away from the effusion – (*if huge effusion*)

❖ Pleural fluid analysis

➤ Appearance

☐ Clear or straw

- Transudative effusion
- Tuberculosis

☐ Hemorrhagic

- Trauma
- Malignancy
- Pulmonary infarction
- Tuberculosis
- Spontaneous pneumothorax

❑ Chylous

- Malignancy
- Trauma to lymphatic vessels
- Tuberculosis
- Thrombosis of the left subclavian vein

❑ Cloudy – bacterial infection, tuberculosis,

- Turbidity
- Cell count
- LDH level
- Protein level
- Glucose level
- Gram stain
- AFB
- Culture
- Cytology study

❖ Blood culture

❖ Ultrasound

- Aids in identification of loculated effusion
- Aids in differentiation of fluid from fibrosis
- Aids in identification of thoracentesis site

❖ CT scan

- Aids in differentiation of
 - ✓ Consolidation Vs effusion
 - ✓ Cystic vs solid lesions
 - ✓ Peripheral lung abscess Vs loculated empyema

- Aids in identification of
 - ✓ Necrotic areas
 - ✓ Pleural thickening, nodules, masses
 - ✓ Extent of tumour

❖ Pleural biopsy and cytology

- Closed pleural biopsy
- CT guided cutting needle biopsy
- Video assisted thoracoscopic (VATS) pleural biopsy

Pleural fluid analysis	Transudate	Exudates
<i>Appearance</i>	<i>Clear</i>	<i>Cloudy or purulent</i>
<i>Cell count (per mm³)</i>	<i>< 1000</i>	<i>Often > 50,000</i>
<i>Cell type</i>	<i>Lymphocytes, monocytes</i>	<i>Neutrophils</i>
<i>Lactate dehydrogenase</i>	<i>< 200 U/L</i>	<i>>1000 U/L</i>
<i>Pleural fluid : serum LDH</i>	<i>< 0.6</i>	<i>>0.6</i>
<i>Protein > 3 g</i>	<i>Unusual</i>	<i>Common</i>
<i>Pleural fluid : serum protein</i>	<i>< 0.5</i>	<i>>0.5</i>
<i>Glucose</i>	<i>Normal</i>	<i>Low (<40mg/dl)</i>
<i>pH</i>	<i>Normal</i>	<i><7.10</i>
<i>Gram stain</i>	<i>Negative</i>	<i><1/3 of cases +ve</i>

Management principle

- ✓ Antibiotics
- ✓ Thoracocentesis
- ✓ Chest tube insertion

PNEUMOTHORAX

- ❑ Accumulation of air in the pleural cavity due to leakage of air from the lung or chest wall punctures into the pleural space

Signs

- Tachypnea
- Trachea deviates away from
- Subcutaneous creptation
- Reduced chest expansion
- Reduced tactile fremitus
- Hyperresonance
- Greatly reduced or absent air entry
- Distended neck vein

Types of pneumothorax

- 1) Spontaneous pneumothorax – can be primary or secondary
 - ✓ Visceral pleural ruptures without an external traumatic or iatrogenic cause
- 2) Open pneumothorax
 - ✓ Also known as “sucking chest wound”
 - ✓ Defects in the chest wall large enough to exceed the laryngeal cross-sectional area allows air to enter from the exterior into the pleural cavity and results in lung collapse.
- 3) Tension pneumothorax
 - ✓ A clinical diagnosis defined as “any pneumothorax with cardiorespiratory compromise or collapse”
 - ✓ Air continues to get into the pleural space but cannot exit = the operation of one way valve system
 - ✓ Usually patients are hypotensive

Investigations

- CXR
 - ❑ Expected to be seen
 - Hypo dense (extremely dark) on the affected side
 - Visceral pleural edge is visible
 - Absent bronchovascular markings peripherally
 - Loss of lung volume on the affected side
 - Mediastinal shift to opposite side
 - Tracheal shift to opposite side

- CT scan

Management principle

- ✓ Thoracocentesis
- ✓ Chest tube insertion

LUNG COLLAPSE

- ❑ Partial or complete loss of lung volume

Signs

- Trachea - deviates towards the collapsed lung
 - ✓ Due to compensatory hyperinflation on the opposite lung
- Reduced chest expansion
- Relative dullness
- Decreased air entry / absent
- Signs of hyperinflation on the opposite side

Investigations

- ❖ CXR

▶ Depends on :

- ✓ Mechanism of collapse
- ✓ Degree of collapse
- ✓ Presence or absence of consolidation and
- ✓ Preexisting state of the pleura

▶ Signs of collapse can be direct or indirect

- ✓ Indirect signs are as a result of compensatory changes in response to volume loss

□ **Direct signs**

- Displacement of interlobular fissure
- Loss of aeration of the lung
- Vascular and bronchial overcrowding

□ **Indirect signs**

- Elevation of the hemidiaphragm
- Mediastinal displacement
- Hilar displacement
- Compensatory hyperinflation

❖ CT scan – (*see at p. effusion*)

❖ Differential diagnosis

- Cystic fibrosis
- Foreign body
- Hilar lymphadenopathy
- Bronchogenic carcinoma
- Asthma

WHEEZE

Wheezing can be:

❖ Polyphonic

- When there is wide spread narrowing of the airways, causing various pitches or levels of obstruction to air flow as seen in asthma

❖ Monophonic

- Wheezing referring to a single pitch sound that is produced in the larger airways during expiration, as in distal tracheomalacia or bronchomalacia

Differential diagnosis

➡ Bronchial asthma

- Clinical features
 - ✓ Diffused, bilateral wheeze
 - ✓ Afebrile
 - ✓ Recurrent symptoms
 - ✓ Response to bronchodilators
 - ✓ May have family history
- Types of wheezers(Tucson Respiratory children's study)
 1. Transient wheezer →Wheeze at <3 years old but no at 6 yrs old
 2. Persistent wheezer →Wheeze at <3 years old and at 6 years old
 3. Late onset wheezer →No wheeze <3 yrs old but wheeze at 6 yrs
 4. Never ->No wheeze by 6 years of age

➡ Foreign body

- ✓ Chocking episode history
- ✓ Localized, uni- or bilateral
- ✓ Afebrile
- ✓ No response to bronchodilators

➡ Congenital anomalies

- ☐ Malecia of the larynx, trachea and/or bronchi
 - ✓ Symptoms since birth
 - ✓ Relief of symptoms in prone position
- ☐ Tracheoesophageal fistula (H – type)
 - ✓ Recurrent pneumonia
 - ✓ Risk factors during pregnancy like polyhydramnios
- ☐ Vascular ring
 - ✓ Symptoms since birth
 - ✓ Noisy breathing
 - ✓ Tachypnea
 - ✓ Opisthotonic position

➡ Infections

- ☐ Bronchiolities
 - ✓ Diffused or scattered, bilateral wheeze
 - ✓ Febrile
 - ✓ First episode of wheeze during infancy
 - ✓ Poor /no response to bronchodilators
- ☐ Pneumonia
 - ✓ Sign of consolidation
 - ✓ Scattered, unilateral wheeze

- ✓ Coarse creptation

- ☐ Endobronchial tuberculosis

- ✓ Contact history to a known TB patient

- ✓ Hilaradenopathy on CXR

➡ Miscellaneous

- ☐ Pulmonary edema

- ✓ Bilateral or unilateral wheeze

- ✓ Signs of CHF

- ✓ Auscultory cardiac findings

- ✓ Bilateral postero-basal rales

- ☐ GERD

- ✓ Vomiting since early infancy

- ✓ Failure to thrive

- ☐ Mediastinal mass/tumor

- ✓ Mediastinal widening on CXR

- ✓ Other systemic signs

Investigations

- WBC with differentials

- CXR

- ✓ Hyperinflation (asthma)

- ✓ Foreign body

- ✓ Cardiomegaly

- ✓ Mediastinal mass

- ✓ Consolidation / pulmonary edema
- Pulmonary function test
- PPD test – bronchial tuberculosis
- Esophagogram - Vascular ring, GERD

STRIDOR

Differential diagnosis of stridor	
New born	Older infants& children
▪ <i>Choanal atresia</i> ▪ <i>Pierre robin sequence</i> ▪ <i>TreacherCollins syndrome</i> ▪ <i>Cystic hygroma</i> ▪ <i>Laryngomalecia</i> ▪ <i>Subglottic stenosis</i> ▪ <i>Laryngeal web, cysts</i> ▪ <i>Tracheomalecia</i> ▪ <i>Vascular ring</i>	▪ <i>Macroglossia</i> ▪ <i>Angioedema</i> ▪ <i>Adenotonsillar hypertrophy</i> ▪ <i>Croup</i> ▪ <i>Bacterial trachieties</i> ▪ <i>Acute epiglottities</i> ▪ <i>Retropharyngeal abscess</i> ▪ <i>Laryngeal papilloma</i> ▪ <i>Foreign body</i> ▪ <i>Caustic ingestion</i> ▪ <i>Inhalational burn</i> ▪ <i>Anterior mediastinal mass</i>
<u>Investigations</u> ✓ <i>WBC count with differentials</i> ✓ <i>Lateral neck x-ray</i> ✓ <i>Neck x-ray (anterio-posterior view)</i> ✓ <i>Chest x-ray</i>	

CVS EXAMINATION

Questions

- ✓ *Do CVS examination*
- ✓ *Do precordial examination = only the precordium*
- ✓ *Asses signs of congested heart failure*
- ✓ *Ddx& investigations*

CARDIOVASCULAR SYSTEM EXAMINATION

Look for

- ✓ *General examination*
- ✓ *Arterial examination*
- ✓ *Veineous examination*
- ✓ *Precordial examination*

GENERAL EXAMINATION

- Peripheral cyanosis
- Clubbing

ARTERIAL EXAMINATION

- Feel for all peripheral arteries bilaterally at the same time except carotid aa
 - Carotid
 - Temporalis
 - Brachial
 - Radial
 - Femoral
 - Popliteal
 - Posterior tibial and
 - Dorsalispedisarteries
- ✓ If asymmetric, think of ↔ Shock, Arteritis, obstruction
- Evaluate for
 - Rate
 - Rhythm

- Character
- Volume
- Radio-femoral delay
- Pulse deficit

1) Rate & rhythm

- Radial artery is commonly used
- Rate – count pulse rate for one minute
- Rhythm
 - ✓ Regular
 - ✓ Regularly irregular – ectopic beats,
 - ✓ Irregularly irregular – atrial fibrillation

2) Pulse character (amplitude & counter)

- Is best assessed in the carotid artery, except
 - ✓ In collapsing pulse where radial artery is preferred

3) Pulse volume

- Provides crude indications of stroke volume
- Small in systolic heart failure
- Large in hyper kinetic heart disease (bounding pulse)
 - ✓ Anemia
 - ✓ Aortic regurgitation (AR)
 - ✓ Patent ductus arteriosus (PDA)
 - ✓ Pregnancy
 - ✓ Thyrotoxicosis

4) Pulse deficit

- Difference of heart beat rate and peripheral arterial rate
- Often occurs in atrial fibrillation

- ✓ Due to failure in conducting all central beats to peripheral arteries

5) Radio – femoral delay

- Press both radial and femoral artery at the same time
- Notice for arterial pulse delay at femoral artery compared to radial artery
- Usually observed in coarctation of the aorta
- Normally, femoral pulse is slightly faster than the radial pulse
 - ✓ B/c of femoral artery is a direct branch of abd aorta

VEINEOUS EXAMINATION

- Distended neck vein
- Jugular Veineous pressure (JVP)
 - The normal upper limit is 3 cm vertically above sternal angle
 - ✓ This is about 8 cm above right atrium,
 - ✓ Corresponding to a JVP of 8 mmHg
 - JVP > 3 cm vertically above sternal angle is considered elevated
 - If the internal jugular vein pulsation is not visible
 - ✓ Measure the vertical distance of the point above which the external jugular veins appear to be collapsed from the sternal angle

- Jugular vein Vs carotid artery pulsations

Internal jugular vein pulsation	Carotid artery pulsation
▪ <i>Rarely palpable</i> ▪ <i>Two pulsations per heart beat</i> ▪ <i>Pulsation is eliminated by light pressure over the vein</i> ▪ <i>Level of pulsation changes with position, dropping as the pt becomes more upright</i> ▪ <i>Level of pulsations usually descends with inspiration</i>	▪ <i>Palpable</i> ▪ <i>A single pulsation per heart beat</i> ▪ <i>Not eliminated by pressure</i> ▪ <i>Unchanged with position</i> ▪ <i>Not affected by inspiration</i>

➤ **Kussmaul's sign**

- Increase rather than the normal decrease in the JVP during inspiration, which is observed in
 - ✓ Constrictive pericarditis
 - ✓ Right ventricular infarction
 - ✓ Severe right ventricular failure

➤ **Hepatojugular reflex test**, positive if there is;

- An increase in JVP during firm, mid-abdominal compression for 10 seconds followed by a rapid drop in JVP of 4 cm on release of the compression

PRECORDIAL EXAMINATION

- **Inspection**
- **Palpation**
- **Auscultation**
- **Percussion**
 - ✓ Has little significance
 - ✓ If u suspect dextro-cardia or huge cardiomegaly

INSPECTION

Look for

- ✓ ***Bulging*** – indicates chronicity or cardiomegaly
- ✓ ***Precordial activity***
- ✓ ***Apical impulse***
- ✓ ***Surgical scar***

Precordial Activity

- Active precordium
 - Visible one or two pulsations
- Hyperactive precordium
 - More than two visible pulsations or a pulse involving > 2.5 cm

- Shows hyper dynamicity
- Volume /pressure overload
- Quiet precordium
 - No visible pulsation
 - Thick chest wall
 - Massive pericardial effusion
 - Dilated cardiomyopathy

Apical Impulse

- Is the apical impulse visible or not?
 - If visible - determine the location in inter costal space in relation to left mid-clavicular line
 - If not – thick chest wall, pericardial effusion, ...
- The apex beat is defined as
 - The lowest and most lateral point at which the cardiac impulse can be palpated
- Apical impulse is due to the recoil of heart as blood is ejected
- The normal left ventricular impulse is located at or medial to the left mid-clavicular line at the 4th or 5th inter costal space
- Down ward and laterally displaced apical impulse suggests
 - left ventricular enlargement
 - ✓ Volume overload – displaced lateral & downward
 - ✓ Pressure overload – displaced down ward

PALPATION

Look for

- ✓ *Shock (Palpable heart sound)*
- ✓ *Point of maximal impulse*
- ✓ *Thrill*
- ✓ *Heave*

Shock (palpable heart sound)

- Feel for palpable heart sounds at
 - ✓ Apex
 - ✓ Left upper & lower sternal border and
 - ✓ Right upper sternal border

Point of maximal impulse/intensity (PMI)

Look for

- Location, amplitude, duration
- Localized or diffused
 - ☐ Diffuse PMI when
 - ✓ Occupies >2.5 cm in diameter
 - ✓ Palpable b/n 2nd & 4th finger tips while off the 3rd finger tip from the precordial area or
 - ✓ Occupies more than one intercostal space
- Sustained or non-sustained

- ✓ Sustained PMI occupies more than 2/3 of cardiac cycle
- Thrusting or tapping

Thrill

- **Look for**
 - ✓ *Location (site)*
 - ✓ *Timing of the thrill*
- Palpable, low frequency, vibrations associated with heart murmurs
- Palpate the ***apex, left upper & lower sternal border and right upper sternal border*** with **palm** of examining hand, and feel for thrill as purring of a cat
- Timing of the thrill – Systolic or diastolic
 - ✓ If the thrill coincides with the carotid pulse – systolic thrill
 - ✓ If the thrill comes after the carotid pulse – diastolic thrill

Heave (lift)

- Can be ***apical*** or ***parasternal*** heave
- Put ulnar border of the hand over apical or left sternal area
 - ✓ Look for lift or heave of ur hand
- Apical heave suggests
 - ✓ Left ventricular hypertrophy or severe MR
- Left parasternal lift suggests
 - ✓ Right ventricular hypertrophy or severe TR with giant right atrium

AUSCULTATION

Look for

- *Heart sounds: S1 & S2*
- *Added heart sounds: S3 & S4*
- *Friction rub*
- *Murmur*

Heart Sounds: S1 (Lub) and S2 (Dub)

First heart sound (S1)

- Corresponds to mitral and tricuspid valve closure at the onset of systole
- Increased intensity of S1 at mitral area occurs in MS and pregnancy
- While reduced intensity occurs in mitral regurgitation (MR)

Second heart sound (S2)

- Corresponds to aortic and pulmonary valve closure following ventricular ejection
- Increased intensity of S2 at pulmonic area occurs in pulmonary hypertension

Added Heart Sounds: S3 and S4

- Use the bell of stethoscope to listen 3rd and 4th heart sounds

Third heart sound (S3)

- Is a low frequency sound
- Occurs due to rapid ventricular filling in early diastole, causing sudden stretching of chordae tendinae and papillary muscles
 - Physiologic S3 occurs in *young adults <40 yr and pregnancy*
 - Pathologic S3 often noticed in *anemia, thyrotoxicosis and severe MR*

Fourth heart sound (S4)

- Is a low frequency sound
- Occurs due to vigorous atrial contraction against a non-compliant ventricle in late diastole
- S4 is usually pathological and occurs in stiff non-compliant hypertrophied ventricle due to;
 - ✓ Hypertensive heart disease
 - ✓ Aortic stenosis
 - ✓ Hypertrophic cardiomyopathy

NB. *Opening of any normal valve is not audible*

Friction Rub

- High-pitched scratching sound audible at any part of cardiac cycle
- Heard best at left lower sternal border in maintained expiration & leaning forward
- Observed in acute pericarditis

HEART MURMURS

- Result from vibrations set up in the blood stream and the surrounding heart and great vessels as a result of turbulent blood flow

Characterizing the murmur

1. *Location of maximal intensity*
2. *Timing*
3. *Intensity/grading of the murmur*
4. *Shape (configuration)*
5. *Pitch*
6. *Quality*
7. *Radiation*
8. *Maneuvers*

Location of maximal intensity

- Signifies origin of murmur

Timing

- ➡ Identify whether the murmur is *systolic, diastolic or continuous* in comparison to carotid pulse
 - ✓ Systolic murmurs coincide with carotid pulse, while the diastolic not

➤ **Systolic murmurs**

1. Mid-systolic murmur (MSM)
 - ✓ Begin after S1 and stops before S2
 - ✓ Murmur of **AS, PS**
2. Pansystolic murmur (PSM)
 - ✓ Starts with S1 and stops at S2
 - ✓ Murmur of **MR, VSD, TR**
3. Late systolic murmur (LSM)
 - ✓ Starts in mid or late systole and persists up to S2
 - ✓ Murmur of Mitral valve prolapse (**MVP**)

➤ **Diastolic murmur**

1. Early diastolic murmur (EDM)
 - ✓ Starts after S2 and fades into silence before next S1
 - ✓ Murmur of **AR, PR**
2. Mid diastolic murmur (MDM)
 - ✓ Starts after S2 and fade away or merge into a late diastolic murmur
 - ✓ Murmur of **MS, TS, ASD**
3. Late diastolic (presystolic) murmur (LDM)
 - ✓ Starts late in diastole and continuous up to S1
 - ✓ Murmur of **MS or TS** in sinus rhythm

➤ **Continuous murmur**

- ✓ Begin in systole, peak at S2, and continue into all or part of diastole
- ✓ Murmur of **PDA** (patent ductus arteriosus)

Grading of intensity

- Systolic murmurs (have 6 grades)
 - Grade I- faint murmur, heard only with special efforts
 - Grade II-quiet but heard murmur
 - Grade III- moderately loud murmur
 - Grade IV - loud murmur accompanied by a thrill
 - Grade V- very loud, heard with a stethoscope partly off the chest
 - Grade VI - heard with the stethoscope entirely off the chest
- Diastolic murmurs have only four grades

Shape (configuration)

- Crescendo
 - Murmur grows louder
 - Pre-systolic murmur of MS,
- Crescendo-decrescendo
 - Murmur that grows louder and then fall
 - Mid-systolic murmur of AS,
- Decrescendo
 - Murmur grows softer and slowly falls
 - Early diastolic murmur of AR,
- Plateau
 - Murmur has same intensity throughout
 - Pansystolic murmur of MR,

Pitch

- High
- Medium
- Low

Quality

- Blowing
- Harsh
- Rumbling
- Musical

Radiation

- Signifies direction of blood flow

Maneuvers

- ☐ Respiration
 - ✓ ***Left***-sided murmurs increase with ***expiration***
 - ✓ ***Right*** - sided murmurs increase with ***inspiration***
- ☐ Leaning forward
 - ✓ Increases the intensity of AR
- ☐ Leaning to left lateral position
 - ✓ Increases the intensity of MR
- ☐ Hand grip exercise
 - ✓ Increases the intensity of MR and AR

COMMON VALVULAR LESIONS

❑ Mitral regurgitation (MR)

- ✓ Laterally and down ward displaced apical impulse
- ✓ Muffled S1
- ✓ Prominent third heart sound
- ✓ Medium to high-pitched
- ✓ Blowing type
- ✓ Plateau configuration
- ✓ Soft to loud pansystolic murmur
- ✓ Best heard at apical area
- ✓ Radiating to axilla or over the precordium
- ✓ Accentuated by leaning to left lateral position

➤ Differential diagnosis

▶ Acute

- Endocarditis
- Papillary muscle rupture
- Trauma
- Chordal rupture/leaflet fail (MVP, IE)

► Chronic

- Rheumatic fever
- Rheumatic heart disease
- Dilated cardiomyopathy
- Myxomatous (MVP)
- Mitral annular calcification
- Endocarditis (healed)
- Congenial (defect on AV canal)

❑ **Aortic regurgitation (AR)**

- ✓ Laterally and down ward displaced apical impulse
- ✓ Wide pulse pressure
- ✓ Absent S2
- ✓ High-pitched
- ✓ Soft to loud blowing
- ✓ Decrescendo configuration
- ✓ Early diastolic
- ✓ Best heard at the 2nd to 4th left interspace (Erb's point)
- ✓ Accentuated by expiration and leaning forward
- ✓ Austin flint murmur at the apex – due to back regurgitation flow

➤ **Peripheral signs of AR**

- **Demusset's sign** – Head bob occurring with each heart beat
- **Quincke's pulses** – Capillary pulsations in the fingertips or lips
- **Mueller's sign** – Systolic pulsations of the uvula
- **Becker's sign** – Visible pulsations of the retinal arteries & pupils
- **Hill's sign** – Popliteal cuff systolic pressure exceeding brachial pressure by more than 60 mmHg
- **Mayne's sign** – More than a 15 mmHg decrease in diastolic blood pressure with arm elevation from the value obtained with the arm in the standard position
- **Traube's sign** – A pistol shot pulse (systolic and diastolic sounds) heard over the femoral arteries
- **Duroziez's sign** – A systolic and diastolic bruit heard when the femoral artery is partially compressed
- **Rosenbach's sign** – Systolic pulsations of the liver
- **Gerhard's sign** – Systolic pulsations of the spleen

➤ **Differential diagnosis**

▶ **Leaflet abnormalities**

- Rheumatic fever
- Endocarditis
- Trauma
- Bicuspid aortic valve

- Rheumatoid arthritis
- Myxomatous degeneration
- Ankylosing spondylitis

▶ Aortic root or ascending aorta problems

- Systemic hypertension
- Aortitis (e.g. Syphilis)
- Trauma
- Marfan syndrome
- Dissecting aneurysm
- Reiter's syndrome
- Ankylosing spondylitis
- Ehler's – Danlos syndrome

❑ **Tricuspid regurgitation (TR)**

- ✓ Low to high-pitched
- ✓ Blowing type
- ✓ Pansystolic murmur
- ✓ Best heard at the left lower sternal border
- ✓ Radiating to the epigastric area
- ✓ Accentuated by deep inspiration (positive caravello's sign)

❑ Mitral stenosis (MS)

- ✓ Accentuated S1
- ✓ Opening snap following S2
- ✓ Low-pitched
- ✓ Rumbling type
- ✓ Crescendo
- ✓ Mid-diastolic murmur
- ✓ Limited to apex
- ✓ Increased intensity of murmur with exercise and left lateral positioning

➤ Differential diagnosis

- Rheumatic fever
- Congenital mitral valve stenosis
- Mitral annular calcification with extension onto the leaflets (degenerative)
- Left atrial myxoma
- Infective endocarditis with large vegetations
- Carcinoid heart disease
- Endomyocardial fibrosis
- Systemic lupus erythematosus
- Rheumatoid arthritis

❑ Aortic stenosis (AS)

- ✓ Anacrotic arterial pulse
- ✓ Laterally displaced thrusting and sustained apical impulse
- ✓ Low- pitched
- ✓ Rasping(rough) type
- ✓ Crescendo – decrescendo
- ✓ Mid-systolic
- ✓ Best heard at aortic area
- ✓ Radiating to the carotid arteries
- Differential diagnosis
 - Rheumatic heart disease
 - Congenital (bicuspid or unicuspid aortic valve)

❑ VSD

- The most common among CHD
- If small defect
 - ✓ Loud harsh holosystolic murmur best heard at left lower sternal border
 - ✓ Frequently accompanied by a thrill

- If the defect is large
 - ✓ May have precordial bulge
 - ✓ Palpable parasternal heave
 - ✓ Laterally displaced apical impulse
 - ✓ Apical thrust and a systolic thrill
 - ✓ Less harsh but more blowing holosystolic murmur
 - ✓ Early systolic murmur at pulmonic area
 - ✓ Rumbling low pitched mid-diastolic murmur at the apex when shunt ratio >2:1
 - ✓ Accentuated 2nd heart sound

❑ PDA

- Asymptomatic in small PDA
- If the PDA is large
 - ✓ Wide pulse pressure
 - ✓ Bounding pulse
 - ✓ Continuous machinery murmur at 2nd left intercostal space
 - ✓ May radiate to
 - Down the left sternal border
 - Left clavicle or

- Toward the apex
- ✓ Often well heard over the left back

❑ ASD

- Most are asymptomatic
- Hyperactive precordium
- Left parasternal heave at left lower sternal border
- Widely split & fixed 2nd heart sound
- Mid systolic murmur best heard at the pulmonic area
- Rumbling mid-diastolic murmur at tricuspid area
- Holosystolic murmur at mitral area in OP (ECD)

SIGNS OF CONGESTED HEART FAILURE

- Consider generalized body swelling due to heart failure

General appearance

- Cardiopulmonary distress
- Edematous

HEENT

- Facial puffiness

Vital signs

- Tachycardia
- Tachypnea

Respiratory system

- Basal rales /creptation

Cardiovascular system

- Distended neck vein
- Raised JVP
- S3 gallop
- Cardiomegaly

Abdomen

- Tender hepatomegaly
- Positive Hepatojugular reflex

- Ascites

Musculoskeletal system

- Peripheral edema

INVESTIGATIONS

➤ CBC – as a baseline

➤ CXR

- Cardiomegaly (cardiothoracic ratio > 60% for children)
- Increased pulmonary vascular bed
- Pulmonary edema
- Pleural effusion

➤ ECG

- Chamber hypertrophy
- Rhythm disorders
- Conduction defect
- Myocardial infarction or ischemia
- Shows findings that suggest specific etiologies : eg
 - ✓ Low-voltage QRS morphologic characteristics with ST-T wave abnormalities may suggest myocardial inflammatory disease **or** pericarditis

➤ Echocardiography

- Gives information about ventricular size, function and valvular abnormalities
 - Size of all four chambers
 - Ventricular function
 - ✓ Fractional shortening (ESD-EDD/EDD)
 - » Normally 28-40%
 - ✓ Reduced ejection fraction
 - » Normally 55-65%
 - ✓ Pre-ejection:ejection period ratio
 - » Normally <40%
 - ✓ Calculation of COP
 - Structural valve abnormalities – regurgitation and stenosis
 - Congenital defects – VSD, ASD, ...
 - Pericardial effusion
- Other investigations are based on the precipitant factors

MANAGEMENT PRINCIPLE

➤ Supportive measures

- ✓ Bed rest
- ✓ Semi upright position
- ✓ Fluid restriction
- ✓ O₂ administration

- ✓ No added salt diets– do not mean total absence of salt
- ✓ Increase daily calories
- ✓ Restriction to competitive and strenuous sport activities
- Treat congestive state
 - ✓ Digoxin
 - ✓ Diuretics– Lasix, spironolactone, ...
 - ✓ After load reducing agents – ACE inhibitors , ...
- Treat the precipitant factor
- Treat the underlying cause
 - ✓ Surgery for valvular heart disease (congenital or acquired)
 - ✓ Implantation of cardiac prosthesis
 - ✓ Cardiac transplantation

ABDOMINAL EXAMINATION

Question

- ✓ *Do abdominal examination*
- ✓ *Differential diagnosis for your findings*
- ✓ *Investigations for your differential diagnosis*
- ✓ *Management principle*

Approaches

- 1. Inspection**
- 2. Palpation**
- 3. Percussion**
- 4. Auscultation**
5. DPR examination
 - ✓ Ask the examiner if s/he wants you to do it

Abdomen has

- Four quadrants (URQ, LRQ, ULQ, LLQ)&
- Nine regions (right & left hypochondrium, epigastrium, right and left lumbar, umbilical, right and left iliac and hypogastrium or suprapubic)

NB. Before starting your examination

- Check for good lighted room
- Communicate with the patient
- The patient should have an empty bladder
- Make the pt lying in supine position with his/her arms & hands in anatomical position
- Expose the abdomen from the nipple line to the mid-thigh

INSPECTION

Look for

- ✓ *Symmetry of the abdomen*
- ✓ *Movement with respiration*
- ✓ *Shape of the abdomen*
- ✓ *Flank fullness*
- ✓ *Contour of umbilicus*
- ✓ *Visible peristalsis*
- ✓ *Visible abdominal vein*

- ✓ *Skin pigmentation*
- ✓ *Scars & striae*
- ✓ *Hernia sites*
- ✓ *Diastasis recti*

❑ 1st stand in front & then on the right side of the patient to see the listed parameters

1) Symmetry of the abdomen

- Symmetry – normal, food, ascites, flatus or faces
- Asymmetry – mass, or fetus

2) Movement with respiration

- Markedly diminished or absent abdominal movement → generalized peritonitis

3) Shape of the abdomen

- Scaphoid → SAM (cachexic), hernia
- Flat → normal
- Full → normal or 5f's
- Distended → 5f's (food, fluid, flatus, faces or fetus)
- Protuberant → mass or extremely distended abdomen

4) Flank fullness

- Full flanks with full or distended abdomen – ascites
- Full flanks without abdominal distension
 - ✓ Psoas abscess
 - ✓ Hydronephrosis

- ✓ Perinephric abscess
- ✓ Wilms tumor

5) Contour of umbilicus

- Normally inverted with circular slit, slightly horizontal in adults
- Everted umbilicus – huge ascites or mass
- Horizontal slit – ascites or bilateral flank mass
- Vertical slit – mass

6) Visible peristalsis

- Vigorous visible peristalsis – pylorus or bowel obstruction

7) Visible abdominal vein

- Due to portal hypertension/Inferior vena-caval obstruction
- Determine direction of flow during palpation by milking the distended veins using the 2nd & 3rd fingers
 - ✓ draining away from umbilicus – PHTN
 - ✓ draining towards umbilicus – IVC obstruction

8) Skin pigmentation

- Hypo or hyper pigmentation

9) Scars & striae

- Scar – previous surgery or trauma
- Striae – are wrinkled linear marks due to gross stretching of the skin or rupture of the elastic fibers of abdominal wall
 - ✓ Striae alba or atrophica in ascites
 - ✓ Purple striae in Cushing's syndrome

- ✓ Striae gravidarum in pregnancy

10) **Hernia sites**

- Support and lift the pt's head with the shoulder with your left arm and let him/her cough repeatedly while observing sites of hernia (for cooperative pts)
- Observe sites of hernia while crying (for uncooperative pts)
- Sites of hernia
 - ✓ Umbilical and periumbilical hernia - at or around the umbilicus
 - ✓ Epigastric hernia - at epigastric area
 - ✓ Inguinal and femoral hernia - in groin region
 - ✓ Incisional hernia - at surgical scar site

11) **Diastasis recti**

- Is a separation of the two rectus abdominis muscles, through which abdominal contents bulge to form a midline ridge when the pt raises head and shoulders
 - ✓ Protrusion of the umbilicus

PALPATION

- ☐ Be on the right side of the patient, unless you are left sided

I. Superficial palpation

Check for

- ☐ *Abdominal tenderness*
 - ☐ *Superficially palpable abdominal masses/organs*
 - ☐ *Abdominal resistance*
-
- Communicate with the patient before you starting to palpate
 - Ask if there is abdominal area which is painful
 - ✓ If there is, start far away from the tender site & palpate the tender site at last
 - ✓ If there is no, start at left lower quadrant and move quadrant-by-quadrant in clock-wise or anti-clockwise rotation
 - Look at the patient's face while you palpate
 - If there is superficially palpable mass, raise the patients head & shoulder to see if the mass is protruded;i.e.
 - ✓ Lipoma
 - ✓ Abdominal muscle abscess
 - ✓ Dermoid cyst

II. Deep palpation

Check for

- ✓ *Guarding*
- ✓ *Rigidity*
- ✓ *Rebound tenderness*
- ✓ *Organomegaly (enlarged liver or spleen)*
- ✓ *Abdominal mass*
- ✓ *Ballotability (if huge ascites)*

Guarding

- Involuntary reflex contraction of abdominal wall muscles overlying an inflamed viscus
 - ✓ Classically seen in uncomplicated acute appendicitis

Rigidity

- Involuntary reflex rigidity of abdominal wall muscles
 - ✓ Board-like rigidity is seen in diffuse peritonitis

Rebound tenderness

- Deeply and slowly palpate the abdomen, and check for the presence of sudden pain while the examiner releases his hand from the abdomen

Abdominal Organ and Mass Palpation

A. Spleen

- The spleen has to be 2-3 times its normal size to be palpable
- Enlargement of the spleen takes place in a superior and posterior direction before it becomes palpable
- Once palpable, the direction of growth is down wards and towards the umbilicus (along the splenic growth line)
- Start palpating from the right iliac fossa and move to the left hypochondrium
- Palpate while the pt breathe in and move ur fingers while he/she breath out
- If not palpable, turn the pt to half on to the right side (with right leg extended and left leg flexed at hip and knee joint) & repeat the examination as above
- Enlarged spleen (splenomegaly) Vs enlarged left kidney

Splenomegaly	Enlarged left kidney
▪ <i>Grows along the splenic growth line</i> ▪ <i>Moves with respiration</i> ▪ <i>Has medial notch</i> ▪ <i>Unable to go beneath the costal margin</i> ▪ <i>Not bimanually palpable</i> ▪ <i>Dull to percussion</i>	▪ <i>Grows antero-posteriorly</i> ▪ <i>Moves with respiration</i> ▪ <i>Has no medial notch</i> ▪ <i>Able to go beneath the costal margin</i> ▪ <i>Bimanually palpable but not always</i> ▪ <i>Colonic resonance to percussion</i>

- If there is splenomegaly, characterize
 - Size along splenic growth line
 - Tenderness
 - consistency - soft, firm, hard
 - Surface - smooth, nodular
 - Edge - sharp, round
 - Border – regular, irregular

- Splenomegaly
 - Tipped < 3cm
 - Moderate – b/n 3 & 7 cm
 - Massive >7 cm

DDx

Tipped splenomegaly	Huge splenomegaly
▪ <i>Malaria</i> ▪ <i>DTB to spleen</i> ▪ <i>IE</i> ▪ <i>ALL</i> ▪ <i>Infectious mononucleosis</i> ▪ <i>Typhoid fever</i> ▪ <i>Typhus</i> ▪ <i>SLE</i> ▪ <i>Sepsis</i> ▪ <i>Sarcoidosis</i> ▪ <i>Relapsing fever ...</i>	▪ <i>VL</i> ▪ <i>HMS</i> ▪ <i>HSS</i> ▪ <i>NHL</i> ▪ <i>CML</i> ▪ <i>B Thalasemia major</i> ▪ <i>Autoimmune hemolytic anemia</i> ▪ <i>Gaucher's disease</i> ▪ <i>Niemann- pick disease ...</i>

NB

➡ *All retroperitoneal organs except kidney & adrenal gland, have no movement with respiration. Why kidney & adrenal gland moves with respiration?*

Investigation for splenomegaly

- Abdominal U/S
- CBC
- rk39
- Splenic aspiration
 - Contraindication
 - ✓ Tipped splenomegaly
 - ✓ Huge splenomegaly
 - ✓ Huge ascites
 - ✓ Tender spleen
 - ✓ Site infection
 - ✓ Thrombocytopenia < 40,000
 - Visible amastigote on microscope – in case of VL
- Others based on your differential diagnosis?

B. Liver

- Ask the pt to breathe in deeply & start palpation from right lower abdomen towards the right hypochondrium with the right hand below & parallel to the right costal margin
- The liver edge is often palpable in normal pts
- Characterize the enlarged liver
 - Size below the right costal margin
 - Tenderness
 - Consistency - soft, firm, hard
 - Surface - smooth, nodular
 - Edge - sharp, round
 - Border – regular, irregular

NB.

Never say hepatomegaly by palpation, unless you found a size below right costal margin is larger than the upper limit of normal liver.

Causes that will push down the liver

- ✓ *Huge right pleural effusion*
- ✓ *Right hyperinflated lung*

Ddx = Isolated hepatomegaly Vs Tender hepatomegaly Vs Hepatosplenomegaly

Isolated hepatomegaly	Tender hepatomegaly	Hepatosplenomegaly
▪ <i>Hepatoblastoma</i> ▪ <i>Kwashiorkor</i> ▪ <i>Acute viral hepatitis</i> ▪ <i>DTB to liver</i> ▪ <i>Amoebic liver abscess</i> ▪ <i>Hydatid cyst</i> ▪ <i>CHF</i> ▪ <i>Pericardial effusion</i> ▪ <i>Constrictive pericardities</i> ▪ <i>Storage disease</i> ▪ <i>Leukemia</i>	▪ <i>Amoebic liver abscess</i> ▪ <i>Pyogenic liver abscess</i> ▪ <i>CHF</i> ▪ <i>Pericardial effusion</i> ▪ <i>Constrictive pericardities</i> ▪ <i>Acute viral hepatitis</i> ▪ <i>Infected hydatid cyst</i> ▪ <i>DTB to liver</i>	▪ <i>HSS</i> ▪ <i>NHL</i> ▪ <i>DTB</i> ▪ <i>CML</i> ▪ <i>VL</i> ▪ <i>B - Thalassemia</i> ▪ <i>Autoimmune hemolytic anemia</i> ▪ <i>Gaucher's disease</i> ▪ <i>malaria</i> ▪ <i>ALL</i> ▪ <i>AML ...</i>

Investigations for hepatomegaly

- ✓ Abdominal U/S
- ✓ CBC
- ✓ Viral markers
- ✓ LFT
- ✓ CT scan, MRI

C. Right Kidney

Bimanual palpation

- Stand on the right side of the patient
- Place the right hand horizontally in the right lumbar region anteriorly with the left hand placed posteriorly in the right loin
- Push forwards with the left hand, ask the patient to take a deep breath in, and press the right hand inwards and upwards
- The lower pole of the right kidney is commonly palpable in thin patients and is felt as a smooth, rounded swelling which descends on inspiration
- It helps to differentiate *enlarged right kidney* from *enlarged gallbladder*

D. Left Kidney

Bimanual palpation

- Stand on the left side of the pt
- Same technique as that of the right kidney (left hand anteriorly)
- Not usually palpable unless it is either low in position or enlarged

E. Urinary bladder

- Normally not palpable
- When it is full and the patient cannot empty it (retention),
 - ✓ A smooth firm regular oval-shaped swelling will be palpated in the suprapubic region and its upper border may reach as far as the umbilicus

- The lateral and upper borders can be readily made out, but it is not possible to feel its lower border (i.e. The swelling is 'arising out of the pelvis')
- The fact that this swelling is symmetrically placed in the suprapubic region beneath the umbilicus, that it is
 - ✓ Dull to percussion, and that pressure on it gives the patient a desire to micturate
- In women, think of other differential diagnoses
 - ✓ Gravid uterus
 - ✓ Ovarian cyst ...

ABDOMINAL MASS PALPATION

Look for

- ✓ *Site/ location*
- ✓ *Size & shape*
- ✓ *Surface, edge & consistency*
- ✓ *Mobility & attachment*
- ✓ *Bimanually palpable or pulsatile*

Site and Location

- Specify the quadrant
- Is it abdominal wall or intra-abdominal mass?
 - Feel the mass while the pt lifts her/his head and shoulders off the pillow to tense the abdominal wall
 - ✓ Prominently protruded mass - abdominal wall mass
 - ✓ If the mass disappears- intra-abdominal mass
- Identify whether abdominal or pelvic origin

Surface, Edge & Consistency

- A swelling that is
 - Hard, irregular in outline and nodular is likely to be *malignant*
 - Regular, round, smooth, tense swelling is likely to be *cystic*
 - Solid, ill-defined and tender mass suggests an inflammatory lesion

Mobility & Attachment

Mobility

- Mass arising from the liver, spleen, kidneys, gall bladder and stomach moves down ward during inspiration
- Mass arising from the small bowel, transverse colon, mesentery and greater omentum are not usually influenced by respiratory movement
- Side-to-side movable lower abdominal mass favors swelling of uterine origin but not from urinary bladder arising mass

Attachment

- When the mass is completely fixed it usually signifies one of three things:
 - A mass of retroperitoneal origin (e.g. pancreas)
 - Part of an advanced tumour with extensive spread to the anterior or posterior abdominal walls or abdominal organs
 - A mass resulting from severe chronic inflammation involving other organs (e.g. diverticulitis of the sigmoid colon or a tuberculousileocecal mass).

Pulsatile Mass

- Decide whether the pulsation comes from the mass or is transmitted
- Pulsatile and expansible abdominal mass favors abdominal aneurysm

PERCUSSION

To detect organomegaly, mass, ascites or intestinal obstruction

Look for

- ✓ *Percussion Notes*
- ✓ *TVLS*
- ✓ *Shifting Dullness*
- ✓ *Fluid Thrill*

Percussion Notes

- Tympanic – normal or intestinal obstruction
- Dull – organomegaly, mass, ascites or cystic mass
- Colonic resonance – enlarged kidney

Liver (TVLS)

- Start percussion at the right 2nd intercostal space over the midclavicular line down ward till you get relative dullness
- And then, start percussion at RLQ upward till you get dullness
- Measure the distance between the two points = “Total Vertical Liver Span”
- Total vertical liver span (TVLS)

- At 1 week of age
 - ✓ 4.5 – 5 cm
- At 12 year of age
 - ✓ 7 -8 cm
- At adult pediatric age group
 - ✓ = 10 cm ± 2 $\leftrightarrow$ normal
 - ✓ >12 cm $\leftrightarrow$ hepatomegaly
 - ✓ <8 cm $\leftrightarrow$ liver is shrunk

Spleen

- If not palpable & not found by percussion, we use maneuvers
 - Nixon's method
 - Castell's method
 - Traube's method

DETECTION OF ASCITES

Shifting Dullness

- Start percussion on the midline near the umbilicus and move your fingers laterally towards the flank
- When dullness is detected, keep your fingers in that position and ask the pt to turn towards the other side
- Wait for 15 seconds till peritoneal fluid redistributes
- Confirmed when the percussion note changed from dullness to tympanic at site of 1st dullness detected

Fluid Thrill

- Place one hand flat over the lumbar region of one side
- Ask an assistant to put ulnar side of his hand longitudinally and firmly in the midline of the abdomen to damp transmission of impulse via abdominal wall fat,
- Then tap the lumbar region on your side with the middle finger and feel for the wave

Ddx: Gross Ascites Vs large ovarian cyst

<i>Gross ascites</i>	<i>large ovarian cyst</i>
• <i>Flank fullness and dullness</i> • <i>Everted umbilicus</i> • <i>May have horizontal slit</i> • <i>Positive for shifting dullness and fluid thrill</i> • <i>Chief dullness is over flanks</i> • <i>Umbilical hernia may present</i> • <i>Mainly bulges laterally</i>	• <i>Resonant in flanks</i> • <i>Umbilicus drawn upward</i> • <i>Have vertical slit</i> • <i>Negative for shifting dullness & fluid thrill</i> • <i>Chief dullness is at the center or over the mass</i> • <i>Can't get below the mass (mass originate from pelvis)</i> • <i>Grows antero-posteriorly</i>

Ascites grading

- Have four grades
- Grade

0 = (minimal) ↔ positive puddle's sign

1 = (mild) ↔ shifting dullness (+ve), fluid thrill (-ve)

2 = (moderate) ↔ shifting dullness (+ve), fluid thrill (+ve)

3 = (massive) ↔ shifting dullness (-ve), fluid thrill (+ve)

Ddx – isolated ascites Vs hepatosplenomegaly + ascites

Isolated ascites	Hepatosplenomegaly + ascites
▪ <i>Nephrotic syndrome</i> ▪ <i>Nephritic syndrome</i> ▪ <i>TB peritonitis</i> ▪ <i>Peritoneal carcinomatosis</i> ▪ <i>Protein energy malnutrition</i> ▪ <i>TB pericarditis</i> ▪ <i>Constrictive pericarditis</i> ▪ <i>Protein losing enteropathy ...</i>	▪ <i>HSS</i> ▪ <i>NHL</i> ▪ <i>DTB to liver, spleen & peritoneum</i> ▪ <i>CHF 2^o to IE</i> ▪ <i>PHIN 2^o to ...</i> ▪ <i>VL rarely ...</i>

PORTAL HYPERTENSION

- Elevation of hepatic venous pressure gradient (HVPG) > 5mmHg

Causes

1. Increased intra-hepatic resistance
2. Increased splanchnic blood flow secondary to vasodilatation within splanchnic vascular bed

On the other way, it can be

I. Pre-hepatic

- Portal vein thrombosis
- Splenic vein thrombosis
- Banti's syndrome

II. Hepatic

a. Pre-sinusoidal

- Schistosomiasis
- Congenital hepatic fibrosis

b. Sinusoidal

- Cirrhosis
- Alcoholic hepatitis

c. Post-sinusoidal

- Hepatic sinusoidal obstruction (veno-occlusive syndrome)

III. Post-hepatic

- Budd chiari syndrome
- IVC obstruction
- Cardiac causes
 - ✓ Restrictive cardiomyopathy
 - ✓ Constrictive pericardities
 - ✓ Severe CHF

Complications

- Gastro-esophageal varieces
- Ascites
- Hypersplenism
- Spontaneous bacterial peritonitis

Investigations for ascites

-

AUSCULTATION

Look for

- ✓ *Bowel Sounds*
- ✓ *Succession Splash*
- ✓ *Vascular Bruits*

Bowel Sounds

- Place the stethoscope just to the right of the umbilicus, at the site of ileocecal valve
- Normal sounds consist of clicks and gurgles with a frequency of 6-36 per minute
- Increased frequency of bowel sounds occur in
 - ✓ Diarrhea
 - ✓ Mechanical intestinal obstruction
- Reduced or absent bowel sounds occur in
 - ✓ Paralytic ileus
 - ✓ Generalized peritonitis

Vascular Bruits

Listen for bruits by applying the stethoscope lightly

- Above and to the left of the umbilicus - aorta

- Above and to right/left of umbilicus - renal artery stenosis
- The iliac fossae - iliac arteries
- The epigastrium - coeliac or superior mesenteric arteries
- Over the enlarged liver - increased blood flow in liver tumors
- Over the enlarged spleen for friction rub - splenic infarction

Succession Splash

- Place the patient in supine position and place the diaphragm of the stethoscope over the epigastrium
- Place your hands on the lumbar region of the abdomen, and roll the pt briskly from side to side
- Splashing sound is heard if the stomach is distended with fluid
- Positive test is confirmed if there is splashing sound after 4 hrs of meal intake
- Succession splash is positive in
 - ✓ Gastric outlet obstruction (pyloric stenosis)
 - ✓ Paralytic ileus
 - ✓ Intestinal obstruction with distended bowel loops

NB.

- ✓ *Better, if auscultate before palpation and percussion for “**bowel sounds**” to avoid bowel obstruction/ disturbance during palpation or percussion.*

Ddx for intra-abdominal mass

- Hepatomegaly
- Splenomegaly
- Hepatoblastoma
- NHL
- Neuroblastoma
- Wilms tumor
- Rhabdomyosarcoma
- Teratoma
- Ovarian cyst
- Appendiceal mass
- Appendiceal abscess
- Abdominal TB
- Lymphadenopathy
- Polycystic kidney disease
- Aortic aneurism
- Psoas abscess
- Perinephric abscess
- Hydronephrosis
- Ileocecal TB

INVESTIGATIONS FOR ABDOMINAL FINDINGS

- CBC
- Abdominal U/S
- Serum albumin
- Ascetic fluid analysis
- LVT
- RFT
- Splenic aspiration
- Plain abdominal x – ray
- CT scan
- MRI

MSS EXAMINATION

Que
stio
ns

- *Do musculoskeletal examination*
- *Ddx for ur finding*
- *Investigations for ur ddx*

MSS EXAMINATION

Approaches to examine

- **Look**
- **Feel**
- **Move**
- **Measure**

NB.

- ➡ Talk to the pt before starting the examination
- ➡ Expose both sides with adequate light exposure
- ➡ Compare bilaterally

Look

Look for

- Symmetry/asymmetry between sides
- Skin color, dryness/wetness, hair
- Swelling
- Wound or wound dressing
- Deformity
- Atrophy
- Limb orientation – internally or externally rotated
- Abnormal movement

Feel

- Temperature – with ur back of hand
- Sensation
- Tenderness
- Dryness or wetness
- Muscle bulk
- Pulse
 - ✓ Lower extremity
 - Dorsalispedis arteries
 - Posterior tibialis arteries
 - Popliteal arteries
 - Femoral arteries
 - ✓ Upper extremity
 - Radial arteries
 - Brachial arteries
- Capillary refill
 - ✓ > 3 second → prolonged or abnormal
- Edema – presence/absence, pitting/non pitting

Move (range of motion)

- Active Vs passive
- **Active**
 - ✓ Ask the pt to move the normal one first, and then the affected site
 - ✓ Watch for

- decreased or increased movement of the joint compared to the normal one
 - pain with movement
 - abnormal movement
- ✓ Listen for crepitus or “popping”
- **Passive**
 - ✓ Move the joints passively, comparing the end points to the active
 - ✓ Again note
 - Any decreased or increased movement
 - Pain with the movement
 - Crepitus or “popping”

Measure

- ☐ Apparent length
 - ✓ From xiphisternum or umbilicus to medial malleolus
- ☐ Real length
 - ✓ From greater trochanter of the femur up to the medial malleolus
- ☐ True length
 - ✓ Between two bony prominences of a single long bone

MASS EXAMINATION

Questions

- ✓ *What do you see?*
- ✓ *Differential diagnosis for your findings*
- ✓ *Investigation for your differential diagnosis*
- ✓ *Management principle*

Approach

- *Inspection*
- *Palpation*
- *Auscultation – for bruit (not for all masses)*

Inspection

- Number – single or multiple
- Site/location
- Overlying skin color change
- Discharge or ulceration
- Size (estimate)
- Movement with respiration or swallowing (especially for neck swelling)

Palpation

➤ Mass

- Temperature – compare with the surrounding skin
- Tenderness
- Surface – smooth, nodular
- Border – regular , irregular or ill-defined
- Consistency – soft, firm, hard
- Fixity – to overlying or under lying tissue
- Pulsatility
- Size (measured value)

➤ Soft tissue Swelling

- Temperature (warmness or coldness)
- Tenderness
- Fluctuation
- Transillumination
- Reducibility

- Compressibility
- Pulsatility

NB.

*When you list your differential diagnosis for a mass; think of the structures that are found under/overlying the mass (i.e. skin, connective tissue, adipose tissue, muscles, vessels, nerves, bones, and others surrounding structures) ... then list the benign & malignant disease forms of each structure and do **NOT** forget the **age** of the patient.*

Example

➤ Neck mass

- | | |
|--------------------|--------------------------|
| ▪ Teratoma | ▪ Cystic hygroma |
| ▪ Fibroma | ▪ Castleman disease |
| ▪ Fibrosarcoma | ▪ Lymphoma |
| ▪ Lipoma | ▪ Neuroblastoma |
| ▪ Rhabdomyosarcoma | ▪ Goiter |
| ▪ SCM tumor | ▪ Thyroglossal duct cyst |
| ▪ Hematoma | ▪ Brachial cleft cyst |
| ▪ Hemangioma | ▪ Nasopharyngeal cancer |
| ▪ Lymphadenopathy | ▪ Osteosarcoma |
| ▪ Lymphadenitis | ▪ Ewing sarcoma |

Investigations

- Based on the location and physical findings of the mass
- ✓ CBC
- ✓ Hct or Hgb
- ✓ FNAC
- ✓ Biopsy
- ✓ X – ray
- ✓ Ultrasonography
- ✓ CT scan
- ✓ MRI, ...

WOUND EXAMINATION

Questions

- ✓ *What do you see?*
- ✓ *Ddx for ur finding*
- ✓ *Investigations for ur ddx*
- ✓ *Management principle*

Approach

- **Inspection**
- **Palpation**

Inspection

- Site / location
- Number – single, multiple
- Visible dressing
 - ✓ Clean
 - ✓ Bloody
 - ✓ Any color change, ...
- Discharge / bleeding
- Size (estimate)
- Margin – regular, irregular, round or oval
- Edge
 - ▶ Types of edge
 - ***sloping edge***
 - Signs of healing
 - Has 3 parts
 - ✓ Outer → white– due to scar or fibrous formation
 - ✓ Middle → blue – due to epithelial formation
 - ✓ Inner → red– due a red healthy granulation
 - ***Undermined edge***
 - Seen in a tuberculous ulcer
 - Disease process advances in deeper plane whereas skin proliferates inwards
 - ***Punched out edge***
 - Seen in a gummatous (syphilitic) ulcer
 - Due to endarteritis
 - ***Raised and beaded edge***

- Seen in a rodent ulcer (BCC)
- Beads are due to proliferating active cells
- ***Everted edge***
 - Seen in a carcinomatous ulcer(SCC)
 - Due to spill of the proliferating malignant tissues over the normal skin
- ***Inflamed edge***
 - Red, irregular with inflamed surrounding skin
- Floor / surface of the wound/ ulcer
 - ✓ Discharge
 - ✓ Granulation
 - ✓ Slough
- Surrounding skin
 - ✓ Redness
 - ✓ Pigmentation
 - Dark – typical for varicose ulcer
 - Hypopigmentation – in non-healing ulcer
 - ✓ Swelling

Palpation

- Temperature
- Tenderness
- Size (measured value)
- Floor – see if bleeds/discharge on touch
- Check the involvement of underlying structures

- ***Related examinations***

- Related lymph nodes
- Related arteries, veins and nerves
 - Pulse & capillary refill
- Movement in neighboring joints
 - Restriction to movement indicates muscle involvement or painful inflammation...

NB.

Ulcer has four parts

1. *Margin – a thin & most outer part of the ulcer*
2. *Edge – connects the margin & floor*
3. *Floor – it is the one visible (the visible wound)*
4. *Base – it is the one the ulcer rests, can be bone or soft tissue*

LOWER MOTOR EXAMINATION

Approaches to examine

- ☐ *Inspection*
- ☐ *Muscle tone*
- ☐ *Muscle strength*
- ☐ *Reflexes (deep and superficial)*

Inspection

- Position of the extremities after repositioning
 - ✓ Normal, rotated inward or outward
- Involuntary movements like fasciculation
- Muscle bulk and symmetry
- Compare left to Right extremities
- Look for presence of atrophy

Muscle tone

- Muscle tone is referred as resistance of a muscle to passive stretch
- Ask the patient to relax
- Flex and extend the patient's fingers, wrist, and elbow
- Flex and extend patient's ankle and knee
- There is normally a small, continuous resistance to passive movement
- Observe for *flaccid or spastic tone*

☐ Flaccid

- ✓ Decreased muscle tone
- ✓ Lower motor neuron lesion
- ✓ Loss muscle tone causing the limb to be loose or floppy
- ✓ The affected limb may be hyper extensible or flail like

☐ Spastic

- ✓ Upper motor neuron lesion
- ✓ Increased muscle tone

Muscle strength

- Test strength by having the patient move against your resistance
- Always compare one side to the other
- Have 6 grades (0 – 5) and scored out of 5

- Grading

0/5 = no muscle movement

1/5 = visible flicker muscle movement but no movement at the joints

2/5 = movement at joints but not against gravity

3/5 = movement against gravity but not against added resistance

4/5 = movement against resistance but less than normal

5/5 = normal strength

Reflexes

Deep tendon reflex

- The patient must be relaxed and positioned properly before starting
- Reflex response depends on the force of your stimulus. Use no more force than you need to provoke a definite response
- Reflexes can be reinforced by having the patient perform isometric contraction of other muscles (clenched teeth, hooking up fingers of both hands), and named as Jendrasik maneuver
- Grading scores ranges from 0 to 4
- Grade
 - 0 = absent (by Jendrasik maneuver)
 - 1+ or + = hypoactive (less brisk)
 - 2+ or ++ = normal (brisk)
 - 3+ or +++ = hyperactive without clonus (very brisk)
 - 4+ or ++++ = hyperactive with clonus

Technique

Knee

- Have the patient sit or lie down with the knee flexed
- In sitting positioned patient:
- Place your left pronated arm below the patient's right knee and left hand over the patient's left knee, and strike the right patellar tendon just below the patella
- Place your left supinated arm below the patient's right and left knees, and strike the left patellar tendon just below the patella
- Note contraction of the quadriceps and extension of the knee

Ankle

- Dorsiflex the foot at the ankle
- Strike the Achilles tendon
- Watch for contraction of gastrocnemius muscle & plantar flexion at z ankle

Clonus

- ☐ When there are sustained rhythmic oscillations (≥ 3 rhythmic oscillations)

Ankle clonus

- If the reflexes seem hyperactive, test for ankle clonus
- Support the knee in a partly flexed position
- With the patient relaxed, quickly dorsiflex the foot
- Observe for sustained rhythmic oscillations

Knee clonus

- Partially flex the knee
- Hold the patella and quickly push downwards
- Observe for sustained rhythmic oscillations

Superficial reflex

Plantar reflex

- Stroke the lateral aspect of the sole of each foot up to the 2nd toe with the end of a reflex hammer or key
- Note movement of the toes, normally flexion (withdrawal)
- Extension of the big toe with fanning of the other toes is abnormal
 - ✓ Referred to as a positive Babinski sign (sign of UMNL)
- Alternative methods of eliciting Babinski sign:
 - ❑ Oppenheim's sign→extension of big toe with dorsiflexion of the foot is also obtained by pressing heavily along the medial border of z tibia
 - ❑ Gordon's sign → squeezing the calf or Achilles tendon
 - ❑ Chadok's sign→ stroking the lateral border of foot
 - ❑ Bing sign→ pressing the dorsum of big toe
- No response to plantar reflex indicates lower motor neuron lesion

Key words

Paresis

- A condition typified by a weakness of voluntary movement or partial loss of voluntary movement or by impaired movement

-Plegia

- Used to describe paralysis in which all voluntary movement is lost or a weakness with complete or nearly complete loss

UMN weakness

- Results from disorders that affect the upper motor neurons or their axons in the cerebral cortex, internal capsule, brain stem or spinal cord

LMN weakness

- Results from disorders of cell bodies of the lower motor neurons in the brain stem motor nuclei and anterior horn of the spinal cord or dysfunctions of axons of these neurons

Myopathic weakness

- Results from disorders within the motor unit that affect the muscle fibers or neuromuscular junction

Clinical sign	LMNL	UMNL	Myopathic
<i>Fasciculation</i>	<i>Present</i>	<i>Absent</i>	<i>Absent</i>
<i>Atrophy</i>	<i>Present</i>	<i>None</i>	<i>Mild</i>
<i>Tone</i>	<i>Hypotonia</i>	<i>Hypertonia</i>	<i>Normal</i>
<i>Tendon reflex</i>	<i>Hyporeflexia</i>	<i>Hyperreflexia</i>	<i>Normal</i>
<i>Plantar reflex</i>	<i>No response</i>	<i>Babinski +ve</i>	<i>Normal</i>

Table of LMNL Vs UMNL Vs Myopathic weakness

Differential diagnosis

- ☐ Flaccid paralysis (monoplegia Vs paraplegia)?
- ☐ Spastic paralysis (monoplegia Vs paraplegia)?

Investigations

-

MALNUTRITION

Defn

- Is any condition caused by excess or deficient food energy or nutrient intake or by imbalance of nutrients
- Classified into undernutrition and over-nutrition

Causes of malnutrition (undernutrition)

- Can be
 - Primary – due to inadequate food intake
 - Secondary – secondary to other diseases
- The three layers of the determinantsof nutritional status (UNICEF (6))

1. Immediate Causes:

- ✓ Are causes which act on individuals
- ✓ Include: inadequate dietary intake and infection or disease

2. Underlying Causes

- ✓ Influence households and communities.
- ✓ Include shocks, such as
 - Drought
 - Flooding
 - Inadequate access to health services
 - Household food insecurity, ...

3. Basic causes:

- ✓ They influence communities and societies
- ✓ These include the country's social, economic and political situation

Q. Asses for signs of malnutrition

- Both macro & micro nutrient deficiencies should be assessed

General appearance

- Level of consciousness
- Health status (acute, healthy, chronic)
- Old man appearance or cachexic – skinny and bony appearance
- Emaciated
- Edematous

Vital signs

- Bradycardia or Tachycardia
- Tachypnea
- Hypo or hyperthermia
- Postural hypotension

HEENT

Head

- Hair color & distribution
- Alopecia, pluckability or brittle hair
- Craniotables
- Caput quadratum
- Frontal bossing
- Fontanel size, closure & surface
- Sutures – closed or not

Eye

- Sunkening of eyeball
- Pale conjunctiva
- Bitot's spot
- Icteric sclera
- Periorbital edema
- Discharge

Mouth and throat

- Buccal mucosa (pink, pale, wet, dry)
- Tongue atrophy
- Angular cheilitis/ stomatitis
- Dentition
- Gum bleeding & swelling
- Oral ulcers, OHL, candidiasis

Respiratory system

- Costochondral beading
- Harrison groove
- Pigeon chest deformity

CVS

- Pounding pulse
- S3 gallop

Abdomen

- Protruded abdomen - because of distended stomach or intestinal loops
- Hepatomegaly – due to severe fatty infiltration (fatty liver)

Integumentary system

- Palmar & plantar pallor
- Wet, dry, pink , pale, ... skin
- Warm or cold extremities

- Skin rash (hyper or hypo pigmentation)
 - ▶ Aka Kuash-dermatoses
 - ▶ Often involves the perineum, groin, limbs, ears, & armpits
 - ▶ **Has 3 grades**
 - ❑ **Grade I(Mild)** →discoloration or a few rough patches of skin
 - ❑ **Grade II (Moderate)** →multiple patchy on arms &/ or legs
 - ❑ **Grade III (Severe)**→ flaky paint appearance of skin, fissures

Musculoskeletal system

- Wrist widening
- Double malleoli
- Bow leg
- Joint swelling
- Pitting leg or sacral edema (GBS)
 - ✓ Grading See @ “edema”

CNS

- ❑ Mental status
 - ✓ Consciousness – Conscious, lethargic, comatose
 - ✓ Irritability
 - ✓ Expression of misery and sadness

Investigations

- RBS
- CBC – leukocytosis or leucopenia
- Hct or Hgb – anemia
- Blood film- to rule out malaria as the cause of anemia
- Peripheral morphology – to know the underlying cause of anemia
- PICT – HIV/AIDS is one of our differential diagnosis
- Stool microscopy – to rule out hookworm as the cause of anemia
- U/A & culture - UTI
- CXR – to rule out pneumonia
- Tests for TB
- Serum albumin
 - ✓ Reduced in kwashi patients
- Serum electrolytes
 - ✓ K^+ decreases
 - ✓ Na^+ increase
- RFT- to rule out renal failure as a complication

Admission criteria

☐ For age 0-6 months

- ✓ $W/L < -3$ z score OR
 - ✓ Visible wasting OR
 - ✓ Edema of both feet OR
- } **plus** any one of medical complication

❑ **For age 6-59 months**

- ✓ WFL/H <-3 Z score OR
 - ✓ MUAC <11.5 cm OR
 - ✓ Edema of both feet (+, ++)
- } **plus** any one of medical complication
or failed appetite test

OR

- ✓ +++ edema OR
- ✓ Marasmic kwashiorkor (WFL/H <-3 z score with edema, or MUAC <11.5 cm with edema)

❑ **For age 5 to 18 years**

- ✓ WFL/H <-3 z score OR
 - ✓ MUAC in severe category OR
 - ✓ BMI for age <-3 SD OR
 - ✓ Edema of both feet (+,++)
- } **plus** any one of medical complication
or failed appetite

OR

- ✓ +++ edema OR
- ✓ Marasmic kwashiorkor (WFL/H <-3 z score with edema, OR MUAC in severe category with edema)

Complications

- Hypoglycemia (< 54 mg/dl)
- Dehydration
- Severe anemia (Hgb< 4 g/dl or Hct< 12%)
- Hypothermia (T°< 35 °c)
- Heart failure – due to severe anemia

- Infection – pneumonia, UTI, ...
- Septic shock – due to infection & fluid loss
- Electrolyte disturbance – due to diarrhea, vomiting, poor absorption or intake
- Renal failure – due to DHN or poor perfusion

Management principles

- ✓ Treat life threatening conditions
- ✓ Nutritional rehabilitations
- ✓ Routine medications
- ✓ Follow up

NB.

*When you asked to assess **MICRO NUTRIENT** deficiency; assess for:*

- ✓ *Signs of rickets - vit D deficiency*
- ✓ *Signs of anemia - iron, folate&vit B12 deficiency*
- ✓ *Signs of Vit C deficiency*
- ✓ *Signs of Vit K deficiency, ...*

Do **NOT** asses for **MACRONUTRIENT** deficiency = (signs of protein- energy malnutrition)

ANEMIA

Defn

- Anemia may be defined as a reduction in RBC mass or Hgb concentration
- The threshold for defining anemia is a Hgb or Hct that is more than two standard deviations below the mean for the reference population
- Normal ranges for Hgb or Hct vary substantially with age and sex

Physiologic adjustments of anemia

- Increased cardiac output
- Reduction of mixed venous oxygen tension (increased oxygen extraction by hypoxic tissue)
- Shunting of blood flow toward vital organs and tissue
- Increased 2,3-diphosphoglycerate (DPG) in the RBCs
 - ✓ Causes reduced oxygen affinity, shifting the Hgb-O₂ dissociation curve to the right and thereby enhancing oxygen release to the tissues
- Stimulation of erythropoietin production

Classifications

- Classified on either a physiologic or morphologic basis

Physiologic classification

1. Decreased production
 - Deficiency states- Iron deficiency, Vit B12 deficiency, ...
 - Bone marrow failure
 - Dyshematopoietic anemia
2. Increased destruction
 - Corpuscular - membrane defects, enzyme defects, Hgb defects
 - Extra corpuscular- isoimmune, autoimmune, idiopathic
3. Blood loss – hemorrhage, poorly tied umbilicus, hook worm infection

Morphologic classifications(based on MCV values)

1. Normocytic anemia
 - Anemia of chronic disease (70%)
 - Iron deficiency (early)
 - Hemolytic anemia
 - Malignancy
 - Renal failure
 - Acute bleeding
 - Hypersplenism
 - Microangiopathy- HUS, TTP, DIC
 - Enzymopathies – G6PD, PK deficiencies

2. Microcytic anemia

- Iron deficiency anemia (late)
- Anemia of chronic disease (30%)
- Thalassemia syndrome
- Sideroblastic anemia
- Lead poisoning
- Copper deficiency

3. Macrocytic anemia

- VitB12 deficiency
- Folate deficiency
- Drug toxicity → Zidovudine, Methotrexate
- Hypothyroidism
- Acquired aplastic anemia
- Congenital aplastic anemia

Common Causes of Anemia in Our Set up (Gondar University Hospital)

➤ Micro-nutrient deficiencies

- Iron deficiency
- Folic deficiency
- Vitamin B12 deficiency

➤ Infectious diseases

- HIV

- Hookworm
 - Malaria
 - Trichuris trichuria
 - Visceral leishmaniasis
 - Schistosomiasis
- Blood loss
 - Malignancy and chronic illness

Patient Evaluation

History

- Emphasize on
 - ✓ Age & sex
 - ✓ Perinatal hx (prematurity)
 - ✓ Dietary hx
 - ✓ Blood loss – acute or chronic blood loss
 - ✓ Underlying disease – malignancy, chronic illness
 - ✓ Exposure to drugs
 - ✓ Family hx
 - ✓ Race
 - ✓ Geographical location (residency)
 - ✓ Travel hx (malaria)
 - ✓ Infection (hook worm, malaria, HIV, ...)
 - ✓ Symptoms of anemia and underlying disease
 - Easy fatigability, tinnitus, vertigo, blurring of vision, ...

Physical examination

❑ *Questions = asses for signs of anemia*

General appearance

- Nutritional status

Vital signs

- Tachycardia
- Postural hypotension
- Wide pulse pressure

HEENT

- Pale conjunctiva
- Icteric sclera (hemolysis)
- Mucosal pallor
- Angular stomatitis
- Atrophy of tongue papillae
- Glossitis

Lymphoglandular system

- Lymphadenopathy

CVS

- Tachycardia
- Bounding pulse
- S3 gallop
- Systolic ejection murmur

Abdomen

- Organomegaly
- PR examination – occult blood

Musculoskeletal system

- Bone tenderness

Integumentary

- Palmar pallor and palmar crease involvement
- Skin pallor
- Koilonychias – spooning of nails
- Petechiae, bruising, purpura
- Ulcers on lower extremities

CNS

- Mental status
- Fundoscopy for
 - ✓ Papilledema (acute anemia)
 - ✓ Retinal hemorrhage (severe anemia)
 - ✓ Optic atrophy (cobalamine deficiency)

NB.

- ☐ Pallor usually noticed when Hgb level reaches to 7-8 g/dl
- ☐ If it involves only the palm = some palmar pallor
- ☐ If it involves the palmar crease = severe palmar pallor

Laboratory

- CBC :
 - ✓ RBC count, Hgb/Hct, RBC indices (MCV, MCH, MCHC, RDW), Reticulocyte count, WBC count with differential, platelet count
- Peripheral smear
 - ✓ Size, shape, chromicity of RBCs, Rouloux formation, hypersegmentation of neutrophils
- Bilirubin level - Hemolytic anemia
- Direct antiglobulin or Coombs test - Autoimmune hemolytic anemia
- Hemoglobin electrophoresis –Hemoglobinopathies
- Red cell enzyme studies - G-6-PD, pyruvate kinase
- Osmotic fragility – Spherocytosis
- Iron studies– Iron deficiency anemia
 - ✓ Serum iron, serum transferrin, serum ferritin, TIBC
- Folate, Vit B12 and serum/urine methylmalonic acid
- Blood typing and cross matching to assess possible isoimmune anemia in a neonate and to prepare for transfusion
- Bone marrow aspiration and biopsy
 - ✓ Cellularity, red cell appearance, iron stain, culture, cytogenic studies

- BUN/creatinine levels - To assess renal function
- Thyroxine (T4) or TSH - Hypothyroidism
- Stool microscopy
- Blood film

NB

When we say severe anemia?

- ❖ Hgb < 7 gm/dl or Hct < 21%
- ❖ Malnourished pts : Hgb < 4 gm/dl or Hct < 12%
- ❖ Malaria : Hgb < 5 gm/dl or Hct < 15%
- ❖ Neonate : Hgb <

WHO grading of anemia

- ☐ Mild = Hgb: 10-cutoff point
- ☐ Moderate =Hgb:7-10 gm/dl
- ☐ Severe =Hgb: less than 7gm/dl

Clinical grading

- ☐ Mild = Conjunctival pallor and/or mucous membrane
- ☐ Moderate = Obvious skin pallor
- ☐ Severe = Palmer crease pallor

Management Principles

- Supportive
 - ✓ ABC of life
 - ✓ Blood transfusion
 - ✓ Oxygen
 - ✓ Fluid
 - ✓ Bed rest
- Specific
 - ✓ Iron
 - ✓ Folate and cobalamine
 - ✓ Corticosteroids
 - ✓ Recombinant erythropoietin
 - ✓ Hematopoietic cell transplantation (HCT)
 - ✓ Immunosuppression

Key words

- Hematocrit (Hct)
 - ✓ Is the fractional volume of whole blood sample occupied by RBC, expressed as percent
- Hemoglobin (Hgb)
 - ✓ Is a measure of a concentration of the RBC pigment Hgb in whole blood, expressed as grams per 100 ml (dl) of whole blood
- Mean corpuscular concentration (MCV)
 - ✓ The mean value of the volume of individual RBCs in the blood sample
 - ✓ Values vary with age
 - ✓ Microcytic, normocytic or macrocytic
- Mean corpuscular hemoglobin concentration (MCHC)
 - ✓ Is a calculated index ($MCHC = Hgb/Hct$), yielding a value of grams of Hgb per 100 ml of RBC
 - ✓ Values vary with age
 - ✓ Normochromic or hypochromic
- Red cell distribution width (RDW)
 - ✓ Is a quantitative measure of the variability of RBC sizes in the sample (anisocytosis)
 - ✓ Generally normal value is 12 – 14 %, may slightly vary with age
 - ✓ Helpful in differentiating IDA from thalassemia in microcytic anemia
 - $RDW > 20$ are more likely IDA

IRON DEFICIENCY ANEMIA

Introduction

- IDA is the most common nutritional deficiency in children
- Prevalence is higher in developing countries
- The development & the rapidity of IDA is dependent upon the body's iron store
- Iron store in turn depends on : age, sex, rate of growth and balance b/n loss & absorption
- Peak prevalence occurs during late infancy and early childhood when the following may occur
 - ✓ Rapid growth with exhaustion of gestational iron
 - ✓ Low levels of dietary iron
 - ✓ Complicating effect of cow milk-induced exudative enteropathy

Iron metabolism

- Iron is available in the body as
 - ✓ Functional iron - Hgb, myoglobin, cytochromes and catalase
 - ✓ Transport iron - transferrin
 - ✓ Storage iron –ferritin and hemosiderin
- Dietary requirement of iron is about 8-10 g/dl/day of which 10% will be absorbed
- Dietary iron absorption occurs through out the intestine, but especialy in duodenum and proximal jejunum
- Intestinal iron absorption is a function of three principal factors
 - ✓ Body iron stores (transferrin and ferritin)
 - ✓ Erythropoietic rate
 - ✓ Bioavailability of dietary iron
- Iron absorption is;
 - ✓ Facilitated by - citrates and ascorbic acid
 - ✓ Inhibited by - phytates, tannates, oxalates, phosphates, sulphates and antacids
- Dietary iron exists in two states:
 - ✓ Heme (10%) – found in animal proteins
 - ✓ Nonheme(90%) – vegetarians, found in the form of ferric state

Causes of IDA

- Increased demand for iron and/or hematopoiesis
 - Rapid growth in infancy or adolescence
 - Pregnancy
 - Erythropoietin therapy
- Increased iron loss
 - Chronic blood loss
 - Menses
 - Acute blood loss
 - Phlebotomy as treatment for polycythemia vera
 - Blood donation
- Decreased iron intake or absorption
 - Inadequate diet
 - Malabsorption from disease (sprue, Crohn's disease)
 - Malabsorption from surgery (post- gastrectomy)
 - Acute or chronic inflammation
 - Unmodified cow milk
- In the first 3 – 6 months of life
 - Maternal iron deficiency
 - Prematurity
 - Administration of erythropoietin (EPO) for anemia of prematurity
 - Fetal – maternal hemorrhage
 - Twin – twin transfusion syndrome
 - Other Perinatal hemorrhagic events

Clinical Manifestations

Symptoms

- Anorexia
- Pica and pagophagia – a desire to ingest unusual substances
- Fatigue
- Irritability when Hgb level < 5 gm/dl
- Impaired psychomotor and/ mental development in infants
- Cognitive impairment in adolescents
- Poor development
- Thrombosis
- Impaired exercise performance

Signs

- Pallor
- Blue sclera
- Angular cheilitis
- Atrophic glossitis
- Koilonychia (spooning of nails)
- When the hemoglobin level falls below 5g/dl
 - ✓ Irritability and anorexia are prominent
 - ✓ Tachycardia and cardiac dilation occur
 - ✓ Systolic murmurs are often present
- Plummer – Vinson syndrome: characterized by the combination of
 - ✓ IDA, glossitis, cheilosis and esophageal web

Stages in the development of IDA

1. Depletion of iron stores

- ✓ Decreased level of serum ferritin
- ✓ Normal level of serum iron, TIBC, % saturation and RBC protoporphyrin level
- ✓ Patient is asymptomatic

2. Iron deficient erythropoiesis

- ✓ Decreased level of serum ferritin
- ✓ Reduced serum iron and % saturation
- ✓ Increased TIBC and RBC protoporphyrin level
- ✓ Serum transferrin receptor levels increases
- ✓ No change in Hct level

3. Iron deficiency anemia

- ✓ Microcytic, hypochromic and aniso-poikilocytosis: cigar & pencil shaped RBCs
- ✓ Increased RDW
- ✓ Elevated levels of Free erythrocyte protoporphyrin (FEP)

Laboratory studies

- Hgb – decreased
- RBC count - decreased
- Peripheral smear, RBCs are
 - ✓ Small (microcytic)

- ✓ Pale (hypochromic)
- ✓ Poikilocytosis in the form of small elongated red cells (pencil cells) and anisocytosis
- Red cell distribution width (RDW)– increased
- Free erythrocyte protoporphyrin - elevated
- Iron studies
 - ✓ Serum iron - decreased
 - ✓ Total iron binding capacity - increased
 - ✓ Serum ferritin – decreased
 - ✓ Iron saturation – decreased
- Bone marrow iron stain (**Prussian blue stain**)
 - ✓ The disappearance of stainable iron from mononuclear phagocytic cells is a diagnostic finding

IDA Vs Thalassemia Vs anemia of Chronic illness

Study	IDA	Thalassemia	Anemia of chronic illness
<i>Hgb</i>	<i>Decrease</i>	<i>Decrease</i>	<i>Decrease</i>
<i>MCV</i>	<i>Decrease</i>	<i>Decrease</i>	<i>Normal–decrease</i>
<i>RDW</i>	<i>Increase</i>	<i>Normal</i>	<i>Normal – increase</i>
<i>RBC</i>	<i>Decrease</i>	<i>Normal - ↑</i>	<i>Normal – decrease</i>
<i>Serum ferritin</i>	<i>Decrease</i>	<i>Normal</i>	<i>Increase</i>
<i>TIBC</i>	<i>increase</i>	<i>Normal</i>	<i>Decrease</i>
<i>Transferrin saturation</i>	<i>Decrease</i>	<i>Normal</i>	<i>Decrease</i>
<i>Free erythrocyte protoporphyrin</i>	<i>Increase</i>	<i>Normal</i>	<i>Increase</i>
<i>Transferrin receptor</i>	<i>Increase</i>	<i>Normal</i>	<i>Increase</i>
<i>Reticulocyte Hgb concentration</i>	<i>Decrease</i>	<i>Normal</i>	<i>Normal – decrease</i>

Taken from Nelson text book of pediatrics 19th edition

Management Principle

- Oral supplement
 - ✓ 4-6 mg/kg elemental iron daily in three divided doses for 6–8 weeks after Hgb level and the RBC indices return to normal
 - ✓ For atleast 3 months
- Parenteral iron therapy ; indications
 - ✓ Unable to tolerate oral iron
 - ✓ Who need iron on an ongoing basis, usually due to persistent GI blood loss
 - ✓ Chronic hemorrhage
 - ✓ Malabsorption
 - ✓ Refractoriness to oral therapy
 - ✓ Patient unwilling to take oral therapy
- Blood transfusion
 - ✓ When the anemia is severe or decompensated
 - ✓ Cardiovascular instability
 - ✓ Continued and excessive blood loss
 - ✓ Patients requiring immediate intervention

Response to iron therapy

Time after iron administration	Response
12 – 24 hours	Replacement of Intracellular iron enzyme (subjective improvement: increased appetite, decrease irritability)
36 – 48 hours	Initial bone marrow response; erythroid hyperplasia
48 – 72 hours	Reticulocytosis ; peaking at 5 -7 days
4 – 30 days	Increase in haemoglobin level
1 -3 months	Repletion of stores

Prevention

- Appropriate nutrition
- Health education
- Supplementation for those with increased demand
- Fortification of formulas, cereals...
- Diagnose and treat underlying conditions
- Quality antenatal and neonatal care
- In term infants, breastfeeding exclusively for first four to six months
- In breastfed preterm or low birth weight infants, elemental iron supplementation starting at one month of age and is continued until 12 months of age

RICKETS

Q. Asses for signs of rickets

Look for

- ✓ Craniotables
- ✓ Frontal bossing
- ✓ Caput quadratum
- ✓ Fontanel size & closure
- ✓ Dentition
- ✓ Rachitic rosary
- ✓ Harrison groove
- ✓ Pigeon chest deformity
- ✓ Protruding abdomen
- ✓ Wrist widening
- ✓ Double malleoli
- ✓ Bowlegs of knock knees

Defn

- Is a disease of growing bone that is due to unmineralized matrix at the growth plates and occurs in children only before fusion of the epiphyses
- Because growth plate cartilage and osteoid continue to expand but mineralization is inadequate, the growth plate thickens

Causes

- Poor sun light exposure
- Inadequate dietary intake
- Breast feeding – contains low calcium
- Malabsorption of vit.D from intestine
- Defects in the metabolism of vit.D
- What is adequate sunlight exposure?
 - ✓ Expose in the morning with in 2:00 – 4:00 local time
 - ✓ For 15 – 20 minutes
 - ✓ No external ointment
 - ✓ Expose all body parts except the eyes & genitalia
 - ✓ Starting from at age of 10 days till he/she starts to ambulate
 - Why not before the age of 10 days?

Clinical Features

Usually appears towards the end of 1st year and during the 2nd year of life,
but

- Mother with vit.D deficiency
 - Prematures and
 - Infants on drugs like phenytoin
- manifest as early as 2 months*

General

- Failure to thrive
- Listlessness
- Protruding abdomen
- Muscle weakness
- Fracture

Head

- Craniotabes
 - ✓ The first sign of rickets
 - ✓ Softening of the cranial bones
 - ✓ Detected by applying pressure at the occiput or over the parietal bones
 - ✓ Sensation is similar to the feel of pressing into a Ping-Pong ball and then releasing
- Frontal bossing
- Caput quadratum – box like structure
- Wide fontanel
- Delayed anterior fontanel closure (> 2 yrs)
- Delayed dentition
- Craniosynostosis

Chest

- Rachitic rosary (costochondral beading)
 - ✓ Widening of the costochondral junctions
 - ✓ Feels like the beads of a rosary as the examiner's fingers move along the costochondral junctions from rib to rib
 - ✓ Non tender & blunted, where as scurvy is tender & sharp
- Harrison groove
 - ✓ The horizontal depression along the lower anterior chest
 - ✓ Due to pulling of the softened ribs by the diaphragm during inspiration
- Pigeon chest deformity

Abdomen

- Protruding abdomen

Back

- Scoliosis
- Kyphosis
- Lordosis

Extremities

- Enlargement of the wrists and ankles
 - ✓ Due to growth plate widening
- Deformity of the pelvis
- Bowlegs of knock knees
- Double malleoli
- Greenstick fractures

Complications

- Bone deformity with pathological fracture
- Infection – recurrent pneumonia
- Hypocalcaemia with tetany
- Cor pulmonale
- Laryngeal spasm

Investigations

- ✓ Wrist x-ray → widening, cupping, fraying
- ✓ Serum Ca level → normal or low
- ✓ Serum phosphorus level → < 4 mg/kg
- ✓ Serum alkaline phosphatase → elevated
- ✓ Urinary cAMP level → elevated
- ✓ Serum 25-hydroxycholecalciferol → low
- ✓ CBC
- ✓ CXR

Differential diagnosis

- *See at HEENT*
- Costochondral beading
 - Rickets
 - Scurvy
 - Chondrodystrophy

- Cytomegalic inclusion bodies
 - Syphilis
 - Copper deficiency
 - Rubella
- Pigeon shaped chest
- Rickets
 - Congenital
 - Skeletal dysplasia
 - Emphysema
 - Marfan's syndrome
 - Noonan syndrome

DEHYDRATION

Defn

- Refers to a deficit of total body water with an accompanying disruption of metabolic process
- Occurs when free water loss exceeds free water intake

Clinical manifestations in well-nourished patients

Symptoms

- Thirst
- Restlessness
- Irritability
- Sunken eye ball
- Diminished level of consciousness
- Decreased urine volume and frequency
- Decreased tear while crying

Signs

- Sunken eye ball
- Sunken fontanel(infants)
- Decreased tear→ if you see when he/she crying (do not make him/her cry)
- Dry mucosal membrane
- Decreased skin turgor
- Low urine output
- Cool extremities
- Rapid and fible pulse
- Peripheral cyanosis

Questions = Asses for signs of dehydration

☐ Assess for signs of the following four important parameters

1. Mental status
2. Eye ball
3. Drinking ability
4. Skin turgor

Classification

- | | | |
|-----------------------|---|--|
| A. No dehydration | } | ≥ 2 of the above parameter signs are needed |
| B. Some dehydration | | |
| C. Severe dehydration | | |

Parameter	No DHN	Some DHN	Severe DHN
Mental status	<i>Alert</i>	<i>Restless, irritable</i>	<i>Lethargic or unconscious</i>
Eye ball	<i>No sunken eyes</i>	<i>Sunken eyeball</i>	<i>Sunken eyeball</i>
Drinking	<i>Normal</i>	<i>Eagerness to drink</i>	<i>Unable to drink</i>
Skin turgor	<i>Normal</i>	<i>Skin pinch returns slowly</i>	<i>Skin pinch returns very slowly</i>

Dehydration in severe acute malnourished patients

- All signs of DHN in normal child are present in SAM with no DHN
- Diagnosis is mainly based on the history
- Ask the mother if the child has :
 - ☐ History of recent fluid loss
 - ✓ Watery diarrhea or vomiting
 - ✓ Recent increment in frequency or volume of fluid lost
 - ☐ History of recent change in child's appearance
 - ✓ Irritability, loss of consciousness ...
 - ☐ History of recent Sunkening of the eyeball
- No classification of DHN in malnourished patients

Why P/E is not done to assess DHN in malnourished patients?

- Kwashiorkor patients will be falsely negative
 - ✓ Edematous children are over-hydrated but they are frequently hypovolemic due to dilation of blood vessels with low cardiac output
- Marasmic patients will be falsely positive, because
 - ✓ SAM patients will have all signs of DHN with no DHN
 - ✓ They already lost their protein and subcutaneous fat which leads to sunkening of eyeball & slow return of skin pinch

NB.

➡ When a patient presented with diarrhea of more than 2 weeks with:

- ☐ No signs of DHN = ***Persistent diarrhea***
- ☐ Signs of DHN = ***Severe persistent diarrhea***

Management principles

Management of dehydration for well-nourished patients

1. No Dehydration (plan A)

- Loss is estimated to be < 5% of body weight
- Treat diarrhea at home
- Rules of 3 'Fs'

1. Give extra **Fluid**
2. Continue **Feeding**
3. When to come for **Follow up**

❑ **Fluid** :-in addition to the usual fluid intake give:

- More fluids than usual – breast milk, food based fluids (soup, rice water, yoghurt) or clean water
- ORS
 - ✓ 10ml/kg OR
 - ✓ 50-100ml for those < 2yrs
 - ✓ 100-200ml for those ≥2yrs

per each bowl motion

❑ **Feeding**

- Depends on age, food preference, pre illness feeding pattern
- Frequent small feeding are better tolerated
- Frequent breast feeding, cow's milk or formula milk
- Continue other foods if he/she started

❑ **Zink supplementation**

- Reduce duration and severity of diarrhea

- Increase use of ORS and reduce inappropriate use of antimicrobials
- 10 mg/d for infants < 6 mon of age and 20 mg/d for those age ≥ 6 mon for 10–14 days

❑ **Follow up**

- See him in 2 days
- Come back immediately if the child
 - ✓ Starts to pass many watery stools
 - ✓ Has repeated vomiting
 - ✓ Eating and drinking poorly
 - ✓ Develops a fever
 - ✓ Has blood in the stool or
 - ✓ The child does not get better in 3 days

2. **Some Dehydration (plan B)**

- Loss is estimated to be 5 – 10 % of body weight
- Treat with ORS
 - ✓ 75ml/kg over 4 hours
 - ✓ Continue breast feeding
 - ✓ If vomiting, wait for 10minutes
 - ✓ Give zink
 - ✓ After 4hours, reassess and classify DHN
 - ✓ If no DHN →send home the child with ORS packet for two days

3. Severe DHN (plan C)

- Loss is estimated to be >10% of body weight
- Start IV fluid immediately
- If the patient can drink, give ORS by mouth till the drip is setup
- Volume is 100ml/kg Ringer's lactate or NS
 - Divided in to two doses (30 ml/kg & 70 ml/kg)
 - ❑ For infants (< 12 months old)
 - 1st give 30 ml/kg over 1 hr
 - Then give 70 ml/kg over 5 hr
 - ❑ For children (≥ 12 months old)
 - 1st give 30 ml/kg over 30 minutes
 - Then give 70 ml/kg over 2 ½ hr

➡ Monitoring

- Reassess the patient every 1-2 hours
 - ✓ If hydration is not improving, give IV fluid more rapidly
- After 6 hours (infant) or 3 hours (older) evaluate the patient and choose appropriate treatment plan (A, B or C)
 - ✓ If signs of severe DHN are still present, repeat IV infusion as a child in treatment plan C
 - ✓ If still shows signs of some DHN, discontinue the IV infusion and give ORS solution for four hours, as specified in Treatment Plan B
 - ✓ If no signs of DHN; follow treatment plan A and observe the child for 6 hours before discharge

Management of dehydration with SAM

- ❑ For conscious and no signs of shock, we give ReSoMal solution
 - 5 ml/kg every 30 minutes for the first 2 hours then
 - 5-10 ml/kg every hour for up to 10 hours
- ❑ For unconscious child with no signs of shock, we give IV fluid
 - Darrow's solution with 5% glucose OR
 - Ringer lactate with 5% glucose OR
 - Normal saline with 5% glucose at 15ml/kg for the 1st hour and reassess
 - ✓ If improving → give 15ml/kg for the 2nd hour
 - ✓ If conscious → give ReSoMal by NG tube
 - ✓ If not improving → Septic shock
- ❑ Only rehydrate until the weight deficit is corrected and then stop.
- ❑ After rehydration
 - Non-edematous children
 - ✓ < 2 years → give 50-100 ml after each watery stool
 - ✓ 2 years and older → give 100-200 ml after each watery stool
 - Edematous children → give 30ml after each watery stool

What is ReSoMal?

- ReSoMal is a rehydration solution for children with SAM
- ReSoMal = Rehydration solution for malnutrition
- Contains less sodium, more sugar and more potassium than standard ORS
- Is intended for severely malnourished children with diarrhoea
- It should be given by mouth or by NG tube

Contents

- Water = 2L
- WHO – ORS = one 1-litre packet
- Sugar = 50 gram
- Mineral mix solution = 40 ml or one leveled scoop combined vitamins & minerals

Preparation

- ✓ Wash hands
- ✓ Empty one 1-litre standard ORS packet into container that holds more than 2 liters
- ✓ Measure and add 50 grams of sugar. (It is best to weigh the sugar on a dietary scale that weighs to 5 g.)
- ✓ Measure 40 milliliters or one leveled scoop of CMV in a graduated medicine cup or syringe; add to other ingredients
- ✓ Measure and add 2 liters of cooled boiled water
- ✓ Stir until dissolved
- ✓ Use within 24 hours

DHN may be due to;

- ☐ Ongoing fluid loss
- ☐ 3rd space fluid loss
- ☐ Inadequate fluid intake
- ☐ Defective intestinal absorption

EDEMA

Defn

- ✓ Edema is a clinically apparent increase in the interstitial fluid volume.

Questions = assess for signs of edema

Sites/ locations

- Pedal – on bony prominence of dorsum of the foot
- Ankle – on bony prominence of medial malleoli
- Pretibial – anteromedial shaft of the tibia, 1/3 below the tibial tuberosity
- Sacral – for infants & bed ridden pts, on the sacral area

Pitting Vs non pitting

- Gently compressing the area for at least 15 sec with the thumb & release

Grading

Based on location

❖ For malnourished patients

- ☐ Grade I (mild) → both pedal / ankle
- ☐ Grade II (moderate) → both feet + lower legs, hands or lower arm
- ☐ Grade III (severe) → generalized bilateral pitting edema including both feet, legs, arms & face

❖ For well-nourished patients

- ☐ Grade I → pedal and pretibial edema
- ☐ Grade II → leg and thigh edema
- ☐ Grade III → abdominal wall & sacral edema
- ☐ Grade IV → anasarca or generalized body swelling

Based on time

❖ By duration of refill of the indented skin (pitting edema)

- ☐ Grade I → returns within 5 seconds
- ☐ Grade II → returns within 5-10 seconds
- ☐ Grade III → returns within 10-15 seconds
- ☐ Grade IV → returns after 15 seconds

Differential diagnosis- (age dependent)

➤ Pitting edema (GBS)

- Kwashiorkor or edematous malnutrition

- CHF
- Nephrotic syndrome
- AGN
- HUS
- HSP
- TB pericardities
- Protein losing enteropathy
- Fulminant hepatic failure
- Congenital hepatic fibrosis
- Cirrhosis
- Budd – chiari syndrome

➤ **Non pitting edema**

- Lymphatic obstruction

Investigations

- ✓ RBS
- ✓ Serum albumin
- ✓ U/A
- ✓ ECG, Echo
- ✓ RFT
- ✓ LFT
- ✓ Abdominal U/S
- ✓ Chest x – ray

DOWN SYNDROME

Q. Asses for Down syndrome

Look for

➤ **Craniofacial**

- Brachycephaly with flat occiput*
- Upward slanted palpebral fissures*
- Epicanthal folds*
- Delayed fontanel closure*
- Flat nasal bridge*
- Protruding tongue*
- open mouth*
- High arched palate*
- Small & low set ear*

➤ **CVS**

- Endocardial Cushings defects (AVSD)* - the most common
- Ventricular septal defect*
- Atrial septal defect
- Patent ductus arteriosus
- Aberrant subclavian artery
- Pulmonary hypertension

➤ **MSS**

- Joint hyperflexibility*
- Short neck, redundant skin*
- Short metacarpals and phalanges*
- Short 5th digit with clinodactyly*
- Single transverse palmar creases (Simian creases)*
- Wide gap between 1st and 2nd toes* (sandal gap)
- Pelvic dysplasia
- Short sternum*

➤ **GIT**

- Duodenal atresia
- Annular pancreas
- Tracheoesophageal fistula
- Hirschsprung disease
- Imperforate anus

➤ **CNS**

- Hypotonia
- Mental retardation
- Developmental delay
- Seizures
- Autism spectrum disorders
- Behavioral disorders (disruptive
- Depression
- Alzheimer disease

- ☐ At least 8 criteria should be full filled to diagnose Down syndrome
- ☐ If criteria are less than 8 = mosaic down syndrome

Complications

- Hearing and vision disorder
- Congenital heart disease
- Failure to thrive
- Mental retardation
- Recurrent upper respiratory tract infections
- Hypothyroidism
- Acute lymphoblastic leukemia
- Megakaryoblastic form of acute myeloid leukemia, ...
- Risk of solid tumors are reduced

Screening

❖ First trimester

- ☐ Fetal nuchal translucency (NT) thickness alone or in conjunction with
- ☐ Maternal beta hCG and
- ☐ Pregnancy associated plasma protein A (PAPP-A)
 - NT alone can detect $\leq 70\%$ of down syndrome pregnancies
 - NT + beta hCG + PAPP-A = detects up to 87%

❖ Second trimester

- ☐ Beta hCG = ↑
 - ☐ Unconjugated estriol = ↑
 - ☐ Inhibin = ↑
 - ☐ Alpha fetoprotein = ↓
- } also known as ***quad screen***

- Can detect up to 80% of down syndrome pregnancies
 - If both the 1st & 2nd trimester screens are combined, detects up to 97%
- ▶ Known as ***integrated test***

❖ Free cell fetal DNA test

- ✓ Blood sample is taken from the mother by venipuncture
- ✓ Can be done starting from 10th week of gestational age
- ✓ Detection rate is 96-100%

Investigations

- CBC
- Otoscopy
- Ophthalmoscopy
- CXR
- ECG
- ECHO
- Abdominal ultrasound, ...

Differential diagnosis

- Patau syndrome
- Edward syndrome
- Congenital hypothyroidism

BURN

Defn

A burn is a traumatic injury to the skin or other organic tissue primarily caused by thermal or other acute exposure.

Types of burns

1. Thermal
 - a. Scaled burn – the most common type in children
 - b. Flame burn – the most common type in adults
2. Electrical
3. Chemical – acid, alkali
4. Inhalational
5. Radiation–e.g. Sunburn

Classifications

- Classified according to the depth of tissue injury
 - ☐ **1st degree** (superficial or epidermal)
 - ☐ **2nd degree** (partial thickness)
 - ☐ **3rd degree** (full thickness)

First degree burn

- Involves only the epidermis
- Characterized by
 - ✓ Swelling , erythema, pain
 - ✓ Can blench or bleed
 - ✓ No blister
 - ✓ Minimal or no edema
 - ✓ Pain resolves within 48-72hrs
 - ✓ Healed within 2-5 days with no scaring

Second degree burn

- Involves the entire epidermis & variable portion of dermal layer
- Superficial vs deep
- Characterized by
 - ✓ Blisters, bullae, serous fluid
 - ✓ Underlying tissue is mottled pink and white, with fair capillary refill
 - ✓ Can bleed
 - ✓ Edematous
 - ✓ Superficial
 - Is extremely painful,
 - ✓ Due to large number of remaining viable nerve endings is exposed
 - Healed in 7-14 days as the epithelium regenerates in the absence of infection
 - ✓ Deep

- Pain is less than superficial burns
 - ✓ Due to fewer nerve endings remain viable
- Healed in 21-35 days spontaneously with no infection
- If infected, it would converted to 3rd degree burn

Third degree burn

- Involves the entire epidermis & dermis
- May include fat, subcutaneous tissue, muscle & bone
- Leaving no residual epidermal cells to repopulate the damaged area
- The wound cannot epithelialize
 - ✓ can heal only by wound contraction or skin grafting
- Characterized by
 - ✓ Absence of painful sensation and capillary filling
 - Due to loss of nerve and capillary elements
 - ✓ No blenching or bleeding
 - ✓ White, yellow, brown leathery appearance

Estimation of Total Body Surface Area (TBSA)

1. Rule of nine– for children > 14 years old

- Each leg represents = 18 % of TBSA
- Each arm represents = 9 %
- Anterior & posterior trunk, each represents = 18 %
- The head represents = 9 %
- Perineum represents = 1 %

2. Rule of palm—for small burn (BSA < 10%)

- Each of the patient palm **excluding** fingers represent = 1 % of TBSA

3. Lund & Brower chart (age to body ratio)– for children < 14 years old

Body parts	Newborn	3 year	6 year	12+ year
Head(%)	18	15	12	6
Trunk (%)	40	40	40	38
Arms (%)	16	16	16	18
Legs(%)	26	29	32	38

NB.

- ✓ Subtract 1% from head for each year above one year of age
- ✓ Add ½% to each leg for each year over one year of age

Indications for admission

- Burns affecting >10% BSA
- Burns >10-20% of BSA in adolescents/adults
- 3rd degree burn
- High tension electrical burn or lightening
- Inhalational injury
- Chemical burn
- Burns to the face, hand ,feet, perineum, genitalia or major joints
- Suspected child abuse
- Inadequate home or social environment
- Associated injuries (fracture)

- Pregnancy

Diagnostic studies

- ✓ CBC
- ✓ Electrolytes
- ✓ BUN
- ✓ Creatinine
- ✓ U/A – may detect myoglobin showing muscle injury
- ✓ Carbon mono oxide level

Management principles

Acute treatments of burns

- First aid, including washing of wounds and removal of devitalized tissue
- Fluid resuscitation
- Provision of energy requirement
- Control of pain
- Control of bacterial wound flora = topical antibiotics
- Use of dressing to close the wound
- Reconstruction and rehabilitation

First aid measures

- Extinguish flames by roll, but don't run
- Check for ABC
- Cover the child with blanket, coat or carpet

- Remove clothes, rings , bracelets
- Brush off any remaining chemical
- Cover the burned area with clean & dry sheeting
- Apply cold (not iced) wet compresses to small injuries
- Administer analgesic medication

Fluid resuscitation

- ***Parkland formula***
 - ✓ 4ml lactated ringer/kg/%BSA burned
 - ✓ Half of the fluid is in the 1st 8hours
 - ✓ The remaining ½ is given over the next 16 hours
 - ✓ Used to replace fluid deficit
- ***Maintenance fluid***
 - ✓ *See at shock management*
 - ✓ Add glucose to maintenance fluid for children < 5years old

Complications

Acute complications

- Infection
- Renal failure
- ARDS
- Dysrhythmia
- Loss of consciousness
- Motor paralysis etc.

Long term disability

- ▶ Skin and soft tissue
 - Contracture
 - Alopecia
 - Hypertrophic scar
- ▶ Orthopedic disabilities
 - Amputations
 - Osteoporosis
 - Heterotopic ossification
- ▶ Psychiatric & neurologic disabilities
 - Sleep disorder
 - Depression
 - Post-traumatic stress syndrome

Ddx of burn

- TEN
- Steven Johnson syndrome

SHOCK

Defn

- Shock is a physiologic state characterized by a significant systemic reduction tissue perfusion resulting in decrease tissue oxygen delivery
- Shock may be present with normal or decreased blood pressure
- Physiologic parameters that the body can manipulate to compensate for compromised perfusion are: $COP = SV \times HR$ (SV = preload, cardiac contractility & after load) and systemic vascular resistance.

Phases of shock

1. Compensated

- The body's haemostatic mechanism rapidly compensate for decreased perfusion
- Characterized by
 - ✓ Normal blood pressure and cardiac output

- ✓ Adequate tissue perfusion
- ✓ Vital organ functions are maintained
- ✓ Signs of peripheral vasoconstriction (cool skin, decreased pulse, oliguria)
- Tachycardia, with or without tachypnea, may be the first or only sign of early compensated shock

2. Decompensated

- Compensatory mechanisms are overwhelmed
- Signs & symptoms of organ dysfunction
- The child's condition usually deteriorates rapidly
- Characterized by
 - ✓ Hypotension
 - ✓ Low cardiac output and
 - ✓ Inadequate tissue perfusion

3. Irreversible

- Multiorgan system dysfunction with end organ injury
- Characterize by cell death and is refractory to medical treatment

Classifications

1. Hypovolemic shock

- The most common cause of shock in children worldwide
- Caused by any condition that results in decreased circulating blood volume, such as hemorrhage or dehydration (e.g., from AGE)

- The amount of volume loss determines the success of compensatory mechanisms, such as endogenous catecholamines, in maintaining blood pressure and cardiac output
- Volume losses greater than 25% result in decompensated shock
- Potential etiologies
 - ✓ Blood loss → hemorrhage
 - ✓ Plasma loss → burns, nephrotic syndrome
 - ✓ Water/electrolyte loss → diarrhea, vomiting
- Clinical manifestations
 - ✓ Often manifests initially as orthostatic hypotension
 - ✓ Dry mucous membranes, dry axillae,
 - ✓ Poor skin turgor
 - ✓ Decreased urine output
 - ✓ Normal or slightly cool extremities
 - ✓ Peripheral or even femoral pulse may be N, ↓ or absent

2. Distributive shock

- Results from decreased SVR with abnormal distribution of blood flow within the microcirculation and inadequate tissue perfusion
- Typically caused by anaphylactic or neurogenic shock, or as a result of medications or toxins
- Anaphylactic shock is characterized by
 - ✓ Acute angioedema of the upper airway
 - ✓ Bronchospasm
 - ✓ Pulmonary edema
 - ✓ Urticaria, and

- ✓ Hypotension because of extravasation of intravascular fluid from permeable capillaries
- **Neurogenic shock**
 - ✓ Characterized by a total loss of distal sympathetic cardiovascular tone with hypotension resulting from pooling of blood within the vascular bed
 - ✓ Typically secondary to spinal cord transection or injury

3. Cardiogenic shock

- Occurs when cardiac output is limited because of primary cardiac dysfunction
- Potential etiologies
 - ✓ Dysrhythmias (e.g., supraventricular tachycardia)
 - ✓ CHD (e.g., any lesion that impairs LV outflow)
 - ✓ Cardiac dysfunction after cardiac surgery
 - ✓ Cardiomyopathies
 - ✓ Ventricular fibrillation
- Clinical features
 - ✓ Because of decreased cardiac output and compensatory peripheral vasoconstriction, the presenting signs of cardiogenic shock are
 - Tachycardia, tachypnea
 - Cool extremities
 - Delayed capillary filling time
 - Poor peripheral and/or central pulses
 - Declining mental status, and
 - Decreased urine output

- Signs of CHF

4. Obstructive shock

- Stems from any lesion that creates a mechanical barrier that impedes adequate cardiac output
- Potential etiologies
 - ✓ Pericardial tamponade
 - ✓ Tension pneumothorax
 - ✓ Pulmonary embolism
 - ✓ Anterior mediastinal masses
 - ✓ Critical coarctation of aorta
- Often manifests as inadequate cardiac output due to a physical restriction of forward blood flow
- The acute presentation may quickly progress to cardiac arrest

5. Septic shock

- Occurs secondary to an inflammatory response to invading microorganisms and their toxins and results in abnormal blood distribution
- Septic process involves more complex interaction of distributive, hypovolumic and cardiogenic shock
- There are two clinical stages:
 - ❑ Hyperdynamic stage; characterized by
 - ✓ Normal or high cardiac output with bounding pulses
 - ✓ Warm extremities, and
 - ✓ A wide pulse pressure
 - ❑ Decompensated stage

- ✓ Follows the hyperdynamic stage if aggressive treatment has not been initiated
- ✓ Characterized by
 - Impaired mental status
 - Cool extremities, and
 - Diminished pulses

Approach to a patient

- ABCD of life
- Asses the circulatory problem
 - ✓ Weather hands are warm or cold
 - ✓ Capillary refill
 - ✓ Pulse
- Asses the child weather malnourished or not

Diagnosis of shock

- Recognition of shock may be difficult because of the presence of compensatory mechanisms that prevent hypotension until 25% of intravascular volume is lost
- Therefore, the index of suspicion for shock must be high

I. **Historic features** that may suggest the presence of shock include:

- Severe vomiting and diarrhea

- Trauma with hemorrhage
- Febrile illness, especially in an immunocompromised patient
- Symptoms of CHF
- Exposure to a known allergic antigen
- Spinal cord injury

II. Physical examination

- Blood pressure may be normal in the initial stages of hypovolemic and septic shock
- Tachycardia almost always accompanies shock and occurs before blood pressure changes in children
- Tachypnea may be present as a compensatory mechanism for severe metabolic acidosis
- Mental status changes may indicate poor cerebral perfusion
- Capillary refill may be prolonged with cool and mottled extremities
- Peripheral pulses may be bounding in early septic shock
- Cold extremities

III. Laboratory studies should include

- RBS → to rule out hypoglycemia
- CBC → to assess for blood loss and infection
- Electrolytes → to assess for metabolic acidosis and electrolyte abnormalities
- BUN and creatinine → to evaluate renal function and perfusion
- Coagulation factors → to evaluate for DIC, which may accompany shock

- Toxicology screens → to evaluate for a poisoning, which could cause shock

Management principles

- Aggressive treatment within a few hours
- Initiate treatment immediately before cause of shock is identified
- Assess response to each intervention and cause of shock while treating the child
- Specific therapy must be initiated as soon as cause of shock is identified

General measures

- If the child has any bleeding; stop by applying pressure
- Give oxygen
- Make sure the child is warm- cover with blanket
- Select appropriate site for administration of fluid
- Draw blood for emergency laboratory

Shock with no SAM

- Give 20ml/kg of IV fluid (RL or NS) as rapidly as possible
- If no response → Give another 20 ml/kg as quickly as possible
- Reassess again if no response → Give another 20ml/kg
- If still no improvement
 - ✓ Give 20ml/kg of **blood** over 30 minutes unless there is profuse watery diarrhea

- ✓ In this case → repeat ringer lactate

Shock with SAM

❑ Four key steps to manage shock

1. Give oxygen(for infant 0.5-1L/min and older children 1-2L/min)
2. Give 10% glucose 5ml/kg IV
3. Keep the child warm
4. Give IV fluid as follows
 - Check starting PR and RR
 - Infuse 15ml/kg over 1hour
 - ✓ RL with 5% glucose OR
 - ✓ 0.9% NS with 5% glucose
 - If either of these are used, add sterile potassium chloride (20 Mmol/l) if possible

Follow up

- Observe the child and check PR and RR every 10 min
- If PR increase by 25 and RR increase by 5 breath per min; stop IV rehydration and assume septic or cardiogenic shock
- If PR & RR are slower after 1hour; the child is improving
 - ✓ Repeat the same amount of IV fluid for another hour
 - ✓ Continue to check PR & RR every 10 minute
- After 2hrs of IV fluid switch to oral or NG rehydration with ReSoMal
 - ✓ Give 5-10ml/kg ReSoMal in alternate hours with F-75 for 10hrs or till fully rehydrated

Maintenance fluid

Method 1—maintenance fluid per 24 hour period

- ☐ Weight $\leq$ 10 kg
 - 100 ml/kg
- ☐ Weight between 10 and 20 kg
 - 1000 ml for 1st 10 kg
 - Plus 50 ml/kg for any increment of weight over 10 kg
- ☐ Weight between 20 and 80 kg
 - 1500 ml for 1st 20 kg
 - Plus 20 ml/kg for any increment of weight over 20 kg
 - Maximum = 2400ml/day

Method 2 – maintenance fluid per hourly basis

- ☐ Weight $\leq$ 10 kg
 - 4 ml/kg/hr
- ☐ Weight between 10 and 20 kg
 - 40 ml for the 1st 10 kg
 - Plus 2 ml/kg/hr for any increment of weight over 10 kg
- ☐ Weight between 20 and 80 kg
 - 60 ml for the 1st 20 kg
 - Plus 1 ml/kg/hr for any increment of weight over 20 kg
 - To a maximum of 100 ml/hr
 - Up to a maximum of 100ml/hr x 24 =2400 ml/day

Management of septic shock

- Correct hypoglycemia and hypocalcemia
- Broad spectrum antibiotics
 - ✓ The choice of antibiotics depends on age of the patient, immune status and comorbid conditions

General principles for antimicrobial coverage for sepsis

- All children with septic shock should receive coverage for MRSA
- Coverage for enteric organisms should be added whenever clinical features are suggestive
- Treatment for pseudomonas should be included for children who are immunosuppressed
- *L. monocytogenes* and HSV are important pathogens in infants ≤ 28 days

Septic shock in SAM

- ❑ Diagnosis of septic shock in SAM
 - Fast weak pulse with
 - Cool peripheries
 - Disturbance of consciousness
 - Absence of signs of HF
 - Failure to improve after 1hr of IV fluid for mgt of shock
- ❑ Management of septic shock in SAM
 - All patients with septic shock
 1. Give broad spectrum antibiotics

2. Keep warm to prevent hypothermia
3. Prevent hypoglycemia
4. Little physical disturbance – no washing, excess examination
5. Never transfer to other facility
 - ✓ Stress of transportation leads to dramatic deterioration
6. Blood transfusion
 - When blood is available
 - Stop all oral intakes and IV fluids
 - Give diuretics to make room for blood then transfuse
 - ✓ 10ml/kg over 3hrs if SAM
 - If HF; give packed cells instead of whole blood

Key words

SIRS

- ❑ Two out of four criteria, one of which must be abnormal temperature or abnormal leukocyte count
 1. Core temperature $>38.5^{\circ}\text{C}$ or $<36^{\circ}\text{C}$
 2. Mean heart rate >2 SD above normal for age
 3. Respiratory rate >2 SD above normal for age
 4. Leukocyte count elevated or depressed for age (not secondary to chemotherapy) or $>10\%$ immature neutrophils

Sepsis= SIRS plus a suspected or proven infection

Severe sepsis = Sepsis plus one of the following

- ❑ Cardiovascular organ dysfunction, defined as:
 - Despite >40 ml/kg of isotonic intravenous fluid in 1 hour
 - Hypotension < 5th percentile for age or systolic blood pressure <2 SD below normal for age OR
 - Need for vasoactive drug to maintain blood pressure OR
 - 2 of the following
 - ✓ Unexplained metabolic acidosis: base deficit > 5 mEq/
 - ✓ Increased arterial lactate: >2 times upper limit of normal
 - ✓ Oliguria: urine output < 0.5 ml/kg/hr
 - ✓ Prolonged capillary refill: >5 sec
 - ✓ Core to peripheral temperature gap >3 °C
- ❑ Acute respiratory distress syndrome (ARDS) as defined by the presence of a Pao₂/Fio₂ ratio ≤300 mm Hg, bilateral infiltrates on chest radiograph, and no evidence of left heart failure;

OR
- ❑ Sepsis plus 2 or more organ dysfunctions (respiratory, renal, neurologic, hematologic, or hepatic)

Septic shock = Sepsis plus cardiovascular organ dysfunction as defined above

Multiple organ dysfunction syndromes (MODS)

- ❑ Presence of altered organ function such that homeostasis cannot be maintained without medical intervention

POISONING

Introduction

- It is intentional or unintentional exposure to toxic agents
- Poisoning can occur at two peak ages
 - ❑ Less than 6 years
 - It is unintentional or accidental
 - Occur because young children put everything to their mouth
 - Young children are hyperactive, they want to explore their environment
 - They do not know what is safe or dangerous
 - ❑ Adolescent (13 – 19 years old)
 - It is intentional (suicidal, abuse, or misuse)
 - Results more severe toxicity
 - Can be because of
 - ✓ Challenging issue in their life
 - ✓ Family problems – death of family members
 - ✓ Psychiatric illness
 - Females are higher risk

Roots of exposure

- Ingestion – the most common
- Dermal
- Inhalational
- Ophthalmic
- Parenteral

Common substances ingested

☐ Drugs

- Analgesics
- Topical preparations
- Vitamins
- Minerals

☐ Non drug substance

- Cosmetics
- Personal care items
- Cleaning solutions
- Plants
- Foreign bodies

Approach to a poisoned patient

- The initial approach is like any acutely sick child
- We follow ABCD of life
 - ❑ **Airway**
 - Remove excess secretion
 - Look for any inhalational injury that worsen with time
 - Check signs of burn around mouth
 - Consider intubation if severe airway compromise
 - ❑ **Breathing**
 - Look for signs of respiratory distress
 - Give oxygen if indicated
 - ❑ **Circulation**
 - Check signs of shock and treat accordingly
 - ❑ **Coma**
 - Check whether a child is comatose or not
 - If comatose; determine RBS and give dextrose
 - ❑ **Dehydration**
 - Asses for signs of dehydration and treat accordingly

When do we suspect poisoning?

History

- Apparently healthy child presented with sudden onset of loss of consciousness or respiratory distress
- If any medication, caustic agent, pesticide, insecticide around the patient
- Name specific ingredients and concentration of the toxin if known
- If pills:- shape, color, quantity
- Where the child was found – at home, work place, laundry, bathroom, ...
- Timing of ingestion
- How much was ingested – count pills or measure the remaining volume of liquid
- If inhalational, ocular, dermal
 - ✓ Concentration of the agent
 - ✓ Length of contact time
- History of psychiatric illness
- The child's social environment

Physical examination

- **General appearance**
 - Level of consciousness – depressed
 - Acutely sick looking

➤ **Vital signs**

☐ **Pulse rate**

- Tachycardia – atropine, antidepressants, caffeine, ...
- Bradycardia – beta blockers, calcium channel blockers, ...

☐ **Respiratory rate**

- Tachypnea – amphetamine, carbon mono oxide, caffeine, ...
- Bradypnea – opioids, alcohol, barbiturates, ...

☐ **Blood pressure**

- Hypertension – amphetamine, anticholinergics, ...
- Hypotension – beta blocker, calcium channel blocker, opioids

☐ **Temperature**

- Hyperthermia – selective serotonin inhibitors, lithium
- Hypothermia – morphine, heroin, ...

➤ **HEENT**

☐ **Eye**

- Miosis – opioid, organophosphate
- Mydriasis – atropine, cocaine, amphetamine
- Lacrimation – organophosphate

☐ **Mouth and throat**

- Salivation – organophosphate, ketamine
- Oral burn – corrosives

➤ **Respiratory system**

- Signs of respiratory distress – opioids , alcohol, barbiturates

➤ **Integumentary**

- Diaphoresis – organophosphate, salicylates
- Alopecia – thallium, arsenic
- Erythema – CO, elemental mercury
- Cyanosis – amidarone

➤ **CNS**

- ❑ Ataxia – alcohol, barbiturates, CO, anticonvulsants
- ❑ Coma – opioids, barbiturates
- ❑ Seizure – organophosphate, antidepressants,
- ❑ Delirium/psychosis – anticholinergics, lithium, steroids
- ❑ Peripheral neuropathy – organophosphate, lead, arsenic, mercury

Investigations

- RBS
- CBC
- Serum electrolytes
- LFT
- RFT
- ECG
- Chest X- ray

Management

❑ General management principle

1. Decontamination

- Ipecac induced vomiting
- Cathartics
- Gastric lavage
- Activated charcoal
- Whole bowel irrigation

2. Enhanced elimination

- Multiple dose of activated charcoal
- Urinary alkalization
- Dialysis

3. Giving antidote

4. Supportive

1. Decontamination

- The goal is to prevent absorption of toxic substance
- The specific methods depend on the properties of the toxin and route of exposure

- Regardless of the methods used the efficacy decrease with increasing the time since exposure especially after one hour of ingestion
- GI decontamination strategies are effective in the first hour after ingestion
- GI absorption may be delayed after
 - ✓ Ingestion of agents that slow GI motility
 - ✓ Massive pill ingestion sustained release preparation
 - ✓ Ingestion of agents that can form pharmacologic bezoars
- So GI decontamination after one hour of ingestion may be considered in patients ingesting toxins with the above properties

Methods of GI decontamination

A. Syrup of Ipecac

- Contains two emetic alkaloids that work both in CNS and locally in GIT to produce vomiting
- Currently it is not used because of multiple adverse effect
- The only indications in out of hospital are
 - ✓ If there is delay of a child to reach emergency medical facility for greater than an hour after toxic ingestion
 - ✓ If there is substantial risk of serious toxicity to the patient
 - ✓ If there is no alternative therapy to decrease GI absorption
 - ✓ If there are no contra indications for it
 - ✓ If the use of it do not adversely affect definitive therapy that may be provided at hospital
- All of these conditions must be fulfilled to use it

B. Gastric lavage

- It involves placing a tube into stomach to aspirate contents followed by flushing with a liquid usually normal saline
- Useful if it is given with 1 hour of ingestion of poison
- However it is time consuming
 - ✓ Can induce bradycardia
 - ✓ Delay administration of more definitive treatment
 - ✓ Under best circumstance only removes fraction of gastric contents
- These in most clinically scenarios are no longer recommended

C. Single dose activated charcoal

- Charcoal is activated through heating to extreme temperature creating an extreme network of pores that provide a very large absorptive surface area
- Many substances are adsorbed on to its surface; these prevent absorption from GI tract
- Effective when given with 1 hour of ingestion
- Substance poorly adsorbed by activated charcoal: - alcohols, caustics, cyanides, heavy metals, hydrocarbons, iron, lithium
- It should be avoided after ingestion of caustic substances
- Before administration air way should be intact
- Dose
 - ✓ Children = 1gm/kg
 - ✓ Adolescent = 50-100 gm

- Amount of water 8-10 times amount of charcoal
- If possible, give the whole amount at once but can be given divided doses if the child cannot tolerate it
- Can be mixed with caffeine free diet, coca or juice

D. Whole bowel irrigation

- It involves installing large volume of polyethylene glycol electrolyte solution to cleans entire GIT
- Dose
 - ✓ Children = 35ml/kg/hr
 - ✓ Adolescents = 1-2 L/hr
- It may have some success after
 - ✓ Ingestion of slowly absorbed substance
 - ✓ Substance not well absorbed by charcoal
 - ✓ Trans dermal patches and drug packets
- Can be combined with activated charcoal
- Administered through NG tube
- Has advantage because of having balancing electrolytes
- Asses airway and examine abdomen before initiating it
- Complications – vomiting, abdominal pain, bezoars formation

Skin decontamination

- Remove all closing and personal effect
- Thoroughly clean all exposed areas with large amount of water
- Use soap and water for oily substances
- Take care to protect your self

Eye decontamination

- Rinse eye for 10-15 minutes with clean running water or normal saline
- Evert the eyelid and insure that all surface are rinsed
- Examine eye for signs of corneal damage
- If conjunctival or corneal damage – refer to ophthalmologist

Decontamination of inhaled poison

- Remove from the source
- Give oxygen supplementation if
 - ✓ Respiratory distress or cyanosis
 - ✓ Oxygen saturation $\leq 90\%$

2. Enhanced elimination

- ❑ Only useful for few toxins and it is lifesaving intervention

A. Multiple dose activated charcoal

- It enhances elimination via two proposed mechanism
 - ✓ Interruption of enterohepatic recirculation
 - ✓ GI dialysis
 - ▶ Uses intestinal mucosa as a dialysis membrane and pull toxin from the tissue into intra luminal space

- Dose = 0.5 gm/kg every 4-6 hours (for ≤ 24 hours)
 - ✓ Continue until there is significant clinical improvement
- Consider in drug poisonings like Phenobarbital, carbamazepine, phenytoin, digoxin, salicylates, theophiline
- Examine airway and abdomen before each dose

B. Urinary alkalization

- It enhances elimination some drugs that are weak acids by forming charged particles
- Accomplished with continuous infusion of sodium bicarbonate containing IV fluids with goal of urinary PH 7.5-8.0
- Serum PH should be closely monitored because serum PH > 7.55 is potentially dangerous for cellular function
- Most useful in managing salicylates and methotrexate toxicity

C. Dialysis

- Few drugs or toxins are removed by dialysis
- Toxins amenable for dialysis have the following properties
 - ✓ Low volume of distribution ($< 1L/kg$)
 - ✓ Low molecular weight
 - ✓ Low degree of protein binding
 - ✓ High degree of water solubility
 - ✓ Such toxins are – methanol and ethyl glycol
- Can also use for large symptomatic ingestion of salicylates, theophiline, bromide and lithium
- Correct severe electrolyte disturbance and acid base derangement

3. Antidote

- Available for few toxins
- Early and appropriate use is key element in managing poisoned child

Poison	Antidote	Route
<i>Acetaminophen</i>	<i>N-Acetylcysteine</i>	<i>PO or IV</i>
<i>Anticholinergics</i>	<i>Physostigmine</i>	<i>IV or IM</i>
<i>Benzodiazepine</i>	<i>Flumazenil</i>	<i>IV</i>
<i>Beta blockers</i>	<i>Glucagon</i>	<i>IV</i>
<i>Calcium channel blockers</i>	<i>Insulin or calcium salt</i>	<i>IV</i>
<i>Carbon monoxide</i>	<i>Oxygen</i>	<i>Inhalational</i>
<i>Iron</i>	<i>Deferoxamine</i>	<i>IV</i>
<i>Isoniazid</i>	<i>Pyridoxine</i>	<i>IV</i>
<i>Organophosphate</i>	<i>Atropine or pralidoxine</i>	<i>IV or IM</i>
<i>Opioids</i>	<i>Naloxone</i>	<i>IV</i>
<i>Salicylates</i>	<i>Sodium bicarbonate</i>	<i>IV</i>
<i>Sulfonylurea</i>	<i>Octreotide</i>	<i>IV or subcutaneous</i>
<i>Tricyclic antidepressant</i>	<i>Sodium bicarbonate</i>	<i>IV</i>

Taken from Nelson text book of pediatrics 20th edition

4. Supportive

- The goal is to support the vital function of the patient until the patient can eliminate the toxin from the body
- In any poisoned patient excellent supportive care and frequent clinical assessment are the key to effective management and improved outcome

- These include
 - ✓ Airway support
 - ✓ Ventilator management
 - ✓ Appropriate and timely management of seizure, dysrhythmia, conduction delay, electrolyte and metabolic derangement
 - ✓ Blood pressure support

SECTION – TWO

COMMON

PEDIATRIC

PROCEDURES

PERIPHERAL IV INSERTION

Indications

- Administration of fluids and electrolytes
- Administration of intravenous medications
- Administration of blood and blood products
- Blood sampling

Contraindications

- Absolute
 - Infected site
 - Burned site
 - Injured site
- Relative
 - Paralyzed extremity
 - Massive edematous extremity
 - Distal to an injured organ
 - ✓ Do not use lower extremities when treating abdominal injuries

Equipments

- Gloves
- Tourniquet or rubber band
- Tape and occlusive transparent dressing
- Alcohol wipes
- Povidone or chlorhexidine
- Syringe filled with injectable saline
- Gauze pads
- IV device: catheter or butterfly of appropriate size to fit the patient & the task
- Topical anesthetic cream
- Ultrasound guiding equipment (if available and if trained in its use)

Common sites

- Dorsum of the hand
- Radial side of the hand (cephalic vein)
- Antecubital fossa (cephalic vein – common site for age >2 months)
- Saphenous vein at the medial aspect of the ankle
- Dorsum of the feet (dorsal Vein plexus)
- External jugular vein
- Scalp veins
 - ✓ The frontal superficial, temporal posterior, auricular, supra-orbital and posterior facial veins can be used.
 - ✓ Often for ages <2 years but best in young infants

Procedures

- ✓ *Wash your hands thoroughly*
- ✓ *Assemble the equipment at the bedside*
- ✓ *Observe universal precautions*
- ✓ *Follow the patient's specific isolation instructions*

Upper and Lower Extremities

- ✓ *Apply tourniquet*
- ✓ *Identify the blood vessel by palpation, visualization, trans illumination, or ultrasound*
- ✓ *Release the tourniquet, cleanse the site*
- ✓ *Inspect the integrity of the catheter/stylet assembly*
- ✓ *Flush the catheter and the connecting tube with saline (omit this step if you intend to draw blood through this catheter)*
- ✓ *Reapply the tourniquet*
- ✓ *Use your nondominant hand to apply traction on the skin linearly or circumferentially in order to stabilize the vein*
- ✓ *Enter the skin at a 30- to 45-degree angle proximal to or alongside the vein*
- ✓ *Reduce the angle as you advance the catheter and enter the vein*
- ✓ *Watch for blood flashback in the hub of the catheter*
- ✓ *Stabilize the catheter with the thumb and middle finger of your dominant hand and advance the catheter over the stylet using the tip of your index finger*
- ✓ *Remove the stylet*
- ✓ *Do not reinsert the stylet once it has been removed; it may damage the catheter*
- ✓ *Release the tourniquet*
- ✓ *Connect the extension tubing and saline-filled syringe to the catheter*

- ✓ Gently flush the catheter; observe for swelling, mottling, or color changes in the extremity
- ✓ Secure the IV with occlusive transparent dressing and tape
- ✓ Make a small loop in the IV tubing and tape it across
- ✓ Attach the line to an IV infusion assembly and turn the pump on
- ✓ Dispose of all sharp instruments in the proper secure container

External Jugular Vein

- ✓ Bundle infant or child
- ✓ Have an assistant position the patient in the Trendelenburg position with the head toward you
- ✓ Turn the patient's head away from the jugular vein you intend to use
- ✓ Elevate the patient's shoulders and neck (15 - 30°)
- ✓ Apply traction with your nondominant hand to the skin over the jugular vein
- ✓ Nick the skin with a large bore needle at a shallow angle below or alongside the vein in order to facilitate the catheter's entry
- ✓ Insert the catheter through the puncture and advance subcutaneously a few millimeters before entering the vein
- ✓ Synchronize your entry with the child's breathing.
- ✓ Enter the vein during exhalation in a spontaneously breathing patient to avoid air embolus
- ✓ Enter the vein during a positive pressure breath if the patient is on positive pressure ventilation
- ✓ Watch for blood flashback in the hub
- ✓ Stabilize the catheter assembly and advance the catheter over the stylet
- ✓ Withdraw the stylet and occlude the catheter with your gloved thumb to avoid air embolus
- ✓ Connect the tubing and saline-filled syringe, draw back and flush

- ✓ *Secure the line with occlusive transparent dressing and tape*
- ✓ *Reposition the child and retest the line*
- ✓ *The jugular vein has a number of valves that can obstruct the catheter when the neck position is altered*
- ✓ *Traction may be needed or the catheter may need to be withdrawn slightly to ensure proper function*

Scalp IV

- ✓ *Find a suitable scalp veins*
- ✓ *Shave the area and clean with antiseptic*
- ✓ *Palpate for pulse in order to avoid inadvertent arterial cannulation*
- ✓ *Apply traction with your nondominant hand*
- ✓ *Fill the butterfly set tubing with saline*
- ✓ *Disconnect the syringe and leave the end of the tubing open.*
- ✓ *Introduce the butterfly needle as described above.*
- ✓ *Blood flowing back slowly through the tubing indicates that the needle is in the vein.*
- ✓ *Secure the IV with transparent occlusive dressing and tape*
- ✓ *Attach to an infusion assembly*

Complications

- Infection
- Hematoma
- Extravasation
- Compartment syndrome (6p's)
 - ✓ Pain
 - ✓ Pallor
 - ✓ Pulselessness
 - ✓ Paresthesia
 - ✓ Paralysis and
 - ✓ Pershingly cool
- Severe vasoconstriction
 - ✓ If vasoactive medications are infused through a peripheral IV & extravasate
- Venous thrombosis
- Embolization of air or catheter fragment
- Local ischemia

Monitoring

Peripheral IV in Extremity

- Compare extremity's color and temperature; watch for congestion & swelling
- Watch for signs of occlusion
- Palpate pulses
- Ensure skin integrity
- Check tightness of dressing
- Inspect IV tubing for blood or precipitation
- Monitor IV pump for increased resistance that may indicate clotting

- Watch for pump malfunction

Jugular Vein

- Monitor for swelling
- Assess for signs of occlusion

Scalp IV

- Monitor for swelling and blanching
- Watch for signs of occlusion
- Assess skin integrity

LUMBAR PUNCTURE

subarachnoid space

Indications

➤ Diagnostic

- Suspected CNS infections
- Suspected subarachnoid haemorrhage
- Metabolic studies e.g. → Aminoacidopathies
→ Neurotransmitter disorders
- Aminoacidopathies
- Neurotransmitter disorders
 - ✓ Undiagnosed movement disorders
 - ✓ Undiagnosed infantile or pediatric epilepsy
- Demyelinating disorders – *multiple sclerosis*
- Contrast media instillation

➤ Therapeutic This is called intrathecal chemotherapy

- Instillation of chemotherapy or spinal anaesthesia
- Removal of CSF in the treatment of intracranial hypertension

Contraindications

➤ Absolute

- ✓ Raised intracranial pressure (with a pressure gradient across the CNS compartments, papilledema)
- ✓ Focal neurologic deficit

- ✓ Local developmental abnormality, e.g., myelomeningocele
- **Relative**
 - ✓ Cardiopulmonary instability
 - ✓ Bleeding diathesis (Platelet count < 50,000)
 - ✓ International normalized ratio (INR) >1.4
 - ✓ Local skin infection
- Children with the following conditions should have CT before LP
 - ✓ Altered mental status
 - ✓ Focal neurologic signs
 - ✓ Recent seizure
 - ✓ Risk of brain abscess (immunocompromised, CHD with right - left shunt)
 - ✓ Papilledema

Equipments

- Sterile gloves
- Sterile drapes
- Povidine – iodine solution
- Sterile sponges
- Manometer (typically used in patients >2 years old)
- Lidocaine 1% without epinephrine and topical anaesthetic
- Syringe
- 22 gauge spinal needle: 0.5 inch for neonate, 1.5 inches <2 years old, 2.5 inches for 2-12 years old, 3.5 inches for > 12 years old

- Manometer (for children older than 2 yrs)
- Sterile collection tubes (sufficient number for studies)
 - ✓ Needs at least three tube – for biochemical, culture and cytology
- 3-way stopcock
- Flexible tubing

Patient preparation

- Sterile technique
- Povidone-iodine preparation
- Sterile drape with fenestration over mid lumbar spine
- Sedation, if needed
- Connect 3-way stopcock to flexible tubing and manometer at 90 degrees from each other
- Free end of tubing will connect to hub of needle

Positioning

- **Lateral recumbent**
 - ✓ Back arched in extreme lordosis
 - ✓ Spine should be as perfectly horizontal as possible
 - ✓ Sacral plane should be as vertical as possible
- **Sitting**
 - ✓ For children who have the potential for developing respiratory compromise because of hyper flexion of the neck in the lateral recumbent position
 - ✓ Does not permit accurate measurement of opening pressure

Procedures

- ✓ *Palpate for L4 spinous process using iliac crests as landmarks*
- ✓ *Place lumbar puncture needle between interspaces L4-5*
- ✓ *Angle the needle tip approximately 15-30 degrees from perpendicular to plane of back in rostral direction, aiming toward umbilicus*
- ✓ *Needle remains fixed in horizontal plane to back*
- ✓ *Advance needle slowly until light resistance (a pop) is felt*
- ✓ *Remove stylet and check for CSF flow*
- ✓ *If no CSF flows, continue to advance the needle slowly*
- ✓ *If CSF flows, connect flexible tubing to hub of lumbar puncture needle*
- ✓ *Allow CSF to flow through tubing into manometer*
- ✓ *Hold base of manometer and stopcock at level of heart*
- ✓ *Straighten the patient's back and legs*
- ✓ *When CSF stops advancing along manometer, measure opening pressure at meniscus*
- ✓ *Collect CSF for studies*
- ✓ *Measure closing pressure, if needed.*

Complications

- Post puncture headache
- Infection
- Cerebral herniation
- Spinal Hematoma
- Epidermoid tumor
- Back pain

Follow up

- ✓ Sterile dressing
- ✓ Older adolescents should rest in bed for 1-3 hours

CSF Analysis

yellowish discoloration of CSF

- Look for appearance (crystal clear- normal, hemorrhagic, xanthochrom – raised protein, turbid, ...)& opening pressure
- Routine studies include protein, glucose, and cell count and differentials
- Studies are selected on clinical suspicion and indication; examples include the following:
 - ✓ Cultures – to identify the organism and drug sensitivity
 - ✓ Bacterial antigen studies
 - ✓ Herpes polymerase chain reaction
 - ✓ India ink stains
 - ✓ Tuberculosis studies
 - ✓ CSF neurotransmitters (movement disorders and epilepsy)
 - ✓ Amino acids (for aminoacidopathies, especially glutaricaciduria)
 - ✓ Lactate (mitochondrial and energy metabolism disorders)
 - ✓ Myelin basic protein (demyelinating diseases)
 - ✓ Oligoclonal bands (demyelinating diseases)
 - ✓ Cytology (neoplasm, usually requires larger volume)
- Opening pressure (normal is < 20 cm H₂O):
 - ✓ Direct measure of intrathecal pressure and indirect measure of intracranial pressure
 - ✓ Patient must be relaxed and back and legs are extended.
- Protein (normal values are based on age)

- ✓ Elevations suggest - inflammatory or degenerative process
- ✓ Bacterial meningitis is usually significantly elevated
- ✓ Newborns may have elevations of protein at baseline
- Glucose (normal is approximately two-thirds of serum)
 - ✓ Decrease in CSF glucose is highly suggestive of - active bacterial infection b/c metabolism rate is increased and bacteria uses for itself
 - ✓ Decrease in CSF glucose also seen in glucose transporter defect (epilepsy)
 - ✓ Normal or a slight decreament in viral infection , since viruses have no metabolic machinery – they cannot use it
- Leukocytes
 - ✓ Normal
 - Neonate: ≤ 30 cells with lymphocyte dominant
 - Infants: ≤ 5 cells with lymphocyte dominant
 - ✓ Elevations suggest inflammatory process (often infective)
 - ✓ Elevations of polymorphonuclear leukocytes suggest - acute bacterial infection or early viral infection
 - ✓ Elevations of lymphocytes suggest TB, viral, fungal or protozoal infection
- Erythrocytes (normal is 0): if elevated
 - ✓ Hemorrhage (not etiologically specific, acute or subacute)
 - ✓ Hemorrhagic infection (herpes)
 - ✓ Evaluate spun sample for hemochromatosis (subacute hemorrhage)

BONE MARROW ASPIRATION AND BIOPSY

Indications

- Pancytopenia
- Unexplained anemia, leucopenia, or thrombocytopenia
- Acute or chronic leukaemia (aspiration only)
- Fever of unknown origin
- Myelodysplasia
- Myeloproliferative disease
- Non-Hodgkin or Hodgkin lymphoma
- Childhood solid tumours (including sarcoma, Wilms tumour, neuroblastoma, germ cell tumour)
- Bone marrow failure (including acquired aplastic anaemia, Fanconi anaemia, Diamond- Blackfan syndrome)
- Storage disease
- Monitoring during chemotherapy or following stem cell transplantation (aspiration only)

Contraindications

Relative

- Congenital factor deficiency or acquired coagulation defect
- Anticoagulation with warfarin or heparin
- Severe thrombocytopenia
- Infection or prior radiation at sample site

Common sites

- Posterior superior iliac spine
- Anterior iliac crest in obese patients
- Anterior-medial surface of the tibia in infants < 3 months of age

Equipments

➤ Site Preparation

- ✓ 10% povidone-iodine
- ✓ Alcohol preparation pads or swabs
- ✓ Sterile gloves, gown, and drape
- ✓ Spinal and subcutaneous needles, 20 to 26 gauge
- ✓ 1% lidocaine hydrochloride, injection
- ✓ 8.4% sodium bicarbonate, injection, USP

➤ Marrow Aspiration and biopsy

- ✓ Sodium heparin, injection, 1000 USP units/ml, preservative free
- ✓ Bone marrow aspiration needles (15 & 18 gauge, adjustable lengths)
- ✓ Bone marrow biopsy needles (11 & 13 gauge, 4 or 2 inches in length)
- ✓ Sterile syringes, 10 to 20 ml
- ✓ Container with fixative for trephine biopsy specimen
- ✓ Vacutainers; one for sodium heparin and one for ethylenediaminetetraacetic acid (EDTA)
- ✓ Gauze sponges
- ✓ Bandages
- ✓ Slides – at least 3 slides are needed

Patient positioning

- If the **posterior iliac crest** is used, the patient is placed in the right or left **decubitus position**, with the hips flexed and the knees drawn up
- If the **anterior iliac crest** is used, the patient is placed in the **supine position** with the hips and knees flexed
- Occasionally, thin patients who do not receive general anesthesia may be placed in the prone position

Procedures

Bone marrow aspiration

- ✓ *Clean the site with povidone-iodine followed by alcohol swab*
- ✓ *Place sterile drape*
- ✓ *Inject buffered lidocaine intradermally with a subcutaneous needle to produce a small wheal*
- ✓ *Use a larger bore needle to push through the skin and subcutaneous tissue and inject 2-3 ml (maximum 3g/kg/dose along the periosteum*
- ✓ *Hold the bone aspirate needle horizontally using the index finger near the tip of the needle for control*
- ✓ *Advance the needle through the skin, subcutaneous tissue, and the surface of the cortical bone with steady pressure and a twisting motion*
- ✓ *An abrupt decrease in resistance occurs when the needle penetrates the cortex and enters the spongy marrow cavity*
- ✓ *Advance the needle 1 cm more before the stylet is removed*
- ✓ *Attach a 10-mL or 20-mL syringe to the end of the needle and pull the plunger back quickly to aspirate approximately 0.25 ml of bone marrow*

- ✓ *If an aspirate is not obtained, replace the stylet and advance or reposition the needle*
- ✓ *This first pull contains the marrow particles or spicules that should be used for preparing initial smears*
- ✓ *A heparinized, larger syringe (30 ml) may be used to obtain additional marrow for cytogenetic analysis, flow cytometry, and other studies.*

Bone marrow biopsy

- ➡ *The trephine biopsy is the preferred method to evaluate cellularity and detect bone marrow metastasis in lymphoma and many childhood solid tumors*
- ➡ *Biopsy specimen is obtained through the same incision site*
- ✓ *Hold the biopsy needle in the same manner as the aspiration needle but angle it to sample a different area from the aspiration*
- ✓ *Advance the needle with steady pressure to the periosteum and twist into the surface of the cortical bone*
- ✓ *Remove the obturator and push the needle through the cortex using a rotating, twisting motion until decreased resistance is met*
- ✓ *Advance the needle another 1-2 cm*
- ✓ *Reinsert the obturator until resistance is met to gauge the length of the specimen*
- ✓ *Rotate the needle 360 degrees vigorously several times while moving it back and forth vertically and horizontally to break the biopsy core off the surrounding bone*
- ✓ *Carefully remove the needle and insert a separate blunt obturator into the distal end of the needle to force the core out through the hub onto a glass slide*
- ✓ *Touch preparations of the core biopsy should be performed*
- ✓ *Specimen should be at least 1.5 to 2 cm in length for optimal processing*

- ✓ *If the specimen is inadequate or consists mostly of cartilage or cortical bone rather than core marrow, which appears dark red with a fine, white trabecular network, attempt additional biopsies*
- ✓ *The specimen should be placed in an appropriate fixative*
- ✓ *Apply direct pressure to the site for at least 5 minutes once the procedure is completed and the needle removed*
- ✓ *Place a pressure dressing.*

How do we know when we reach at the site?

- ✓ Pop sound
- ✓ Foamy blood

Complications

- Bleeding at any site, with or without development of a hematoma, is rare if adequate pressure is applied
- Bleeding risk has been reported to be increased in adults with osteoporosis or extensive bony involvement by disease, such as multiple myeloma
- Retroperitoneal hemorrhage, osteomyelitis and needle breakage have also been rarely described
- Infection (rare)

Follow up

- Multiple evaluations are often required to monitor efficacy and recovery following chemotherapy or stem cell transplantation

- Patients should lie on their backs for an additional 15–20 minutes for procedures performed on the posterior iliac crest
- Patients should be reminded that a dull ache may be felt for several days following the procedure
- Routine post-procedural monitoring should be performed when heavy sedation or general anesthesia is administered

Interpretations and monitoring

- ✓ Prepare the initial aspirate at the bedside immediately
- ✓ Various techniques for spreading the film have been described, including the traditional wedge technique, particle crush method, and cover slip preparation
- ✓ Slides are then dried, fixed in methanol and stained by the Wright-Giemsa technique
- ✓ Systematic analysis of the aspirate slides includes assessment of the adequacy of spicules, cellularity, megakaryocyte count as well as maturation and morphologic features of other cell lineages
- ✓ The biopsy specimen, once fixed, is decalcified
- ✓ Further processing includes hematoxylin and eosin (H&E), reticulin, and immunohistochemical staining when necessary
- ✓ Comprehensive analysis of the biopsy specimen should include evaluation of the specimen's adequacy, cellularity, and bone structure and detection of focal lesions or metastatic disease.

INTRAOSSEROUS INFUSION

Indications

- Emergent temporary vascular access during cardiopulmonary resuscitation or during the treatment of uncompensated shock when unable to insert an intravenous line for :
 - ▶ Volume resuscitation
 - ▶ Administration of blood and blood products
 - ▶ Administration of fluids and electrolytes
 - ▶ Administration of medications
 - ▶ Infusion of inotropes and pressors
 - ▶ Sampling of blood and bone marrow
- The method is safe if the needle is left in place no longer than 6-8 hours

Contraindications

Absolute

- Recently fractured bone
- Infected site or burn
- Osteogenesis imperfecta
- Osteopetrosis

Relative

- Osteoporosis or osteopenia
- Cystic bones

Equipments

- Intraosseous needle (18 or 20 gauge) or bone marrow aspiration needle
- Povidone, chlorhexidine, and alcohol wipes
- Gauze
- Tape
- T-connector
- Syringe (5ml syringes(2) filled with NS)
- Iv infusion equipments
- Sterile gloves

NB. All equipments should be latex free

Common sites

Infants and children

- Antero-medial surface of tibia 1-2 cm below tibial tuberosity
 - ✓ Direct the needle caudally to avoid the growth plate
- Distal tibia

Adolescents

- Antero-medial surface of tibia 2 cm below tibial tuberosity
- Distal femur 2 cm above the lateral condyle

- Distal tibia proximal to medial malleolus
- Proximal humerus approximately 2cm below acromion process
- Anterior superior iliac spine
- Posterior superior iliac spine
- Iliac crest

Patient preparation and positioning

- Choose the most appropriate site
- Inject local anaesthetics if the patient is conscious
- Support the site of insertion over a firm surface
- Hold the extremity above and below the insertion site
- Position the patient with the selected site closest to where you are standing

Procedures

- ✓ *Sanitize or wash your hands thoroughly and don gloves*
- ✓ *Cleanse the site with antiseptic solution*
- ✓ *Support the leg on a firm surface and have an assistant support the leg above and below the insertion site*
- ✓ *Ensure no hand is under the site*
- ✓ *Administer local anesthetic if the patient is conscious*
- ✓ *Align needle bevel and stylet*
- ✓ *Insert needle assembly through the skin and advance to bone cortex*
- ✓ *Hold the needle slightly angled (10 to 15 degrees) the bone, and direct it away from the nearest joint*
- ✓ *Use a twisting motion when advancing the needle*

- ✓ *Stop advancing the needle when you feel a sudden decrease in resistance*
- ✓ *Unscrew the cap, remove the stylet, and attempt to aspirate bone marrow (looks like blood)*
- ✓ *The unsupported needle should remain upright if properly placed*
- ✓ *Flush the line with normal saline*
- ✓ *Watch for increased resistance to the flush or swelling of the extremity*
- ✓ *If no marrow is aspirated but you think you are in the bone marrow, attempt to flush*
- ✓ *If you encounter resistance, advance needle assembly and reattempt aspiration*
- ✓ *Fluid should flow freely through the needle and the line should flush without resistance*
- ✓ *Use tape and gauze to secure the line*

Monitoring

- Watch for signs of compartment syndrome
 - ✓ Pain, pallor, pulselessness, paresthesia, paralysis and pershingly cool
- Look for swelling, redness, blanching, and leakage
- Assess for vasoconstriction
- Stop the intraosseous infusion as soon as venous access is available

Complications

- Extravasations of fluids or medications into subcutaneous tissue
- Compartment syndrome

- Infection : subcutaneous abscess, osteomyelitis, and bacteraemia
- Epiphysis injury and fracture
- Fat embolus

FEMORAL VENOUS CATHETERIZATION

Indications

- Any situation that requires central venous access or venous access that cannot be obtained peripherally
- An emergency resuscitation requiring administration of large amounts of fluids
- The need for central venous pressure monitoring
- Placement of a pulmonary artery catheter
- The need for frequent blood draws
- Infusion of hyperalimentation, concentrated solutions (i.e. KCl, dextrose concentrations greater than 12.5%, chemotherapeutic agents, hyperosmolar saline)
- Infusion of vasoactive substances (i.e. dopamine and norepinephrine) that can extravasate and cause soft-tissue necrosis
- The need for hemodialysis

Advantages of Catheter Placement at Femoral Site

- It does not interfere with procedures or monitoring involving the head, neck, or chest (such as cardiopulmonary resuscitation)

- Pressure can be applied easily in the event of femoral artery puncture or catheterization
- It leaves the patient's neck free of devices

Disadvantages of Catheter Placement at Femoral Site

- It is a relatively “dirty” area (though this can be managed with good sterile technique and dressing changes)
- Placement of a long line is required for central venous pressure monitoring
- It can be challenging to place a pulmonary artery catheter through a femoral venous catheter

Contraindications

Absolute

- Severe abdominal trauma (provided that adequate venous access can be obtained elsewhere)

Relative

- A patient with distorted anatomy or landmarks
- Risk factors for excessive bleeding, such as thrombocytopenia, coagulopathy, and anticoagulant or thrombolytic therapy
- Skin lesions (such as cellulitis, burns, abrasions, or dermatitis)
- Conditions that predispose the patient to sclerosis or thrombosis (such as vasculitis)
- Known thrombus of the femoral vein

Equipments

- The catheter
- An appropriate size guidewire (at least 2 times the length of the catheter)
- An appropriate size introducer needle
- A tissue dilator if the catheter is larger than 3F
- Two or three 3- to 5-ml syringes
- 1% lidocaine and a 26-gauge needle to inject the lidocaine
- Skin preparation solution (either 2% chlorhexidine-based preparation for patients older than 2 months or 10% povidone-iodine)
- Sterile drapes
- Scalpel blade
- Suture (i.e., 3.0 silk)
- Sterile gauze pads

Patient preparation and positioning

- Place the patient in the supine position
- Raise the hips slightly to flatten the inguinal area
- Position the patient with his or her legs extended or with the hips and knees slightly flexed in the “frog” position
- In the case of a child whose respiratory status is compromised in the supine position, it may be necessary to obtain definitive airway control (i.e., intubate the patient) prior to the procedure
- Children who can comfortably lie supine may require conscious sedation so that they remain still throughout the procedure

- To avoid aspiration during intubation or conscious sedation, the procedure should be delayed 6 hours after the ingestion of solid food and 4 hours after the ingestion of clear liquids, unless central access is needed emergently

Anatomy review

- The femoral artery and vein run in parallel with the artery lateral to the vein
- The inguinal ligament runs from the anterior superior iliac spine to the pubic tubercle
- Remember the mnemonic “NAVEL” (nerve, artery, vein, empty space, lymph), which describes the structures’ anatomic location from lateral to medial

NB.

- ✓ *Femoral artery, vein and femoral canal are found inside the femoral sheath, but the nerve found outside the sheath.*

Procedures

- ✓ *The operator should wear a cap and mask, be scrubbed, and use a sterile gown and gloves for this procedure*
- ✓ *Prepare the area using either 2% chlorhexidine-based preparation for patients older than 2 months of age or 10% povidone-iodine*
- ✓ *Using the nondominant hand, palpate the femoral artery in 2 to 3 places to get a sense for the path of the artery*
 - ▶ *The femoral vein lies just medial to the artery and typically follows in parallel*
- ✓ *Inject a local anesthetic (1% lidocaine) in the area of the venipuncture site and the tissues deep to the venipuncture site*

- ✓ Always withdraw the plunger slightly before injecting lidocaine to avoid injecting intravascularly
- ✓ Flush all ports of the catheter with normal saline or heparinized saline prior to the procedure
- ✓ The technique for placing the catheter is called the **Seldinger technique**
- ✓ Attach the insertion needle to a syringe partially filled with saline
- ✓ With the bevel of the insertion needle facing up (toward the ceiling) hold the needle and syringe at a 30- to 45-degree angle directing the needle toward the patient's umbilicus
- ✓ Puncture the skin approximately 1-2 cm distal to the inguinal ligament in the location of the femoral vein (just medial to the palpated femoral artery)
- ✓ As the needle is slowly advanced, gently draw back on the syringe
- ✓ When a free flow of blood appears remove the syringe
- ✓ Insert the guidewire into the needle
- ✓ The guidewire should pass easily through the needle into the vessel
- ✓ Leave the distal end of the guidewire exposed
- ✓ If resistance is met, redirect the needle or remove the guidewire and needle, apply pressure until the bleeding stops and begin the process over
- ✓ **Do not force the guidewire into place**
- ✓ Using the scalpel, make a small (approximately 2 mm) incision at the venipuncture site
- ✓ Apply gentle pressure to the venipuncture site and remove the needle, leaving the guidewire in place
- ✓ If the catheter is larger than 3F, thread the tissue dilator over the wire and advance it into the venipuncture site
- ✓ It may be necessary to use a twisting action to advance the tissue dilator

- ✓ *Remove the tissue dilator*
- ✓ *The guidewire should be exposed and visible continuously*
- ✓ *Hold the guidewire with the nondominant hand and thread the catheter over the guidewire until the guidewire protrudes from the distal end of the catheter*
- ✓ *It may be necessary to hold a little pressure at the insertion site if there is a significant amount of bleeding*
- ✓ *Take the end of the guidewire with the non-dominant hand and pass the catheter over the wire and into the vessel*
- ✓ *Remove the guidewire*
- ✓ *There should be good blood flow from all ports*
- ✓ *Flush each port with attention to removing any air bubbles before flushing*
- ✓ *Suture the catheter in place*
- ✓ *Cap off the catheter or attach it to IV tubing*

Monitoring

- If the catheter is correctly placed in the femoral vein, blood flow from the catheter should be steady, but not pulsatile
- Verify correct placement by obtaining a venous gas and documenting an appropriate venous saturation
- However, remember that placement confirmation using a blood gas can be unreliable in patients who have significant cardiopulmonary disease whose arterial oxygen saturation may be abnormally low or in hyperoxygenated patients whose venous oxygen saturation may be abnormally elevated
- A more reliable method for verifying catheter placement is to transduce the catheter and confirm a venous wave form and pressure

Complications

During Catheter Placement

- Bleeding, the main complication, often results from inadvertent artery puncture
- Local hematomas
- Bowel or bladder perforation
- Air embolus
- Catheter embolus
- Creation of an arteriovenous fistula

With Catheter in Place

- Infection
- Current recommendations for limiting infectious complications include the following:
 - ✓ Strict adherence to sterile technique during catheter placement
 - ✓ Use of 2% chlorhexidine-based preparation in patients older than 2 months of age
 - ✓ Use of a catheter with the fewest number of lumens that is essential for treatment
- Swelling of the lower extremity, resulting from impaired venous return
 - ✓ The swelling can often be managed by elevating the leg
 - ✓ Care should be taken to confirm palpable distal pulses in such cases
- Deep venous thrombosis or inferior vena cava thrombosis
- Catheter knotting
- Catheter malposition (i.e., insertion into the lumbar venous plexus, a potentially lethal complication)

Follow up

- ✓ The catheter should be removed as soon as it is not needed
- ✓ When catheter is no longer needed, remove the sutures and pull the catheter slowly and carefully
- ✓ Apply pressure to the insertion site until the bleeding stops

UMBILICAL VEIN CATHETERIZATION

Indications

- Emergency vascular access for fluid and medications when iv access is failed
- Administration of high glucose concentration and total parenteral nutrition
- Central venous pressure monitoring
- Exchange transfusion

Contraindications

Absolute

- Omphalitis
- Omphalocele
- Gastroschisis
- Necrotizing enterocolitis
- Umbilical surgery
- Peritonitis

Equipments

- Sterile catheter
 - ✓ Use 3.5F catheter for patients weighing < 1500 g
 - ✓ Use 5F catheter for patients weighing > 1500 g
- Sterile umbilical catheter tray includes the following
 - ✓ Sterile drapes
 - ✓ Povidone-iodine swabs
 - ✓ Umbilical tie
 - ✓ Toothed iris forceps
 - ✓ 2 curved non-toothed haemostats
 - ✓ Suture scissors
 - ✓ Small needle holder
 - ✓ 3-0 silk suture on small curved needle
 - ✓ 3-way stopcock with Luer-Lok
 - ✓ 3-mL and 1-mL syringes with needles
 - ✓ 2 × 2 gauze
 - ✓ 4 × 4 gauze
 - ✓ Saline solution with heparin 1 unit/ml

Patient preparation and positioning

- Place the infant in the supine position, and secure the upper and lower extremities
- Place the infant on a radiant warmer
- Place chest leads for continuous cardiorespiratory monitoring and a sensor for pulse oximetry monitoring throughout the procedure

Length of the tube inserted

- 2/3 of the distance from tip of the shoulder to umbilicus
- Approximately 5-7 cm
- Until the blood comes through the tube

Procedures

- ✓ *Carefully clean the cord and surrounding skin with povidone-iodine and alcohol solutions*
- ✓ *Place an umbilical tie at the base of the umbilicus to control bleeding*
- ✓ *Drape the infant with sterile drapes with head and feet visible*
- ✓ *Cut the cord horizontally with a scalpel, approximately 1-2 cm above the skin*
- ✓ *Identify vessels- usually 2 arteries and 1 vein. The vein is a large, thin-walled gaping vessel lying superiorly*
- ✓ *Grasp the umbilical stump on either side with the curved haemostats*
- ✓ *Remove visible clots in the lumen with a forceps*
- ✓ *Gently insert the tip of the iris forceps into the lumen of the vein and dilate as needed. In general, minimal dilation is needed*
- ✓ *Insert the heparinized saline-filled catheter attached to a stopcock and syringe into the vessel while applying gentle traction on the cord*
- ✓ *Advance in a cephalad direction to the estimated catheter length*
- ✓ *Aspirate gently. If there is smooth blood flow, secure in place and obtain chest and abdominal radiographs to verify position*
- ✓ *If the catheter meets resistance before achieving its estimated distance, it has most likely entered the portal system or an intrahepatic branch of the umbilical vein*

Monitoring

- The ideal location of the tip of the umbilical catheter is T9-10, just above the right hemidiaphragm and below the heart
- On a radiograph, the catheter will lie to the right of the vertebral column in the inferior vena cava

Complications

- Hemorrhage
 - ✓ From displacement of catheter or perforation of the umbilical artery
- Infection
 - ✓ Especially portal vein thrombophlebitis
- Cardiac arrhythmias, tamponade, perforation, or thrombotic endocarditis
 - ✓ Due to catheter malpositioned in heart or great vessels
- Hepatic necrosis
 - ✓ Due to catheter malpositioned in the portal system, especially if hypertonic solutions are infused into the liver tissue
- Air embolism
 - ✓ If the catheter is inadvertently opened to the atmosphere

NB. = How to differentiate umbilical artery Vs umbilical vein?

Umbilical artery	Umbilical vein
▪ <i>Two in number</i> ▪ <i>Narrow lumen</i> ▪ <i>Thick wall</i> ▪ <i>Cord like to touch</i> ▪ <i>Lies at around 5 and 7 o'clock</i>	▪ <i>Only one in number</i> ▪ <i>Wide lumen</i> ▪ <i>Thin wall</i> ▪ <i>Soft to touch</i> ▪ <i>Visible dark red blood clot</i> ▪ <i>Usually lies superiorly at around 12 o'clock</i>

EXCHANGE TRANSFUSION OF NEW BORN

Indications

- If phototherapy fails to lower UCB
- If risk of kernicterus exceeds risk of procedure
- If newborn has sign of kernicterus
- When UCB is reached to level of exchange transfusion
- To treat coagulopathy due to DIC and life-threatening metabolic disorders
- To treat Polycythemia
- To treat Severe anemia

Contraindications

- When patient is unstable and the risk of the procedure outweighs the possible benefit

Types of exchange transfusion

1. Partial volume exchange transfusion..... → Polycythemia

- Total volume of blood removed is replaced by same volume of NS
- When?

☐ Hct ≥ 65% in symptomatic infants

☐ Hct ≥ 70% in asymptomatic infants

- Volume(ml) to be exchanged

$$= \frac{(\text{observed Hct} - \text{desired Hct}) \times (85 \text{ ml/kg} \times \text{wt(kg)})}{\text{observed Hct}}$$

- The desired Hct should be 50 to 55%
- (85ml/kg x wt(kg)) = blood volume of the neonate

2. Volume to volume exchange transfusion..... → Severe anemia

- Total volume of blood removed is replaced by same volume of Packed RBCs

3. Double volume exchange transfusion → Hyperbilirubinemia

- Twice the total blood volume of the neonate is removed and replaced by donor blood
- ***Volume(ml) to be exchanged = 2 × blood volume of the neonate***
- Replaces 85% of circulating newborn RBC
- Reduces bilirubin level by half of the previous pre-exchange value
- Besides removing bilirubin corrects anemia

Patient preparations

- Discuss with the family
- Stop feedings before the procedure
- Modified whole blood (red cells and plasma) that is type O⁻ cytomegalovirus negative, irradiated, cross-matched against the mother, and compatible with the infant
 - ✓ Should be 2 times the infant's blood volumes (1 blood volume in a term infant is 80 ml/kg) as well as enough volume to prime the tubing and blood warmer is needed.
- As cross-matching to both the mother and infant may require additional time, blood should be ordered before the infant meets criteria for the procedure
- Place infant on warmer with total accessibility and controlled environment
- Restrain infant suitably. Sedation and pain relief are not usually required
 - ✓ Connect physiologic monitors and establish baseline values (temperature, respiratory and heart rates, oxygenation)
 - ✓ Empty infant's stomach
 - ✓ Start IV line for glucose and medication infusion
- Stabilize infant prior to starting exchange procedure

Precautions

- *Do not start exchange procedure until personnel are available for monitoring and as backup for other emergencies*
- *Use blood product appropriate to clinical indication. Use freshest blood available, preferably <5 to 7 days*
- *Check potassium level of donor blood if patient has hyperkalemia or renal compromise*
- *Monitor infant closely during and after procedure*
- *Do not rush procedure*
- *Use only thermostatically controlled blood-warming device that has passed quality control for temperature and alarms. Be sure to review operating and safety procedures for specific blood warmer. Do not overheat blood, i.e., beyond 38°C.*
- *Do not apply excessive suction if it becomes difficult to draw blood from line. Reposition line or replace syringes, stopcocks, and any adapters connected to line.*
- *Leave anticoagulated, banked blood in line or clear line with heparinized saline if the procedure is interrupted.*
- *Clear line with heparinized saline if administering calcium*

Procedures

Push-pull technique: Central access usually through umbilical venous catheter.

Isovolumetric exchange:

- ✓ *With simultaneous infusion of donor blood through venous line and removal of baby's blood through arterial line*

- ✓ *This technique may be better tolerated in sick or unstable neonates because there is less fluctuation of blood pressure and cerebral hemodynamic.*
- ✓ *The technique is also favoured when only peripheral vascular access is available or preferred for various reasons*
 - *Infusion of donor blood may be through umbilical venous catheter or peripheral intravenous catheter*
 - *Removal of baby's blood may be from umbilical arterial or venous catheter, or peripheral arterial catheter, usually a radial arterial line.*

Techniques

- ✓ *Scrub as for major procedure. Wear mask, head cover, sterile gown, and gloves*
- ✓ *Open preassembled equipment tray, using aseptic technique*
- ✓ *Identify positions on special stopcock in clockwise rotation*
- ✓ *Have an assistant document all vital signs, volumes, and other data on the exchange record*
- ✓ *Check peripheral glucose levels every 30 to 60 minutes. Monitor cardiorespiratory status, continuous pulse oximeter. Determine blood gases as often as indicated by pre-existing clinical condition and stability*
- ✓ *Draw blood for diagnostic studies*
- ✓ *Usual rate of removal and replacement of blood during the ET is aliquots of approximately 5 ml/kg over a 2- to 4-minute cycle*
- ✓ *If infant is hypovolemic or has low CVP, start exchange with transfusion of aliquot (5 ml/kg) into catheter. If infant is hypervolemic or has high CVP, start by withdrawing precalculated aliquot*
- ✓ *Remeasure CVP if indicated. Expect rise as plasma oncotic pressure increases, if CVP is low at start*

- ✓ *Ensure that the stages of drawing and infusing blood from and into the infant are done slowly, taking at least a minute each to avoid fluctuations in blood pressure*
- ✓ *Gently agitate the blood bag every 10 to 15 minutes to prevent red cell sedimentation, which may lead to exchange with relatively anaemic blood towards the end of the exchange*
- ✓ *Perform calculated number of passes, until desired volume has been exchanged*
- ✓ *Be sure there is adequate volume of donor blood remaining to infuse after last withdrawal, if a positive intravascular balance is desired*
- ✓ *Clear umbilical line of banked blood and withdraw amount of infant's blood needed for laboratory testing, including re-cross-matching*
- ✓ *Infuse IV fluids with 0.5 to 1 U heparin/ml of fluid through UVC if further ET is anticipated*
- ✓ *Total duration for double-volume ET: 90 to 120 minutes*
- ✓ *Document procedure in patient's hospital record.*

Post-exchange

- Continue to monitor vital signs closely for at least 4 to 6 hours
- Rewrite orders; adjust any drug dosages as needed to compensate for removal by exchange
- Keep infant NPO for at least 4 hours. Restart feeds cautiously if clinically stable. Monitor abdominal girth and bowel sounds every 3 to 4 hours for next 24 hours if exchange has been performed using umbilical vascular lines. Observe for signs of feeding intolerance
- Monitor serum glucose levels every 2 to 4 hours for 24 hours.

- Repeat blood gases as often as clinically indicated.
- Measure serum ionized calcium levels and platelet counts in sick infants immediately after the ET and then as indicated.
- Repeat haemoglobin, hematocrit, and bilirubin measurements approximately 4 hours after exchange, and further as clinically indicated

Complications

Immediate

- Cardiovascular instability - arrhythmia
- Hypoglycemia
- Hypocalcemia
- Hypomagnesemia
- Acidosis, alkalosis
- Hyperkalemia
- Anemia/polycythemia
- Thrombocytopenia
- Infection

Long term

- Necrotizing enterocolitis
- Portal and hepatic vein thrombosis
- Blood-born infection→HIV, Hepatitis B, C, CMV,...

SUBDURAL TAP

Indications

- Evacuations of subdural blood or fluid in young infants, when such collections cause symptoms (i.e., seizures, unilateral paresis) from increased intracranial pressure
- To sample convexity subdural collection for hematologic, microbiologic, and biochemical studies
- To diagnose acute subdural collection over the cerebral convexities (haemorrhage, effusion, empyema)

Contraindications

- Clinical instability when risk exceeds potential benefit
- Uncorrected thrombocytopenia or bleeding diathesis
- Infection in the skin or underlying tissue at or near the puncture site

Equipment's

- Subdural or spinal needle (19 or 20 gauge)
- 10 ml syringe
- Razor blade
- Povidone–iodine solution
- 70% alcohol
- Sterile gauze
- 1% lidocaine with epinephrine

Procedures

- ✓ *Prepare the infant for the procedure in the supine position*
- ✓ *Head should be face up*
- ✓ *Shave the scalp in an area around the lateral boundaries of the anterior fontanel.*
- ✓ *Wearing sterile gloves, palpate the coronal suture at the lateral aspect of the anterior fontanel. If the fontanel opening is extremely small, move several millimetres farther laterally in the coronal suture. Inject local anesthetic (i.e., 1% lidocaine with epinephrine) in the conscious child.*
- ✓ *Grasp a 19- or 20-gauge subdural or spinal needle by the hub and check its patency. Hold it between the thumb and index finger, and rest the heel of the hand against the infant's scalp*
- ✓ *Puncture the skin at a right angle to surface, stretching it slightly to obtain a Z-track. Advance the needle through the puncture site between the edges of the coronal suture until the feeling of resistance lessens.*

- ✓ *Then, remove the stylet to allow fluid or blood to drain; 10 to 15 ml can be safely evacuated from each side.*
- ✓ *Normally, the needle is not advanced more than 5 to 8 mm below the scalp's surface. Some infants may require bilateral taps*

Complications

- Intracranial haemorrhage
- Contusion of the cerebral cortex
- Subgaleal collection of fluid or blood
- Infection

THORACENTESIS

Indications

- Therapeutic drainage of pleural effusion in patient with respiratory compromise when fluid is unlikely to reaccumulate
- Diagnostic evaluation of pleural effusion of unknown etiology
- Therapeutic removal of small pneumothorax

Contraindications

Relative

- Skin infection (eg, herpes zoster) at site of insertion
- Bleeding diathesis, anticoagulant therapy
- Mechanical ventilation

Equipments

- Sterile gloves, mask, and gown
- Iodinated skin preparation with sterile sponges
- Sterile towels

- Local anesthetic (1%lidocaine without epinephrine)
- 5-ml syringe with 25-gauge needle
- 18-gauge 2-inch needle
- 18-20 gauge angiocatheter
- Collection basin
- 3-way stopcock
- 20-60-ml syringe

Common sites

- If pneumothorax, @
 - ✓ 2nd intercostal space over the mid clavicular line or
 - ✓ 4th intercostal space over anterior axillary line
- If pleural effusion, @
 - ✓ 6th or 7th intercostal space just distal to the scapular tip in the midscapular line or posterior axillary line

Patient preparation

- Patient should have intravenous access
- Oxygen should be available
- Monitor oxygen saturation with pulse oximetry
- Younger patients may need sedation for procedure
- Explain procedure in a developmentally appropriate manner before and during procedure

Patient positioning

- Pleural effusion
 - ✓ Sitting upright with arms supported on table in front of patient
 - ✓ Lying in lateral decubitus position with effusion side down
- Pneumothorax
 - ✓ Supine with head of bed up 30 degrees

Procedures

Locate effusion

- Chest radiograph
- Manual percussion to find onset of dullness
 - ✓ Ideal location is 1-2 cm (about 1 intercostal space) below onset of dullness
- Ultra sonogram marked location
 - ✓ Mark location of effusion with the patient in the same position as necessary for procedure
 - ✓ If possible, do not move patient after marking the location because the fluid may shift

Prepare sterile field

- Cleanse area in sterile fashion
- Drape surrounding area with sterile towels

Numb the area

- Use a 25-gauge needle and 5-ml syringe to infiltrate the skin and make a wheal under the skin

- *Change needle to 18 gauge with 2-inch needle*
- *Going over top of sixth rib, infiltrate through wheal, overtop of rib to anesthetize the periosteum, and into pleural space*
 - ✓ *Be sure to aspirate first, and know when you are in the pleural space*
 - ✓ *The parietal pleura needs to be anesthetized, but, to avoid a puncture of the lung, do not advance the needle further*
 - ✓ *When in the pleural space, a 'pop' may be felt and fluid or air will enter syringe*

Removal of Pleural Effusion for Diagnostic Evaluation

- *Remove lidocaine syringe and needle to outside the pleural space, with needle still inserted but outside the pleural space; replace syringe with empty 20-60 ml syringe*
 - ✓ *Reinsert needle into pleural space while applying gentle negative pressure on syringe*
 - ✓ *When in pleural space, a 'pop' may be felt and fluid or air will enter syringe*
- *Remove effusion into syringe*
- *Remove needle and apply bandage to area*

Therapeutic removal of pleural effusion

- *Completely remove needle and syringe filled with lidocaine*
- *Insert angiocatheter into same track and enter pleural space while applying gentle negative pressure*
 - ✓ *When in pleural space, a "pop" may be felt and fluid or air will enter syringe*
- *Remove inner needle, leaving catheter in place*

- ✓ *Ensure that the stopcock is closed to pleural space and chest wall or place a finger over the end of catheter to avoid introducing air into chest wall and creating a pneumothorax*
- *Withdraw fluid*
 - ✓ *Withdraw syringe full of fluid, close stopcock to chest wall and pleural space and drain into collection basin*
 - ✓ *Repeat withdrawal of fluid until desired amount has been removed*
- *Remove angiocatheter and apply bandage to area*

Interpretations

- ➡ *See “pleural effusion analysis” on “respiratory system examinations”*

Complications

- Pneumothorax
- Bleeding: from intercostal vessel creating subcutaneous hematoma or hemothorax
- Hypoxia
- Pulmonary edema
- Puncture of lung, liver or spleen
- Infection

NB.

- ✓ *Insert needle over top of the rib since the neurovascular bundle is on the lower caudal edge of the rib*
- ✓ *The normal pleural effusion found in the pleural space is ≤ 1 ml*

CHEST TUBE INSERTION

Indications

- Pneumothorax
- Hemothorax
- Chylothorax
- Empyema
- For pleurodesis

Contraindications

Relative

- Bleeding diathesis
- Mechanical ventilation
- Presence of adhesions: may require pleurodesis
- Skin infection over the insertion site
- Transudative effusions which can resolve by diuretics

Insertion site

- @ 5th intercostal space over anterior axillary line
- @ 2nd intercostal space over midclavicular line for small pneumothorax or
- @ safe triangle
 - Boundaries
 - ✓ Posterior border of pectoralis major
 - ✓ Superior border of the 5th rib&
 - ✓ Anterior border of latissimus dorsi

Equipments

- Sterile gloves, mask, and gown
- Iodinated skin preparation with sterile sponges
- Sterile towels
- Local anesthetic (1% lidocaine without epinephrine)
- 5-mL syringe with 25-gauge needle
- 18-gauge 2-inch needle
- #10 scalpel with handle
- Chest tube and Kelly clamp for large bore insertion
- Pleurevac or other drainage system, including all connectors necessary to connect to chest tube and to suction
- Suction
- Needle holder
- Suture scissors
- 2-0 silk suture
- 4 × 4 gauze

- Transparent occlusive dressing

Patient preparation

- Patient should have intravenous access
- Oxygen should be available
- Monitor oxygen saturation with pulse oximetry
- Younger patients may need sedation or anesthesia for procedure, especially with large bore chest tube insertion
- Explain procedure in a developmentally appropriate manner before and during procedure

Patient positioning

- Patient lying on bed with head of table elevated 30 degrees with arm above head

Procedures

Prepare Sterile Field

- ✓ *Cleanse area in sterile fashion*
- ✓ *Drape surrounding area with sterile towels.*

Numb the Area

- ✓ *Use 25-gauge needle and 5-mL syringe to infiltrate skin and make wheal under skin*
- ✓ *Change needle to 18 gauge with 2-inch needle*

- ✓ *Infiltrate through wheal, over top of rib, to anesthetize the periosteum, and into pleural space*
 - *Be sure to aspirate first, and know when you are in the pleural space*
 - *The pleura needs to be anesthetized, but, to avoid a puncture of the lung, do not advance the needle further*

Insertion of Chest Tube by Seldinger Technique

- ✓ *Remove syringe from needle*
- ✓ *Pass guidewire through needle into pleural space*
- ✓ *Remove needle (while always maintaining a hold on the guide wire)*
- ✓ *Make small incision at site of insertion (large enough to pass chest tube)*
- ✓ *Starting with smallest dilator, insert dilator over guide wire using a twisting motion (while always maintaining a hold on the guide wire)*
- ✓ *Repeat with larger dilators over guide wire until track is large enough to easily pass chest tube (while always maintaining a hold on the guidewire)*
- ✓ *Insert chest tube over guidewire until all port holes are within the pleural space*
- ✓ *Remove guidewire*
- ✓ *Suture chest tube to chest wall*
- ✓ *Connect tube to drainage device with suction at 15-20 cmH₂O*
- ✓ *Apply sterile 4 x 4 dressing and transparent occlusive dressing*

Insertion of Chest Tube by Blunt Dissection Technique

- ✓ *Remove needle used for local anesthesia*
- ✓ *Using scalpel make ~1-2-cm incision through the skin and subcutaneous tissue (large enough to pass chest tube)*
- ✓ *Insert Kelly clamp and tunnel up 2 intercostal spaces I the subcutaneous space*
- ✓ *Push through the intercostal muscle superior to the rib with the Kelly clamp and enter the pleural space; air or fluid may rush out*

- ✓ *Spread the clamp to widen the area to allow for the chest tube*
- ✓ *Remove clamp*
- ✓ *Insert gloved finger into tract and ensure correct location and lyse any adhesions*
- ✓ *Using the Kelly clamp attached to the chest tube as a guide, insert the chest tube into the pleural space*
 - *If to drain air, guide anterior and superior (turning the clamp so that the curve is upward will assist with this guidance)*
 - *If to drain fluid, guide posterior and inferior (turning the clamp so that the curve is down will assist with this guidance)*
- ✓ *Advance chest tube until all ports are within the pleural space*
- ✓ *Suture chest tube to chest wall*
- ✓ *Connect tube to drainage device with suction at 15-20 cmH₂O*
- ✓ *Apply sterile 4 x 4 dressing and transparent occlusive dressing*

To check functionality

- *If effusion*
 - ✓ *Continuous bubbling in the bottle*
- *If pneumothorax*
 - ✓ *Oscillations in the bottle*

When we remove chest tube?

- *If effusion*
 - ✓ *When the lung is fully expanded*
 - ✓ *Fluid output is less than 200 ml/day*
- *Pneumothorax*

- ✓ When the lung is fully expanded
- ✓ No visible air leak is present and air does not accumulate when suction is removed

Complications

- Improper position
- Subcutaneous emphysema
- Hemorrhage – due to intercostal vessel injury
- Puncture of lung, liver or spleen
- Infection
- Intercostal neurovascular injury

Follow up

- ✓ Obtain chest radiograph to ensure correct placement

PERICARDIOCENTESIS

Indications

- Therapeutic
 - ✓ Impending cardiac tamponade
- Diagnostic
 - ✓ Infectious pericarditis
 - ✓ To rule out an oncologic process
- Compromise in the patient's hemodynamic status

Contraindications

Relative

- Blood dyscrasia
 - ✓ May have a significant bleeding
 - Site infection
 - Effusion due to aortic dissection
 - Significantly elevated diaphragm
 - Grossly enlarged liver
 - Profound ascites
- > may change the standard landmarks of needle insertion*
under such circumstances, use the intercostal approach

Equipments

- Povidone-iodine or equivalent sterilization substrate
- 1% or 2% lidocaine or xylocaine
- 25-gauge, 1.5 inch long needle
- 16 or 18 gauge needle, ≥ 1.5 inch
- Floppy tip wire that can be introduced through the needle
- Pigtail catheter with multiple side holes as well as an end hole
- Scalpel
- 3-way stopcock
- 30 ml or 60 ml syringe and suture kit
- ECG monitor, pulse oximeter, and blood pressure cuff

Patient preparation

- Prepare and drape the subxiphoid area in the usual sterile fashion
- If the subxiphoid approach might be difficult, consider preparing the left sternal border
- All equipment should be readily available and an assistant should be available to help with manipulation of needles, wires, and catheters

Patient position

- Supine position, with 30-45 degrees of reverse Trendelenburg
- Occasionally, the partially sitting position may be required or beneficial
 - ✓ Makes an orthopedic patient more comfortable
 - ✓ May allow for most of the pericardial fluid to position inferiorly or closer to the drainage site

Procedures

- *Prepare the subxiphoid area and left sternal border*
- *Administer 1% or 2% lidocaine approximately 0.5-1 cm below the left costoxiphoid angle using a 25- or 27-gauge 1.5-inch-long needle*
 - ✓ *Infiltration of the lidocaine should be superficial as well as deep, pushing the needle superiorly, posteriorly, and leftward*
 - ✓ *Withdraw fluid each time the needle is passed deeper within the skin and subcutaneous tissues*
- *To allow for easier passage of the needle, precut the skin with the scalpel before introducing the 16- or 18-gauge 1.5-inch to 2.5-inch needle*
- *Insert the larger needle at an approximate 30-45-degree angle with the abdomen with constant negative pressure on the syringe. Depth of the needle in children is 4-5cm*
- *Monitor ECG very carefully for evidence of dysrhythmias or ST segment changes (evidence of coronary or myocardial injury)*
- *Slowly insert the needle until fluid is withdrawn*
 - ✓ *Suspect cardiac perforation if the fluid is grossly bloody*
 - ✓ *Serous fluid confirms that the needle has passed into the pericardial space*
 - ✓ *It is not unusual to feel the needle pass through the pericardium*
- *Fix the needle into position once pericardial fluid is extracted*

- *Pass the floppy tip wire through the needle with the intent of passing the wire deep within the pericardium and into the posterior pericardial space*
- *Because the wire may irritate the epicardium, ventricular ectopy is not uncommon*
- *Once the wire is secured deep within the pericardium, remove the needle*
- *Make a larger incision in the skin adjacent to the wires so that the catheter can be inserted*
- *Insert the soft-tipped multiple side hole or pigtail catheter over the wire and secure it in the posterior pericardial space*
- *Connect the catheter to a 3-way stopcock, and fluid should be extracted slowly, monitoring for blood pressure and ectopy*
- *Continuous echocardiographic monitoring is useful for location of the wire and catheter as well as for monitoring adequate extraction of pericardial fluid*
- *Once catheter position is confirmed, the catheter should be secured with sutures and the entire site covered sterilely to minimize infection*

Interpretations

- Send pericardial fluid for cell count, protein, glucose, lactic dehydrogenase, cytology as well as all other studies for infectious agents
- Normal pericardial fluid is clear to straw colored, scant (< 50 ml) with < 500 white blood cells/mcl

- An elevated white blood cell count suggests either an infectious or inflammatory process
- Protein, glucose, and lactic dehydrogenase can be helpful in differentiating a transudate from an exudate
- Metastasis to the pericardial space or pericardial tumors are frequently exudative, with abnormalities seen on cytology

Monitoring

- Monitor the patient closely for rhythm disturbances and unstable blood pressure
- Patients may require fluid resuscitation if large amounts of fluid are extracted from the pericardial space
 - ✓ Remove the fluid slowly
 - ✓ Replace with isotonic fluid, if possible
 - ✓ Large children with nephrotic syndrome can have as much as 1–2 L of pericardial fluid and be relatively asymptomatic

Complications

- Compromise in the hemodynamic status
- Bleeding
 - ✓ Can be superficial and easily controlled with pressure
 - ✓ Deeper bleeding from either a liver or splenic injury may be less obvious and more difficult to control.

- A coronary injury is rare but potentially catastrophic (as is a cardiac perforation)
 - ✓ Requires emergent intervention by a cardiac surgeon
- Arrhythmias are typically transient
 - ✓ Can be managed by repositioning the needle, wire, or catheter
 - ✓ Occasionally, more persistent rhythm disturbances occur that require antiarrhythmic therapies
- Vasovagal response after successful pericardial decompression or left ventricular dysfunction can occur
- Pneumoperitoneum or small pneumothorax
 - ✓ Requires careful monitoring but can be self-limited
- Pneumopericardium
 - ✓ should resolve as long as the pericardiocentesis catheter is secure, in proper position, and connected to negative pressure

NASOGASTRIC TUBE INSERTION

Indications

- Decompression of upper GIT
- Gastric lavage - for poisoning or overdose of medications
- Enteral feeding – for pts unable to eat by mouth or swallow sufficient diet
- Administration of medications – for pts unable to swallow medications
- To obtain specimen from gastric contents
- After bowel surgery - to promote healing
- To administer radiographic contrast media to GIT imaging

Contraindications

Absolute

- Unstable airway
- Intestinal perforation
- Cervical spine trauma
- Severe mid facial trauma
- Recent nasal surgery

Relative

- Coagulopathy (prothrombin time > 18 seconds)
- Thrombocytopenia (platelet count < 100,000/mcl)
- Recent intestinal tract surgery (< 1 month ago)
- Esophageal stricture

Equipments

- Lubricant gel
- Nasogastric (NG) tube
 - ✓ Larger diameter, polyethylene NG tube for suction and decompression
 - ✓ Smaller diameter, silicone NG tube for enteral feeding
- Water or normal saline at room temperature
- Drainage bag or feeding pump
- 60-ml catheter tip syringe
- Stethoscope

Patient preparation and position

- Explain indication and risks to the patient and parents
- Inform the patient of the intention of the procedure
- Patient should be sitting

Procedure

- ✓ *Measure the length of insertion from the nares to the ear and to the epigastrium - mark it on the tube with an indelible pen*
- ✓ *Lubricate tube with gel*

- ✓ *Insert the tube through the nose*
- ✓ *Ask the patient to cooperate by swallowing while the tube is being inserted*
- ✓ *Advance the tube to the length mark*
- ✓ *Secure tube to the face with tape*

Length of the tube inserted

- Measure length of tube insertion by positioning the tube from the nares or mouth to the ear, then to the umbilicus.

How to check the tube is in the stomach

- Aspirate the tube with 50-ml syringe
 - ✓ Gastric aspirate (pH = 1–3) confirms positioning in stomach
- Insert small amount of air (20–30 ml) via NG tube
 - ✓ While listening to epigastric area of stomach with stethoscope → gurgling sound in the epigastrium → tube is in the stomach
- Chest x-ray

Monitoring

- Monitor intake and output volume
- Evaluate tube position
- Patient symptoms

Complications

- Discomfort
- Sinusitis (caused by long-term NG tube feeding)
- Bleeding
- GERD
- Esophageal
- Malposition (respiratory tree insertion)

Follow up

Follow for the following sign and symptoms

- ✓ Fever
- ✓ Nausea and vomiting
- ✓ Melanotic stool or bright red hematemesis
- ✓ Persistent abdominal pain
- ✓ Abdominal distention
- ✓ Chest pain

ABDOMINAL PARACENTESIS

Indications

- Diagnostic sampling of ascitic fluid : eg
 - ✓ Internal bleeding following blunt abdominal trauma
 - ✓ Chylous ascites after surgery
 - ✓ Rule out malignancy
 - ✓ Identification of infectious organism in spontaneous bacterial peritonitis
- Therapeutic removal of the ascitic fluid : eg
 - ✓ Chylous ascites
 - ✓ Tense ascites
 - ✓ Intestinal lymphangiectasia

Contraindications

Absolute

- Unstable airway
- Hemodynamically unstable patient
- Intestinal perforation

Relative

- Infection of the abdominal wall
- Coagulopathy (prothrombin time > 18 seconds)
- Thrombocytopenia (platelet count <100,000/mcl)
- Recent intestinal tract surgery (< 1 month ago)

Equipment

- Alcohol swabs, povidone-iodine
- 23-gauge and 21-gauge needles or angiocatheters with syringes
- Local anesthetic (eg, 1% lidocaine)
- Large bore needle with plastic catheter
- Sterile containers for fluid collection
- Appropriate culture tubes for microorganisms

Patient preparations and position

- Explain indication and risks to the patient and parents
- Inform the patient of the intention of the procedure
- Supine or side

Site of paracentesis

- The preferred site is in the midline approximately one third of the distance from the umbilicus to the symphysis pubis
 - ✓ Right lower quadrant at McBurney point is the preferred site. Why?

- In infants, the fluid may bulge laterally, and the paracentesis may be obtained laterally to that point

Procedure

- ✓ *The puncture site should be shaved, if necessary, and cleansed with povidone-iodine*
- ✓ *Inject local anesthetic, infiltrating the skin first and then penetrating into deeper layers*
- ✓ *A small 3-mm incision can be made with a scalpel to help insert the needle. Using Z-track technique, insert the tap needle 1-2 inches into the abdomen*
- ✓ *Obtain a sample of fluid or withdraw as much fluid as necessary with a syringe (in case of therapeutic lavage)*
- ✓ *Remove the needle and apply a pressure dressing to the puncture site*
- ✓ *If an incision was made, it may be closed using 1 or 2 stitches*
- ✓ *The ascitic fluid removed may be replaced 1:1 with 5%albumin IV*
- ✓ *Inspect for appearance, turbidity and send for laboratory and cytological study*

Monitoring

- Monitor vital signs
 - ✓ A rapid loss of significant volumes of ascitic fluid may lead to hypotension

Complications

- Pneumoperitoneum
- Perforation to intestine or organ
- Bleeding
- Infection

Follow up

Follow for any of the following sign and symptoms

- ✓ Fever
- ✓ Nausea and vomiting
- ✓ Blood in the stool
- ✓ Abdominal pain
- ✓ Abdominal distention

TIP (TRIADS)

- 1) CHF in infants
 1. Tachypnea
 2. Cardiomegaly
 3. Hepatomegaly
- 2) Sydenham chorea
 1. Chorea
 2. Hypotonia
 3. Emotional liability
- 3) Cardiac tamponade(Beck's triad)
 1. Decreased arterial BP
 2. Raised JVP
 3. Distant heart sound
- 4) Pneumonia
 1. Cough
 2. Fast breathing
 3. Fever
- 5) Croup
 1. Barking cough
 2. Hoarseness of voice
 3. Stridor
- 6) Epiglottitis = 3D's
 1. Drooling
 2. Dysphagia
 3. Dyspnea
- 7) Meningitis
 1. Fever
 2. Neck stiffness
 3. Headache
- 8) Meningitis in infants
 1. Fever
 2. Neck stiffness
 3. Vomiting
- 9) Raised ICP (**Cushing triad**)
 1. Bradycardia
 2. Hypertension
 3. Irregular breathing pattern
- 10) Infectious mononucleosis
 1. Fever
 2. Pharyngitis
 3. Lymphadenopathy
- 11) Congenital syphilis
 1. Skin rash
 2. Hepatosplenomegaly
 3. Lymphadenopathy
- 12) Congenital toxoplasmosis
 1. Chorioretinitis
 2. Hydrocephalus
 3. Intracranial calcification

13) Congenital rubella

1. Microcephaly
2. PDA
3. Cataract

14) Measles

1. Fever
2. Maculopapular rash
3. One of the 3'C's
 - a. Conjunctivitis
 - b. Cough
 - c. Coryza

15) Pellagra = 3D's

1. Dermatitis

2. Diarrhoea

3. Dementia

16) Intussusception

1. Abdominal pain

2. Palpable abdominal mass

3. Blood in the stool

17) Pyelonephritis

1. Fever

2. Vomiting

3. Suprapubic pain

REFERENCES

1. Nelson text book of pediatrics 20th edition
2. Up to date 21.2
3. Hutchison's clinical methods 22nd edition
4. Bate's a guide to physical examination and history taking 6th edition
5. Blue book pediatrics clinical examinations 2011
6. Guidelines for the management of acute malnutrition; (government of Ethiopia FMOH, 2016)
7. Emergency triage assessment and treatment (ETAT): manual for participants (WHO, 2005)
8. The Treatment of diarrhoea : a manual for physicians and other senior health workers - 4th rev (WHO, 2005)
9. Neonatal intensive care unit (NICU) training : participants manual (FMOH of Ethiopia, 2014)
10. Board Review Series (BRS) pediatrics
11. Approach to practical pediatrics 2nd edition
12. Lange : current procedures pediatrics
13. Browse's introduction to the symptoms and signs of surgical disease 4th edition
14. Wikipedia, medscape and different websites
15. Different lecture, seminar, bedside and round notes

**YOU WILL NEVER
WALK ALONE !!!**

Liverpool fc

GOOD LUCK !!!